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Candlesticks and graph patterns
in cryptocurrencies
Bachelor's thesis

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Declaration of Authorship

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Prague, January 7, 2025

Abstract

This thesis investigates the effectiveness of technical analysis, focusing on candlestick patterns, in cryptocurrency markets characterized by high volatility and continuous trading. Using statistical methods, including skewness-adjusted t-test and binomial test, the study evaluates 41 bullish and bearish patterns across five datasets: four datasets covering cryptocurrencies in general (excluding stablecoins) and one specific to stablecoins. Gap-dependent patterns were rare due to the continuous trading nature of cryptocurrency markets. Eight patterns demonstrated predictive potential in the non-stablecoin datasets, though two produced returns contrary to their bearish classification. The most compelling patterns are Hammer Bullish, Rising Window Bullish, On Neck Bearish, and Shooting Star Bearish, which produced returns contrary to its bearish classification, as they appear in three datasets. In contrast, the stablecoin dataset showed Doji Star Bullish and Doji Star Bearish as significant; however, these likely reflect price-stabilization mechanisms rather than intrinsic predictive properties. By leveraging large, diverse datasets and employing modern trend-definition methodology, the study highlights the limited applicability of traditional candlestick patterns and questions the standard labelling of patterns as bullish or bearish in cryptocurrency trading.

JEL Classification G11, G12, G14, G17, C12

Keywords Technical analysis, Candlesticks, Candlestick patterns, Chart patterns, Cryptocurrencies, Bitcoin, Simple moving average, Caginalp-Laurent exit strategy, Skewness adjusted t-test, Hammer, On Neck, Rising Window, Shooting Star

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Abstrakt

Tato práce zkoumá účinnost technické analýzy se zaměřením na svíčkové formace na trzích s kryptoměnami, které se vyznačují vysokou volatilitou a nepřetržitým obchodováním. Pomocí statistických metod, včetně t-testu se zohledněním šikmosti a binomického testu, studie hodnotí 41 býčích a medvědích formací napříč pěti datovými sadami: čtyřmi zahrnujícími kryptoměny obecně (s výjimkou stablecoinů) a jednou specifickou pro stablecoiny. Formace závislé na cenových mezerech byly vzácné kvůli nepřetržitému obchodování na trzích s kryptoměnami. Osm formací prokázalo prediktivní potenciál v datových sadách nezahrnujících stablecoiny, přičemž dvě z nich přinesly výnosy opačné než jejich medvědí klasifikace. Nejvýznamnější formace byly následující: Hammer Bullish, Rising Window Bullish, On Neck Bearish a Shooting Star Bearish, která přinesla výnosy opačné než její medvědí klasifikace, neboť se vyskytly ve třech datových sadách. Naproti tomu v datové sadě stablecoinů se jako významné ukázaly formace Doji Star Bullish a Doji Star Bearish. Pravděpodobně však spíše odrážejí mechanismy stabilizace cen než vnitřní prediktivní vlastnosti. Díky využití rozsáhlých a různorodých datových sad a aplikaci moderní definice trendu studie zdůrazňuje omezenou použitelnost tradičních svíčkových formací a zpochybňuje standardní označení formací jako býčí či medvědí v obchodování s kryptoměnami.

Klasifikace JEL

G11, G12, G14, G17, C12

Klíčová slova

Technická analýza, Svíčkové grafy, Vzory svíčkových grafů, Grafové vzory, Kryptoměny, Bitcoin, Jednoduchý klouzavý průměr, Caginalp-Laurent strategie výstupu, Test t s úpravou na šikmost, Kladivo, Na krku, Stoupající okno, Padající hvězda

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Acronyms

EMH	Efficient Market Hypothesis
SMA	Simple Moving Average
CEX	Centralized Exchange
DEX	Decentralized Exchange
OHLC	Open, High, Low, Close
DJIA	Dow Jones Industrial Average
PF	Profit Factor
MDD	Maximum Drawdown

Chapter 1

Introduction

The effectiveness of technical analysis in classical markets such as stocks, bonds, and commodities as well as in cryptocurrency¹ markets remains a contentious topic within academic circles. Technical analysis entails the study of historical price charts and market data to predict future price movements, premised on the assumption that patterns and trends tend to recur. However, this methodology is met with considerable scepticism, especially from advocates of the Efficient Market Hypothesis, which posits that asset prices reflect all available information, thereby rendering the prediction of future prices based on past data unfeasible due to the influence of unforeseen information (Fama, 2017).

A significant critique of technical analysis is its susceptibility to data mining bias and overfitting. This problem arises when traders recognize patterns in historical data that, despite lacking a robust theoretical foundation, seem effective merely by chance. The extensive datasets available in financial markets facilitate the identification of spurious correlations that may not be sustainable over time or under varying market conditions.

Despite these criticisms, proponents of technical analysis argue for its validity by highlighting the role of psychological and behavioural factors in market dynamics (Shleifer, 2000). They contend that since market movements are influenced by human behaviour, technical analysis can unearth trends in investor sentiment, albeit without guaranteeing consistent prediction of future prices.

¹In this thesis, both "coins" and "tokens" are categorized as cryptocurrencies. Coins, like Bitcoin, operate on their own blockchain and are primarily used as digital money. Tokens, built on existing blockchains like Ethereum, represent various assets or utilities within a platform.

The debate also extends to the distinctions between classical and cryptocurrency markets, which are marked by higher volatility, relatively lower liquidity, and minimal regulation. These characteristics could either undermine or enhance the applicability of technical analysis. The pronounced volatility and speculative nature of cryptocurrencies might lead to more discernible patterns, although the relative infancy of these markets limits the historical data available for analysis. Furthermore, external influences such as regulatory changes or technological advancements can unpredictably affect prices.

Empirical evidence on the efficacy of technical analysis is mixed, with some studies suggesting that certain technical strategies may yield excess returns under specific conditions (Lo et al., 2000), while others find minimal evidence of its predictive capability. This inconsistency contributes to the ongoing debate, indicating that the success of technical analysis may depend on market conditions, the methodologies applied, and the individual practitioner's expertise (Park and Irwin, 2007). Despite these challenges, technical analysis remains widely utilized among traders and investors across various markets, although its academic acceptance is debated due to both theoretical and empirical challenges.

This thesis aims to thoroughly examine candlestick patterns, focusing on those characterized as "bullish" and "bearish." The research is driven by two main objectives. Firstly, it seeks to enrich the existing academic discourse on technical analysis and trading patterns, which currently lacks depth and detail. Despite the widespread use and application of these patterns in financial markets, scholarly exploration of their effectiveness, reliability, and utility for investor guidance is notably insufficient in cryptocurrency markets. This study aims to provide empirical insights and theoretical contributions that significantly advance the field of financial studies.

Secondly, the motivation stems from the observation that platforms like TradingView.com, with a user base exceeding 50 million as reported in 2024 and identified by SimilarWeb (2025) as a top investing website as of December 2024, play a crucial role for investors globally. The prevalent use of this platform underscores the importance of understanding how patterns, especially those labelled with emotionally charged terms like "bullish" and "bearish," influence investor behaviour. This terminology not only affects investment decisions but also may expose investors to increased risks if the predictive efficacy and market implications of these indicators are not comprehensively understood.

Therefore, this thesis endeavours to bridge the existing gap in the literature by providing an in-depth analysis of bullish and bearish patterns within the cryptocurrency market context. It investigates the historical performance of these patterns and their predictive validity, aiming to offer both investors and academics a more nuanced understanding of these technical indicators. By doing so, the study seeks to promote more informed investment practices and improve the practical application of technical analysis in financial decision-making, thus serving the interests of both the academic and investing communities. The focus will primarily be on empirical insights and theoretical advancements without delving into the psychological effects of terminologies on investor decision-making.

This thesis builds upon and expands the foundational work of Ho et al. (2021), serving as a pivotal reference point. It aims to broaden the empirical dataset by incorporating several hundred additional cryptocurrencies, thereby enhancing the understanding of their applicability and efficacy in cryptocurrency trading. The investigation further includes a categorical analysis based on distinct subgroups of cryptocurrencies, delineated primarily by their market capitalization. Through this comprehensive inquiry, the thesis aspires to offer an extensive examination of the role and impact of candlestick patterns, alongside other technical analysis tools, within the dynamic landscape of the cryptocurrency market.

The remaining sections of this thesis are organized as follows: Chapter 2 provides a comprehensive review of the literature relevant to the study. Chapter 3 details the methodology used in this research. Chapter 4 describes the datasets. Chapter 5 presents the research findings and discusses potential limitations. Finally, Chapter 6 concludes the thesis, summarizing the key insights and suggesting avenues for future investigation.

Chapter 2

Literature Review

2.1 Technical Analysis Overview

Technical analysis has its roots in the early trading practices observed in the Netherlands and Japan, where pioneers such as Joseph de la Vega and Munehisa Homma utilized early forms of charting to predict market movements (Lo, 2004; Nison, 2001). The formalization of these practices into a coherent analytical discipline, however, began with the contributions of Charles H. Dow in the late 19th century (Dow, 1920). Dow's introduction of the Dow Theory laid the foundational principles of trend analysis, positing that market prices move in discernible patterns reflective of underlying economic conditions (Edwards et al., 2018).

The subsequent decades saw the refinement and expansion of technical analysis through the works of Richard W. Schabacker, Robert Rhea, and Ralph Nelson Elliott, among others. These contributions introduced new concepts and tools, such as the Elliott Wave Theory and various chart patterns, enriching the analytical toolkit available to market analysts (Schabacker, 2021).

The advent of computational technology in the latter half of the 20th century marked a significant milestone in the evolution of technical analysis. The development of computers facilitated the creation and adoption of quantitative indicators, such as the Relative Strength Index (RSI) and Moving Average Convergence Divergence (MACD), allowing for more nuanced and sophisticated market analyses.

The proliferation of the internet and personal computing in the late 20th and early 21st centuries further democratized access to technical analysis tools, enabling a broader spectrum of investors to engage in market analysis. The integration of machine learning and neural networks in recent years represents the latest frontier in technical analysis, offering unprecedented capabilities in data analysis, pattern recognition, and predictive modeling.

The following literature review is exclusively dedicated to examining the subset of technical analysis that pertains to candlestick patterns.

2.2 Stock Markets

Nison (2001) authored one of the pioneering books on candlestick patterns, which are now widely used in Western countries. In his work, he attempts to explain the underlying logic of these patterns. It is important to note that the definitions of candlestick patterns are somewhat ambiguous, as they were not initially intended for use with pattern recognition algorithms. Nison's book and findings are primarily based on his personal experience with candlestick patterns, which lends a subjective nature to his work. Despite this subjectivity, his insights are valuable and noteworthy. Nison emphasizes that technical analysis using candlestick patterns is inherently subjective due to the lack of standardized definitions. He advises that when applying candlestick patterns, one must consider the current market conditions, specifically whether it is a bull or bear market. He also points out that each market possesses its own unique characteristics, and certain patterns may perform better in specific market types. To enhance the effectiveness of candlestick patterns, Nison recommends integrating them with Western technical signals, such as trendlines, retracement levels, moving averages, oscillators, volume, and open interest.

The study conducted by Caginalp and Laurent (1998) offers one of the first comprehensive analysis of candlestick patterns and their predictive efficacy in financial markets. The research concentrated on eight three-day reversal candlestick patterns, examining the daily prices of S&P 500 stocks from 1992 to 1996. The findings demonstrated notable predictive power: bullish reversal patterns generated an average return of 0.9% per trade, compounding annually to 309%, whereas bearish reversal patterns for short sales produced a 0.27% return per trade, com-

pounding annually to 140%. These results suggest that traders utilizing these patterns could realize substantial profits, thus challenging the Efficient Market Hypothesis.

Marshall et al. (2006) conducted an investigation on stocks included in the DJIA index over the period from January 1992 to December 2002. Their study tested fourteen one-day candlestick patterns and fourteen two or three-day reversal patterns. The findings revealed that candlestick technical analysis is not profitable in the US stock market, as neither bullish nor bearish signals consistently outperformed a buy-and-hold strategy. These results further support the notion that the US stock market is informationally efficient.

Marshall et al. (2007) investigated the profitability of candlestick technical analysis using U.S. equity market data from 1992 to 2002, specifically focusing on all stocks within the Dow Jones Industrial Average (DJIA). The study tested 14 single candlestick lines and 14 reversal patterns. Overall, these candlestick patterns did not provide statistically significant profits when applied to large U.S. stocks. Bullish and bearish patterns such as the Bullish Engulfing and Bearish Engulfing failed to consistently outperform random chance. The results suggest that relying solely on candlestick patterns for trading decisions is not advisable. However, the authors acknowledged the possibility that these patterns might complement other technical trading strategies, though their standalone effectiveness remains questionable.

In a subsequent study, Marshall et al. (2008) conducted a comprehensive investigation into the profitability of candlestick technical analysis on 100 stocks listed on the Tokyo Stock Exchange from 1975 to 2004, including 59 stocks from the TOPIX Large 70 Index and 41 from the TOPIX Mid 400 Index. Dividing the study period into three ten-year sub-periods, the research assessed the impact of market conditions on profitability by considering both bull and bear market phases. Adopting the same methodology and assumptions as in the work by Marshall et al. (2006), the study revealed that candlestick technical analysis is not profitable for large stocks in the Japanese equity market. Neither bullish nor bearish signals consistently outperformed a buy-and-hold strategy across the entire sample, within sub-periods, or during varying market conditions. Notably, candlestick technical analysis was unprofitable even before accounting for transaction costs, strongly suggesting that it offers no value for large stocks in the Japanese equity market.

Goo et al. (2007) analyzed daily data from the 25 component stocks of the Taiwan Top 50 Tracker Fund and the Taiwan Mid-Cap 100 Tracker Fund over the period from 1997 to 2006, examining 26 candlestick patterns. The study revealed that 11 of these patterns achieved a positive mean rate of return on specific holding days, indicating their potential utility for investors. Reversal patterns were found to perform better than one-day candlestick patterns, likely due to their higher informational content. Furthermore, the study suggested that the performance of most candlestick patterns could be enhanced by employing a stop-loss strategy. Among the patterns analyzed, bullish reversal patterns were identified as the most effective, delivering the highest mean rates of return to investors.

Horton (2009) built upon the work of Caginalp and Laurent (1998), incorporating additional statistical tests into his analysis. He utilized daily price data from Commodity Systems Inc. for a sample of 349 companies, randomly selected from the Value Line database, encompassing all major industry groups. Horton examined eight candlestick patterns and ultimately recommended against their use for trading purposes, a conclusion that stands in contrast to the findings of Caginalp and Laurent (1998).

A study by Shiu and Lu (2011) analyzed daily price and volume data for 69 electronic securities listed on the Taiwan Stock Exchange over a ten-year period from 1998 to 2007. The researchers examined six two-day reversal candlestick patterns and found that four of these patterns demonstrated significant predictive power for specific holding periods. Specifically, the bullish harami and bearish harami patterns yielded the most promising results.

Research conducted by Lu et al. (2012) investigated six two-day reversal patterns using the Taiwan 50 component stocks, analyzing data from October 2002 to December 2008. The study determined that bullish reversal patterns generally outperform bearish reversal patterns in terms of profitability. The analysis indicated that the three bullish patterns tested possess significant predictive power in the Taiwan stock market. Unlike previous studies that utilized fixed holding periods, this research implemented a variable holding period strategy, where trades were initiated based on a reversal pattern and held until an opposing pattern signalled an exit.

Lu and Shiu (2012) conducted a comprehensive analysis of 24 two-day candlestick patterns using data from the Taiwan 50 Index component stocks spanning from

January 2002 to December 2009. Their initial findings indicated that 15 out of the 24 patterns demonstrated significant predictive power for at least one of the three holding periods when the prior market trend was disregarded. However, after adjusting for a 1% transaction fee, none of these patterns remained profitable. Further analysis incorporating the prior market trend revealed that three patterns exhibited significant predictive power and profitability in uptrend markets, while one pattern showed similar results in downtrend markets. These findings underscore the importance of considering market trends when employing candlestick patterns for investment strategies.

Lu and Chen (2013) analyzed the effectiveness of candlestick charting in three major European stock markets: the FTSE 100, DAX 30, and CAC 40, using data from 2003 to 2012. The study tested 24 two-day candlestick patterns and found that certain patterns can generate significant profits. Specifically, a bullish continuation pattern, similar to the Flag Consolidation pattern in the DAX 30, and an Engulfing pattern in the CAC 40, produced positive average returns¹ of 1.09% and 1.33%, respectively, over a 10-day holding period. Additionally, a bullish reversal pattern, akin to a Hammer in the FTSE 100, was effective after a downtrend, generating a significant average return of 1.31%. However, the study noted that the profitability of these patterns declined after the 2008 global financial crisis, suggesting that market conditions influence the predictive power of candlestick techniques.

do Prado et al. (2013) conducted a comprehensive analysis of 16 candlestick patterns to assess their effectiveness in predicting stock price movements within the Brazilian stock market from 2005 to 2009. The results revealed that most patterns exhibited limited predictive power, with only a few demonstrating statistical significance. For instance, the Harami-Bullish pattern showed some predictive ability, particularly within the first few days following its occurrence. Conversely, patterns such as the Hanging Man were found to lack any predictive efficacy. The authors concluded that candlestick patterns, as traditionally applied, cannot be universally generalized to the Brazilian market without adaptation, as their performance significantly diverged from studies conducted in the U.S. market. The findings suggest that candlestick techniques may require adjustment to suit local

¹Average Returns: In Lu and Chen (2013), average returns are defined as the percentage change in stock price over a specified holding period. After identifying a candlestick pattern, a trade is simulated by entering at the opening price on the day following the pattern's formation and exiting at the closing price on the last day of the holding period (typically 10 days)

market conditions.

In a separate study, Lu (2014) conducted an analysis of twelve one-day candlestick patterns using data from 151 individual Taiwan stocks over the period from January 1992 to December 2009. Each pattern was assessed in the context of both upward and downward market trends. The results revealed that four out of the 24 combinations (encompassing both bullish and bearish signals) were profitable in the Taiwan stock market. Additionally, the findings suggest that the candlestick approach demonstrates greater efficacy with smaller firms and lower-priced stocks.

Lu et al. (2015) performed an extensive analysis of three trend definitions and four holding strategies across eight three-day reversal patterns. The dataset comprised 26 component stocks of the DJIA index from January 1992 to December 2012. Their findings indicate that irrespective of the trend definition applied, the eight three-day reversal patterns with the Caginalp–Laurent (CL) holding strategy remain profitable, even after accounting for 0.5% transaction fees. The results also support the notion that a three-day holding period is more advantageous than a ten-day holding period, and the CL exit strategy is more effective than the Marshall–Young–Rose (MYR) exit strategy in candlestick trading strategies. Furthermore, the study suggests that candlestick investors can achieve higher profits in more volatile markets.

Chen et al. (2016) examine the predictive efficacy of four widely recognized pairs of two-day bullish and bearish Japanese candlestick patterns within the Chinese stock market from January 2007 to August 2015. The statistical analysis reveals that the predictive power varies among the different patterns. Specifically, three out of the eight patterns demonstrate both short-term and relatively long-term predictive capabilities. Conversely, one pair exhibits significant forecasting power exclusively in the very short-term. The remaining four patterns present inconsistent results across various market value groups. Furthermore, for all four pairs, predictive power diminishes as the forecasting horizon extends, and the efficacy is more pronounced for stocks with medium market value compared to those with large market value.

Utilizing data from the Chinese stock markets from 1999 to 2009, Zhu et al. (2016) conducted a comprehensive analysis of ten bullish and bearish candlestick patterns, with significant findings on their predictive effectiveness. The study

identified bearish harami and cross patterns as strong predictors of downward reversals, especially for low-liquidity stocks. In contrast, bullish engulfing, piercing, and harami patterns were profitable for highly liquid, smaller companies, indicating that market liquidity and company size play crucial roles in the efficacy of candlestick signals. Moreover, the study highlighted optimal holding periods, with bearish signals being most effective over a 5-day period, while bullish signals yielded the highest returns within just one day. The findings further revealed that bearish signals performed well even during bullish market conditions, suggesting their robustness across various environments. Finally, when compared with other technical strategies, such as moving averages, candlestick patterns consistently demonstrated superior predictive power, especially in short-term trading scenarios. These results underscore the importance of market and stock characteristics in determining the success of candlestick trading strategies.

Chin et al. (2017) examined 10 candlestick continuation patterns using data from 420 Malaysian firms between 2000 and 2014. The study found that the Falling Window pattern was the only one to consistently generate significant bearish signals, especially over a 5-day holding period following a downtrend. In contrast, patterns like In Neck and On Neck displayed limited and inconsistent predictive power. These findings suggest that candlestick continuation patterns have weak predictive validity in the Malaysian market, supporting the notion of weak-form market efficiency.

Son et al. (2018) analyzed ten candlestick reversal patterns in the Vietnamese stock market from 2013 to 2018. None of the patterns showed statistically significant predictive power. Despite this, bullish patterns like Bullish Engulfing and Piercing demonstrated some profitability over 4 to 10 day holding periods, with Piercing achieving a 2% return after 6 days. In contrast, bearish patterns generally failed to produce profitable outcomes.

Jamaloodeen et al. (2018) conducted an empirical analysis to assess the predictive validity of Japanese candlestick patterns, focusing on the Shooting Star and Hammer patterns using extensive historical data from the S&P 500 over a 60-year period. Their findings indicate that the predictive power of these patterns is notably influenced by the specific price metric applied. When employing the closing price as a criterion, the patterns did not yield predictive success significantly different from random selection. However, using the high price for the Shooting Star, to signal a market peak, and the low price for the Hammer, to indicate a market

bottom, substantially enhanced predictive accuracy, particularly over short-term intervals of 5 to 10 days. These results suggest that Japanese candlestick patterns may demonstrate enhanced reliability for market forecasting when adapted to specific price metrics rather than relying solely on closing prices.

Cohen (2020) examined the effectiveness of three popular candlestick patterns—Engulfing, Harami, and Kicker—using 10 years of stock price data from 20 U.S. stocks. The results indicated that both the Engulfing and Harami patterns failed to produce positive returns or outperform the Buy and Hold (B&H) strategy. The Kicker pattern showed only marginal gains and was similarly outperformed by the B&H strategy. However, the study introduced a new "Stairs" pattern, which achieved consistent positive gains across all stocks and outperformed the B&H strategy for 16 out of the 20 stocks, highlighting its potential as a more reliable tool for stock trading than traditional patterns.

Heinz et al. (2021) conducted a statistical analysis of Bullish and Bearish Engulfing candlestick patterns using historical data from the S&P 500 index (1950–2020). The study found that Bearish Engulfing patterns have strong short-term predictive power when using the open and high price criteria², signaling market tops effectively. Conversely, Bullish Engulfing patterns showed predictive efficacy for market bottoms when evaluated using the open and low price criteria. Both patterns showed significantly less accuracy when using the close price as a criterion. The analysis also revealed that these patterns were more successful when preceded by an appropriate trend (uptrend for Bearish and downtrend for Bullish), highlighting the importance of market context.

Deng et al. (2022) conducted an empirical investigation into the profitability of Japanese candlestick patterns applied to component stocks of the SSE50 index, analyzing daily data spanning from January 2000 to December 2018. This study evaluates the effectiveness of ten single-day and two-day candlestick patterns, under varying conditions such as trend direction and overbought/oversold states. Results demonstrate that certain bullish patterns, particularly Long White and

²In Heinz et al. (2021), the success of Bullish and Bearish Engulfing patterns is evaluated using three specific price criteria: open, high/low, and close. For the Bullish Engulfing pattern, success is determined if no future closing price falls below the opening or lowest price of the second candle, signaling a strong reversal. Similarly, for the Bearish Engulfing pattern, success is defined when no subsequent closing price exceeds the opening or highest price of the second candle, indicating a bearish trend. However, the close price criterion—where success is based on whether future closing prices stay above or below the second candle's close—was found to be less effective for predicting market movements.

Bullish Gap, generate statistically significant positive average returns over holding periods of up to 10 days, particularly under bullish conditions. In contrast, bearish patterns generally exhibit limited predictive value, with exceptions where patterns like Gravestone Doji yield profitability when employed as contrarian indicators.

2.3 Cryptocurrency Markets

Cohen (2021) explored the application of Engulfing, Harami, and Kicker patterns in Bitcoin trading using data from 2012 to July 2020. The study identified the classical Engulfing pattern as the most effective, particularly for long positions, generating a profit factor³ (PF) of 3.54. By optimizing the Engulfing pattern with a 0.9% strength proxy⁴, the strategy further improved, achieving a higher PF and reduced maximum drawdown⁵. The Harami pattern was found ineffective in its classical form but produced positive results when reversed, with a PF of 3.33. Similarly, the classical Kicker pattern appeared too infrequently to be useful, but a reversed version proved to be a successful long-position strategy, achieving a 74.36% success rate and a PF of 6.92.

Ho et al. (2021) performed a comprehensive analysis of 68 candlestick patterns across 23 leading cryptocurrencies, using data spanning from their earliest available records up to July 2021. The study found that none of the patterns consistently yielded profitable outcomes, with many showing low success rates and some even resulting in negative returns. The authors suggest that the inherent volatility of the cryptocurrency market diminishes the effectiveness of traditional candlestick patterns, making them unreliable indicators for trading in this sector.

³Profit Factor (PF): Profit factor is a key performance metric in trading strategies, defined as the ratio of gross profits to gross losses. A profit factor greater than 1 indicates that the strategy is profitable, as profits outweigh losses. For instance, a PF of 3.54 means that for every dollar lost, the strategy earns \$3.54 in profit.

⁴Strength Proxy: The strength proxy is a percentage-based filter applied to candlestick patterns, such as the Engulfing pattern, to ensure only stronger price movements are considered for trading. It measures the difference between the closing price of the engulfing bar and the opening price of the previous bar. A higher strength proxy indicates that the pattern has more significant price movement, thus potentially improving the signal's reliability.

⁵Maximum Drawdown (MDD): Maximum drawdown represents the largest percentage or dollar loss from the peak to the trough of a trading strategy's performance before it recovers. It is a critical measure of risk, with a lower MDD indicating that the strategy experiences smaller losses, making it preferable for risk-averse investors.

Chapter 3

Methodology

3.1 Patterns

In financial charting, black and white candlesticks are essential elements of candlestick charts, representing price movements over a specific time period. Depending on the chosen timeframe, these candlesticks can reflect various durations, such as minutes, hours, days, or weeks. This study focuses on daily candlesticks, where each one summarizes the price activity of a single trading day. A white candlestick indicates that the closing price was higher than the opening price, while a black candlestick shows that the closing price was lower than the opening price. The rectangular body of the candlestick represents the range between the opening and closing prices. The thin lines extending above and below the body, known as the upper and lower shadows, mark the highest and lowest prices reached during the day. Together, the body and shadows provide a clear visual of the opening, closing, high, and low prices for the day, as illustrated in Figure 3.1.

Candlestick patterns are visual formations created by the arrangement of one or more candlesticks on a chart, believed to reflect shifts in market sentiment and potential price movements. These patterns are thought to encapsulate the psychology of market participants by highlighting changes in buying and selling pressure. Widely used in technical analysis, they are employed by traders as tools to interpret market dynamics and assess the likelihood of potential future trends. However, their predictive reliability can vary depending on the context and other market factors.

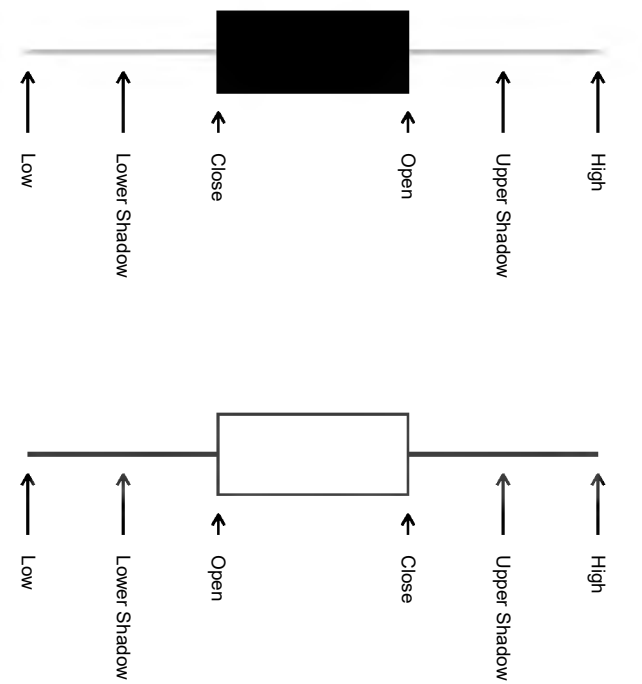


Figure 3.1: Black and White Candlesticks

Candlestick patterns vary in interpretation and the number of days they span, depending on their complexity. Some patterns are formed within a single day, such as the Hammer Bullish or Shooting Star Bearish, which provide immediate insights into market sentiment for that day. Other patterns span two or more days, offering a more comprehensive narrative of price action. For instance, the Engulfing Patterns Bullish or Bearish are formed over two days, while patterns such as the Morning Star Bullish or Three White Soldiers Bullish require three consecutive days to form fully.

Candlestick patterns can be further classified into four categories: bullish reversal, bullish continuation, bearish reversal, and bearish continuation patterns, each offering insights into potential market movement. A bullish reversal pattern occurs after a downtrend and signals a possible shift to an uptrend, suggesting an increase in buying interest. Examples include the Hammer Bullish, Engulfing Bullish, and Morning Star Bullish. A bullish continuation pattern forms during an uptrend and indicates that the upward momentum is likely to persist. Examples include the Upside Tasuki Gap Bullish and Rising Window Bullish. Conversely, a bearish reversal pattern appears after an uptrend and signals a potential shift to a downtrend, reflecting increasing selling pressure. Common examples include the Shooting Star Bearish, Engulfing Bearish, and Evening Star Bearish. A bearish continuation pattern forms during a downtrend, suggesting the downward movement is likely to continue. Examples include the Falling Window Bearish and On

Neck Bearish. These patterns offer traders valuable visual cues for identifying potential turning points or continuations in price trends.

This study aims to evaluate the predictive power of a selected set of candlestick patterns, including both bullish and bearish reversal and continuation patterns. Table 3.1 presents a complete list of the analyzed candlestick patterns, their respective classifications and the number of candlesticks involved. The analysis assesses the performance of these patterns within identified trends, evaluating their profitability using a rigorous statistical framework.

This study aims to evaluate the predictive power of a selected set of bullish and bearish patterns. These patterns include a range of reversal and continuation signals, both bullish and bearish. Table 3.1 presents the complete list of candlestick patterns, along with their respective types and the number of candles involved. The analysis is conducted by evaluating the performance of these patterns within identified trends and assessing their profitability using a rigorous statistical framework.

Table 3.1: Patterns and their Characteristics

Pattern	Number of Candlesticks	Type
Dragonfly Doji	1	Bullish Reversal
Gravestone Doji	1	Bearish Reversal
Hammer	1	Bullish Reversal
Hanging Man	1	Bearish Reversal
Inverted Hammer	1	Bullish Reversal
Long Lower Shadow	1	Bullish Reversal
Long Upper Shadow	1	Bearish Reversal
Marubozu Black	1	Bearish Continuation
Marubozu White	1	Bullish Continuation
Shooting Star	1	Bearish Reversal
Dark Cloud Cover	2	Bearish Reversal
Doji Star	2	Bearish Reversal
Doji Star	2	Bullish Reversal
Engulfing	2	Bearish Reversal
Engulfing	2	Bullish Reversal
Falling Window	2	Bearish Continuation
Harami	2	Bearish Reversal
Harami	2	Bullish Reversal
Harami Cross	2	Bearish Reversal
Harami Cross	2	Bullish Reversal
Kicking	2	Bearish Reversal
Kicking	2	Bullish Reversal
On Neck	2	Bearish Continuation
Piercing	2	Bullish Reversal
Rising Window	2	Bullish Continuation
Tweezer Bottom	2	Bullish Reversal
Tweezer Top	2	Bearish Reversal
Abandoned Baby	3	Bearish Reversal
Abandoned Baby	3	Bullish Reversal
Downside Tasuki Gap	3	Bearish Continuation
Evening Doji Star	3	Bearish Reversal
Evening Star	3	Bearish Reversal
Morning Doji Star	3	Bullish Reversal
Morning Star	3	Bullish Reversal
Three Black Crows	3	Bearish Reversal
Three White Soldiers	3	Bullish Reversal
Tri Star	3	Bullish Reversal
Tri Star	3	Bearish Reversal
Upside Tasuki Gap	3	Bullish Continuation
Falling Three Methods	5	Bearish Continuation
Rising Three Methods	5	Bullish Continuation

3.2 Pattern Identification

For this analysis, code openly available on TradingView (2024) was utilized. The code for each pattern can be accessed by applying the pattern to any chart and clicking on the curly brackets labeled source code next to the pattern. TradingView uses the Pine Script programming language. For this analysis, the code was translated into the R programming language with necessary adjustments, while maintaining the exact same logic for pattern identification as implemented on TradingView. Specifically, version 5 of the code was used for all patterns. The identification logic for each pattern is detailed in Appendix A, which outlines the exact conditions required for each pattern to occur and describes the components necessary for calculating these conditions. Appendix A does not include any R code but focuses on the logic behind the patterns.

3.3 Trend Definition

The trend is determined using a 50-day Simple Moving Average (SMA50), a well-established method in technical analysis for identifying long-term market trends. A distinguishing feature of this study compared to previous research is the use of the SMA50, as prior studies on candlestick patterns in the literature typically employed much shorter time periods for trend detection. The 50-day SMA is computed using closing prices, and its definition goes as follows:

$$\text{SMA}_{50}(t) = \frac{1}{50} \sum_{i=0}^{49} p_{t-i}^c \quad (3.1)$$

Where p_{t-i}^c is the closing price of the asset on day $t-i$. The sum takes the average of the closing prices over the last 50 days, including the most recent day denoted as t .

A Candle is in the uptrend when the closing price is above the SMA50:

$$p_t^c > \text{SMA}_{50}(t) \quad (3.2)$$

A Candle is in the downtrend when the closing price is below the SMA50:

$$p_t^c < \text{SMA}_{50}(t) \quad (3.3)$$

The utilization of trends in individual candlestick patterns varies due to the differing durations of these patterns. For a detailed explanation of pattern definitions and the incorporation of trends, please refer to the appendix. Generally, it is not necessarily true that, for example, all candlesticks within a bearish reversal pattern must occur within a downtrend. This observation is also applicable to bearish continuation, bullish reversal, and bullish continuation patterns.

3.4 Holding Strategy

This study exclusively employs the Caginalp-Laurent (CL) Exit strategy to measure the returns from trades initiated by candlestick patterns. The CL Exit strategy is designed to mitigate price volatility by averaging exit prices over the holding period, thereby providing a more stable assessment of profitability. The holding periods are 1, 2, 3, 5 and 10 days to evaluate both short- and medium-term profitability. The returns are calculated as follows:

$$R_{CL(n)} = \left(\frac{\sum_{i=1}^n \frac{p_{t+i}^c}{n} - p_{t+1}^o}{p_{t+1}^o} \right) \times 100\% \quad (3.4)$$

Where p_{t+i}^c represents the closing price of the asset on day $t+i$ and p_{t+1}^o represents the opening price of the asset on day $t+1$.

Regardless of the exit strategy, returns are multiplied by 1 for bullish patterns and by -1 for bearish patterns. This approach ensures that positive returns indicate the desired outcomes: profit from bullish patterns and avoidance of loss from bearish patterns.

3.5 Transaction Costs

The concept of transaction costs plays a crucial role in understanding the efficiency and functioning of financial markets, and cryptocurrency markets are no exception. In traditional financial markets, transaction costs can significantly influence trading strategies and overall market behavior. In the context of cryptocurrencies, transaction costs present unique challenges due to the diverse structures of centralized and decentralized exchanges, as well as the impact of blockchain technology. These costs include exchange-specific fees, network fees, slippage¹, and various hidden charges, all of which can vary greatly depending on the platform and market conditions. Understanding these costs is essential for evaluating trading efficiency and the overall user experience within the cryptocurrency ecosystem.

Centralized exchanges (CEXs) exhibit distinct characteristics, particularly regarding their fee structures, which generally include maker and taker fees². These fees can vary based on user tiers, with reduced fees frequently offered to users with higher trading volumes or exclusive membership levels. Many CEXs also provide fee discounts and incentives, such as discounts for using native utility tokens like Binance’s BNB, volume-based discounts for high-frequency traders, and time-limited promotional offers. The dynamic nature of CEX fee structures includes tiered systems based on trading volume or token holdings, with fee schedules that may change frequently and unpredictably. The availability of trading pairs may necessitate multiple trades, resulting in accumulated fees. Additionally, the lack of standardization in data reporting complicates accurate fee data collection, with hidden fees often only becoming evident after transactions. Regulatory requirements can further exacerbate these challenges by imposing transaction limits that increase cumulative fees. Furthermore, liquidity issues, particularly on less popular exchanges, can lead to wider bid-ask spreads and increased slippage due to insufficient market depth³.

Decentralized exchanges (DEXs), by contrast, operate on blockchain networks and

¹The difference between the expected price of a trade and the actual price at which it is executed, often due to market volatility or low liquidity.

²Fees paid by traders depending on whether they add (maker) or take (taker) liquidity from the order book. Makers place limit orders that add liquidity to the exchange, while takers execute trades that remove liquidity by matching existing orders. Typically, maker fees are lower than taker fees to incentivize liquidity provision.

³Market depth refers to the ability of the market to sustain large orders without significant impact on the asset’s price. Low market depth implies fewer buy and sell orders, leading to higher volatility and slippage.

are subject to different cost structures. Transaction costs on DEXs are influenced by network fees, such as gas fees on Ethereum, which can vary substantially depending on network congestion and demand. Liquidity remains a significant challenge, as assets with low liquidity tend to exhibit wider bid-ask spreads⁴ and are more vulnerable to price slippage. Additionally, DEX users may need to perform multiple swaps to complete their desired trades, especially when indirect trading pairs are involved, which further increases transaction costs. Cross-chain transactions⁵ also contribute to higher costs, often necessitating the use of bridging solutions. Network congestion⁶ can lead to increased fees and extended transaction times, particularly during peak activity periods, while scalability limitations can result in unpredictable fee spikes. Technological variations, such as protocol upgrades, may also alter fee structures in unforeseen ways. Security-related costs are an additional consideration, as users may need to pay priority fees for faster transaction processing and may incur costs even for failed transactions. The absence of standardized data reporting across protocols complicates accurate fee data collection, while hidden costs, such as fees for failed transactions, further add to the complexity of the fee landscape for DEX users.

Due to the unique characteristics of cryptocurrency markets, as discussed in the preceding paragraphs, and considering the large sample size of 589 cryptocurrencies, transaction costs are disregarded in this study.

3.6 Hypothesis Testing

A candlestick pattern is considered helpful for trading if it achieves a higher winning rate⁷ than a random decision while producing a positive return. In other words, an effective candlestick pattern should satisfy the following four conditions simultaneously:

⁴The difference between the highest price a buyer is willing to pay and the lowest price a seller is willing to accept. A wider spread indicates lower liquidity and higher transaction costs.

⁵Transactions that involve transferring assets across different blockchain networks, often requiring intermediary bridging mechanisms to facilitate the transfer.

⁶A state where the blockchain network becomes overwhelmed with transactions, leading to delays and increased fees due to limited processing capacity.

⁷The winning rate refers to the proportion of trades yielding a positive return, calculated as the ratio of profitable trades to total trades.

- 1) The average return is positive.
- 2) The winning rate is greater than 50%.
- 3) The winning rate is statistically different from 50%.
- 4) The average return is statistically different from 0.

To evaluate the third condition, which requires the winning rate to be statistically different from 50%, a binomial test is utilized. This test is designed to determine whether the observed proportion of successful trades (i.e., the winning rate) significantly deviates from 50%, which represents a scenario of random decision-making without predictive advantage.

In the context of this analysis, the null hypothesis H_0 assumes that the winning rate is 50%, implying no predictive value beyond random chance, while the alternative hypothesis H_A proposes that the winning rate is significantly different from 50%. The binomial test assesses whether the observed success rate of the candlestick pattern is greater than what would be expected by chance alone. To be specific:

$$H_0 : \text{winning rate} = 0.5,$$

$$H_A : \text{winning rate} \neq 0.5.$$

If the p-value derived from the test is less than 0.05, the null hypothesis is rejected. This would suggest that the candlestick pattern's winning rate is statistically different from 50%, providing evidence of its potential effectiveness in predicting profitable trades. Consequently, the binomial test serves as a crucial step in validating the predictive capability of candlestick patterns in trading strategies.

To evaluate the fourth condition, recognizing that financial returns are often skewed, a skewness-adjusted t-test is employed to determine whether the average return from trades is significantly different from zero. This approach is particularly important because traditional t-tests assume that the data is normally distributed. However, financial returns are frequently characterized by skewness, meaning that their distribution is not symmetric. If this skewness is not accounted

for, the results of standard t-tests may be biased, leading to incorrect conclusions about the profitability of trading strategies.

Specifically, when the data is positively skewed, the standard t-test tends to understate the significance of the results because the positive tail inflates the variance, making it harder to detect true differences from zero. As a result, this increased variability may cause the test to fail to reject the null hypothesis when it should, potentially leading to missed opportunities by failing to identify significant returns.

Conversely, when the data is negatively skewed, the standard t-test may overstate the significance of the results. Negative skewness tends to compress the variance, which can make the data appear less variable than it actually is, thereby inflating the t-statistic. This may lead to an increased likelihood of incorrectly rejecting the null hypothesis, giving a false impression of profitability.

These effects occur because the standard t-test relies on the assumption of symmetry in the data distribution. When this assumption is violated, the estimates of the mean and variance become biased, affecting the calculation of the t-statistic and leading to incorrect inferences. By using a skewness-adjusted t-test, we are able to adjust for this asymmetry in the distribution, thereby providing a more reliable and accurate measure of profitability.

The skewness-adjusted t-test is calculated according to the method outlined by Johnson (1978). Specifically, the test adjusts the standardized mean to account for skewness in the data, which helps to mitigate the potential biases introduced by non-normal distributions. The null hypothesis H_0 assumes that the average returns are 0 implying no positive or negative returns, while the H_A proposes that the average returns are statistically different from 0. Specifically,

$$\begin{aligned} H_0 : \mu_r &= 0, \\ H_A : \mu_r &\neq 0 \end{aligned}$$

where μ_r is the mean of returns. H_0 is rejected if the test p-value is less than or equal to 0.05, indicating that the candlestick pattern yields a statistically significant returns, which implies that the pattern has predictive value beyond random

chance.

The skewness-adjusted t-statistic is defined as follows:

$$t_{sa} = \sqrt{m} \left(s + \frac{1}{3}\gamma s^2 + \frac{1}{6m}\gamma \right)$$

where:

$$s = \frac{\bar{r}}{\sigma_r} \quad (\text{standardized mean of returns})$$

$$\gamma = \sum_{i=1}^m \frac{(r_i - \bar{r})^3}{m\sigma_r^3} \quad (\text{estimated skewness of returns})$$

m = number of trades

$\bar{r}$ = mean return of trades

σ_r = standard deviation of returns

r_i = return of trade i

Conditions 1 and 2 are verified by examining the numerical values of the winning rate and the average returns, respectively. Specifically, condition 1 is satisfied if the average return is positive and condition 2 is met if the winning rate exceeds 50%. These conditions are straightforward and can be assessed through direct observation of the data, without the need for statistical testing.

Chapter 4

Data

The datasets utilized in this analysis were sourced from Yahoo Finance (2024). Yahoo Finance offers historical daily cryptocurrency pricing data, including the opening, high, low, and closing prices. This data is originally provided by CoinMarketCap, a leading platform for cryptocurrency market tracking. CoinMarketCap aggregates information from numerous exchanges, ensuring comprehensive market coverage.

The temporal coverage of the datasets varies across cryptocurrencies. Bitcoin and Litecoin exhibit the longest observation periods, with data available from September 17, 2014, encompassing 3,748 days of observation. Conversely, Luckycoin has the shortest observation period, with data available from October 17, 2024, spanning only 65 days. The data collection period for each cryptocurrency extends from the earliest available date on Yahoo Finance to December 20, 2024, ensuring the inclusion of as much historical information as possible.

Five distinct datasets were constructed to address various research objectives. Dataset A includes the top 17 cryptocurrencies by market capitalization and encompasses a total of 37,065 days of data on opening, high, low, and closing prices. Dataset B expands the scope to the top 300 cryptocurrencies, while Dataset C includes the top 589 cryptocurrencies and provides a total of 782,172 days of data on opening, high, low, and closing prices. Dataset D is similar to Dataset C but excludes the top 17 cryptocurrencies, focusing on a broader segment of the market. Finally, Dataset E is centered on 23 stablecoins ranked by market capitalization. These datasets facilitate a detailed analysis of market dynamics across different

tiers of cryptocurrencies.

Certain cryptocurrencies were intentionally excluded from Datasets A, B, C, and D. The first exclusion category encompasses stablecoins, which are designed to maintain a stable value by pegging to an underlying asset, such as the US dollar. Stablecoins achieve price stability through mechanisms fundamentally different from other cryptocurrencies. Consequently, their price movements are less likely to reflect patterns in candlestick charts or other graphical analyses, as fluctuations are primarily driven by arbitrage activities aimed at maintaining their peg.

The second exclusion category includes restaked, wrapped, and bridged cryptocurrencies. These assets closely mirror the prices of their underlying cryptocurrencies. Including them could artificially inflate the occurrence of price patterns, as they represent the same asset under different names. Price deviations for these assets are typically corrected through arbitrage mechanisms, often triggered by market turbulence or extraordinary events. As a result, their price movements, like those of stablecoins, are unlikely to provide meaningful insights through candlestick or graphical analyses.

The construction of these datasets faced several challenges related to data integrity. Missing values were a common issue, particularly in earlier periods. To address this, all observations up to and including the last missing value were removed. Another challenge involved zero values in the data, likely resulting from data entry errors or rounding discrepancies. Yahoo Finance rounds OHLC prices to six decimal places, which, while adequate for stocks, can distort pricing for cryptocurrencies valued in fractions of a cent or smaller. Consequently, zero-value entries necessitated the removal of all affected observations. Additionally, cryptocurrencies with fewer than 50 data points were excluded from the analysis, as this threshold was deemed necessary for reliable trend identification.

Detailed lists of the cryptocurrencies included in each dataset are provided in Appendix B. Dataset A is detailed in Table B.1, Dataset B in Table B.2, Dataset C in Table B.3, Dataset D in Table B.4, and Dataset E in Table B.5. These tables serve as references for the specific assets analyzed, categorized by market capitalization and dataset-specific criteria. The first column of each table contains the symbols used to download data from Yahoo Finance with the help of the `quantmod` package. The second column lists the cryptocurrency names as referenced by Yahoo Finance, and the third column provides their market capitalizations.

Chapter 5

Results and Discussion

5.1 Description of Tables

The results for datasets A through E are presented in Tables 5.1 to 5.5 at the end of this chapter. For each pattern and holding period, the following metrics are calculated and displayed: average returns, win rate, binomial test p-value, and skew-adjusted t-test p-value. The variable N denotes the total number of occurrences for each pattern. Patterns that satisfy conditions 2, 3, and 4, as described in Section 3.5, for at least one holding period are marked with an asterisk (*).

5.2 Preliminary Analysis of Patterns

Before conducting a more comprehensive analysis, it is essential to first examine why the Kicking Bearish, Kicking Bullish, Abandoned Baby Bearish, Abandoned Baby Bullish, and Tri Star Bearish patterns exhibit zero occurrences, while Tri Star Bullish appears exactly once in dataset C and D. The primary explanation lies in the way these patterns are defined, all of which require the presence of a “gap” between consecutive candles. For instance, in the Kicking Bearish pattern, the first candle’s low must exceed the second candle’s high (i.e., previous low $>$ current high), thereby eliminating any overlap between the two candles.

A similar phenomenon applies to patterns such as Downside Tasuki Gap Bearish, Evening Doji Star Bearish, Evening Star Bearish, Morning Doji Star Bullish, Morning Star Bullish, and Upside Tasuki Gap Bullish. Although these patterns are also partly defined by gaps, their criteria are less stringent, focusing on the highs and lows of the candle bodies rather than the absolute extremities of the candles.

Notably, the cryptocurrency market’s continuous trading schedule—24 hours a day, seven days a week—renders such gaps exceedingly rare. This nonstop activity substantially reduces the likelihood of price discontinuities that would otherwise form the basis for these gap-dependent patterns.

5.3 Results

5.3.1 Dataset A (Table 5.1)

None of the patterns in Dataset A satisfy the four criteria outlined in Section 3.5 for any of the examined holding periods. Although some patterns exhibit lower frequencies, this does not apply to all of them. For instance, the Long Lower Shadow Bullish pattern occurs 1063 times. This may be attributable to higher liquidity in these cryptocurrencies, which can reduce the effectiveness of candlestick patterns.

5.3.2 Dataset B (Table 5.2)

In contrast, the results for Dataset B indicate that Hammer Bullish, On Neck Bearish, Rising Window Bullish, and Downside Tasuki Gap Bearish each fulfill the four criteria specified in Section 3.5. Hammer Bullish meets these criteria for the 1-, 2-, 3-, and 5-day holding periods. On Neck Bearish satisfies the criteria for the 3- and 10-day holding periods, Rising Window Bullish for the 10-day holding period, and Downside Tasuki Gap Bearish for the 2- and 5-day holding periods. Shooting Star Bearish meets criteria 2, 3, and 4 from Section 3.5 for the 10-day holding period but exhibits negative average returns, suggesting that it should be reclassified from bearish to bullish. The frequencies of On Neck Bearish and

Downside Tasuki Gap Bearish are below 100, whereas the other three patterns appear more than 100 times.

5.3.3 Dataset C (Table 5.3)

Results for Dataset C differ in certain respects. Hammer Bullish, Harami Cross Bullish, On Neck Bearish, and Rising Window Bullish satisfy the four criteria delineated in Section 3.5. Hammer Bullish meets these criteria only for the 3-day holding period, whereas Harami Cross Bullish does so for the 10-day holding period. On Neck Bearish again satisfies them for the 3- and 10-day holding periods, and Rising Window Bullish once more for the 10-day holding period. Shooting Star Bearish again fulfills criteria 2, 3, and 4 from Section 3.5 for the 10-day holding period, but exhibits negative average returns. Long Upper Shadow likewise meets the same subset of conditions, but across all holding periods, suggesting that both Shooting Star Bearish and Long Upper Shadow should be reclassified from bearish to bullish. All six patterns in this dataset occur more than 100 times.

5.3.4 Dataset D (Table 5.4)

The findings for Dataset D closely mirror those of Dataset C. Although the specific numerical values differ slightly, the same patterns are identified for the same holding periods.

5.3.5 Dataset E (Table 5.5)

Finally, for Dataset E, which comprises stablecoins, Doji Star Bearish satisfies the four criteria for the 1-, 2-, 3-, and 5-day holding periods. In addition, Doji Star Bullish meets all four criteria for all examined holding periods.

5.4 Discussion

Across the first four datasets, only a small subset of candlestick patterns satisfies the four criteria specified in Section 3.5. Patterns that recur across all four

datasets provide the most compelling evidence of robust predictive power, as their consistency reduces the likelihood that any observed signals are merely artifacts of a particular sample. By contrast, patterns that appear exclusively in certain datasets—or do so with a relatively small number of occurrences—may reflect transient market conditions, making it difficult to generalize their apparent significance beyond the specific dataset in which they were identified.

Because no pattern in Dataset A satisfies the four conditions, the reliability of patterns identified in Datasets B, C, and D is thereby called into question. However, this discrepancy may be attributable to the lack of occurrences of these patterns in Dataset A. Focusing on the patterns that fulfill the conditions simultaneously in Datasets B, C, and D reveals the most robust results for Hammer Bullish (3-day holding period), Shooting Star Bearish (10-day holding period, though requiring reclassification to bullish), On Neck Bearish (3- and 10-day holding periods), and Rising Window Bullish (10-day holding period).

Long Upper Shadow Bearish fulfills the conditions for all holding periods in Datasets C and D but also requires reclassification to bullish. Harami Cross Bullish meets the four conditions in Datasets C and D, though only for the 10-day holding period. Their failure to fulfill the condition in Dataset B may suggest that these two patterns perform more effectively on cryptocurrencies with lower market capitalizations.

Meanwhile, Hammer Bullish (for the 1-, 2-, and 5-day holding periods) and Downside Tasuki Gap Bearish (for the 2- and 5-day holding periods) satisfy the four conditions solely in Dataset B, which may indicate that they are more effective when applied to cryptocurrencies with higher market capitalizations.

Within this stablecoin dataset E, Doji Star Bullish (across all holding periods) and Doji Star Bearish (for the 1-, 2-, 3-, and 5-day holding periods) are the only candlestick patterns that fulfil the four conditions. These patterns are generally regarded as reversal signals, but their apparent predictive power may be attributable to the stablecoin mechanism of maintaining a 1:1 peg to the underlying asset. Whenever the price deviates from this peg, it must revert, creating the illusion of a candlestick-driven reversal. In reality, however, this price correction stems from the stablecoin’s inherent design rather than any unique predictive property of the candlestick pattern. Thus, these findings support the rationale for excluding stablecoins from the primary analysis.

5.5 Limitations

Despite the thoroughness of the preceding analysis, several limitations must be acknowledged. First, survivorship bias may influence the findings presented in this thesis. Cryptocurrency assets that failed or were delisted during the study period could have been excluded, potentially skewing the observed performance of candlestick patterns toward those surviving in the sample.

Second, the price data obtained from Yahoo Finance is aggregated via CoinMarketCap and does not necessarily represent the precise trading data from individual exchanges. Consequently, it would not have been feasible to trade every identified candlestick pattern based on these specific data alone. Furthermore, slight price discrepancies among various exchanges mean that the same pattern might emerge differently on each platform, thus diminishing the robustness of the observed results.

Third, the analysis employed a particular definition of trend, specific holding periods, and an exit strategy—any of which could have been defined differently. Alternative approaches to each of these variables may yield diverging results, indicating that the outcomes are sensitive to the methodological choices adopted here.

Fourth, although the Yahoo Finance data are of relatively good quality, the data preparation process resulted in the loss of a nontrivial number of values due to missing or zero entries. Superior data sources with fewer gaps and a longer available history do exist but are often subscription-based, which creates barriers to their use and may have limited the scope of the present investigation.

Fifth, the possibility of seasonality in the occurrence of certain patterns cannot be ruled out. Patterns may appear more frequently in bullish markets and less so in bearish periods (or vice versa). The omission of explicit controls for seasonal or cyclical market conditions may thus restrict the generalizability of the results.

Sixth, an arbitrary decision was made to exclude wrapped and restaked cryptocurrencies. Although a rationale was provided for this exclusion, it is conceivable that the decision could have systematically removed patterns occurring more frequently in such assets, thereby affecting the final results.

Seventh, some candlestick patterns may occur on only a limited subset of cryptocurrencies. Even if a pattern demonstrates statistical significance in this study, it may not necessarily be deemed broadly useful across diverse market segments. Further analysis is required to determine the specific assets or asset classes for which these patterns hold predictive power, including an examination of how frequently they appear in different subsets of the cryptocurrency market.

Finally, a general limitation in this area of research is that, although candlestick patterns are relatively straightforward to code, there can be differences in pattern-identification logic or related parameters. As a result, comparing findings across different studies becomes more challenging, given the variability in how researchers define and detect these patterns.

These limitations highlight the importance of interpreting the findings with caution and underscore the need for continued research using alternative data sources, different methodological choices, and broader sample coverage.

5.6 Result Tables

Table 5.1: Empirical results, dataset A

Patterns		Holding period 1 day					Holding period 2 days					Holding period 3 days					Holding period 5 days					Holding period 10 days				
Pattern	Pattern type	N	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.
Dragonfly Doji	Bullish	170	-0.62	18.24	0.000000	0.5409630	1.61	27.06	0.000000	0.3220974	2.74	34.12	0.0000419	0.1778403	6.28	37.96	0.0000187	0.0869425	8.21	41.76	0.0300637	0.0207173	8.21	41.76	0.0300637	0.0207173
Gravestone Doji	Bearish	168	-2.68	10.71	0.000000	0.0000000	-3.08	17.26	0.000000	0.0000000	-3.20	21.43	0.0000000	0.0000000	-3.70	29.76	0.0000002	0.0000001	-6.98	33.33	0.0000186	0.0001134	-6.98	33.33	0.0000186	0.0001134
Hammer	Bullish	155	-1.13	33.55	0.000512	0.0035813	-0.91	38.06	0.0036930	0.0305807	-0.70	41.94	0.0333376	0.1337130	-0.63	43.87	0.1479976	0.2678050	-0.75	47.74	0.0299976	0.3364559	-0.75	47.74	0.0299976	0.3364559
Hammer	Bearish	108	-0.73	42.59	0.1485763	0.1752599	-1.19	43.52	0.2107658	0.0363980	-1.40	43.52	0.2107658	0.0296002	-2.00	40.74	0.0670104	0.0110245	-3.23	44.44	0.2879382	0.0051178	-3.23	44.44	0.2879382	0.0051178
Inverted Hammer	Bullish	83	-0.76	48.19	0.8264047	0.1991338	-0.55	42.17	0.1874599	0.4567344	-0.87	42.17	0.1874599	0.2868477	-0.73	43.37	0.2726889	0.4337857	-0.52	42.17	0.1874599	0.6607083	-0.52	42.17	0.1874599	0.6607083
Long Lower Shadow	Bullish	1063	0.07	45.58	0.0045580	0.7391426	0.77	46.91	0.4052276	0.0084481	1.27	48.05	0.2176811	0.0002555	2.32	49.57	0.8052835	0.0000265	3.92	51.67	0.2938233	0.0000000	3.92	51.67	0.2938233	0.0000000
Long Upper Shadow	Bearish	899	-0.47	41.38	0.0000003	0.0114695	-0.73	41.82	0.0000011	0.0005706	-0.85	42.60	0.000103	0.0001088	-1.24	46.21	0.0253092	0.0000285	-2.66	47.71	0.1811735	0.0000001	-2.66	47.71	0.1811735	0.0000001
Marubozu Black	Bearish	463	-0.74	33.69	0.0000000	0.0077059	-0.85	36.29	0.0000000	0.0015588	-0.97	37.45	0.0000001	0.0001803	-0.85	42.64	0.0017959	0.0282872	-1.09	44.90	0.0320467	0.0406643	-1.09	44.90	0.0320467	0.0406643
Marubozu White	Bullish	564	0.78	0.000137	0.0000137	0.1678822	0.85	42.73	0.0006352	0.0090265	1.39	44.86	0.0163134	0.0004471	1.89	45.57	0.0389924	0.0002614	3.72	48.85	0.6130786	0.0000021	3.72	48.85	0.6130786	0.0000021
Shooting Star	Bearish	70	0.17	52.86	0.7292028	0.8404052	0.27	50.00	1.0000000	0.7640737	-0.35	51.43	0.9049745	0.6702947	-0.31	55.71	0.4029631	0.7561889	-0.26	55.71	0.4029631	0.8354873	-0.26	55.71	0.4029631	0.8354873
Dark Cloud Cover	Bearish	26	-1.44	38.46	0.3269396	0.1989075	-1.06	42.31	0.5571971	0.3441811	-1.05	42.31	0.5571971	0.4465532	-0.58	46.15	0.8450190	0.7137954	0.09	50.00	1.0000000	0.9574868	0.09	50.00	1.0000000	0.9574868
Doji Star	Bearish	63	-0.12	50.79	1.0000000	0.8598160	-0.92	50.79	1.0000000	0.2994621	-1.44	47.62	0.8013065	0.1679450	-2.01	47.62	0.8013065	0.1015905	-3.44	41.27	0.2073679	0.0241322	-3.44	41.27	0.2073679	0.0241322
Doji Star	Bullish	63	0.40	66.13	0.0151341	0.5827806	0.04	53.23	0.7035367	0.9724067	0.14	53.23	0.7035367	0.8825613	-0.23	45.16	0.5257734	0.8041639	-0.77	43.55	0.3741517	0.4626607	-0.77	43.55	0.3741517	0.4626607
Engulfing	Bearish	419	-0.71	41.05	0.0002902	0.0027993	-1.02	42.24	0.0017356	0.0001505	-1.41	41.83	0.0009967	0.0000132	-2.03	44.44	0.0288723	0.0000170	-3.15	45.99	0.1143542	0.0000661	-3.15	45.99	0.1143542	0.0000661
Engulfing	Bullish	428	-0.07	47.66	0.3584246	0.7589149	0.19	48.36	0.5298046	0.4173881	0.15	45.56	0.0735779	0.5643009	0.01	47.66	0.3584246	0.9597645	-0.06	44.86	0.0375410	0.0847627	-0.06	44.86	0.0375410	0.0847627
Falling Window	Bearish	36	-1.33	30.56	0.0288167	0.1446980	-1.23	50.00	1.0000000	0.2851210	-0.41	53.56	0.6177193	0.7148264	0.33	52.78	0.8679394	0.8002030	1.06	55.56	0.6177193	0.4642897	1.06	55.56	0.6177193	0.4642897
Harumi	Bearish	36	-0.56	52.78	0.8679394	0.5169260	-1.10	58.33	0.4050322	0.3103972	-1.63	63.89	0.1324982	0.2127158	-2.68	58.33	0.4050322	0.1207273	-5.69	33.33	0.0652453	0.0009824	-5.69	33.33	0.0652453	0.0009824
Harumi	Bullish	28	0.11	57.14	0.5715882	0.8703619	0.32	53.57	0.8505540	0.7065050	0.14	57.14	0.5715882	0.8988802	0.33	57.14	0.5715882	0.8029022	-1.69	35.71	0.1849333	0.1750396	-1.69	35.71	0.1849333	0.1750396
Harumi Cross	Bearish	24	2.55	33.33	0.1515896	0.0392928	0.75	41.67	0.5412562	0.7902088	-1.61	37.50	0.3074563	0.4663903	-5.49	33.33	0.1515896	0.1805429	-8.07	45.83	0.8388197	0.1768259	-8.07	45.83	0.8388197	0.1768259
Harumi Cross	Bullish	27	1.57	22.22	0.0059246	0.1950039	2.00	33.33	0.1220781	0.1139774	2.12	37.04	0.2477886	0.1233715	2.23	40.74	0.4420683	0.1693346	3.23	55.56	0.7011080	0.1356678	3.23	55.56	0.7011080	0.1356678
Kicking	Bearish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Kicking	Bullish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
On Neck	Bearish	4	1.31	25.00	0.6250000	0.7078723	1.35	50.00	1.0000000	0.7002190	0.15	50.00	1.0000000	0.9749146	-0.58	50.00	1.0000000	0.8301531	0.93	75.00	0.6250000	0.8250575	0.93	75.00	0.6250000	0.8250575
Piercing	Bullish	14	0.64	35.71	0.4239502	0.5401187	0.99	42.86	0.7905273	0.3822759	1.17	42.86	0.7905273	0.2692882	1.31	42.86	0.7905273	0.2717775	0.70	50.00	1.0000000	0.7087458	0.70	50.00	1.0000000	0.7087458
Rising Window	Bullish	31	-1.28	35.48	0.1496128	0.1955420	-2.00	35.48	0.1496128	0.0410914	-1.52	45.16	0.7201001	0.1808692	0.14	51.61	1.0000000	0.9487455	3.76	54.84	0.7201001	0.2567245	3.76	54.84	0.7201001	0.2567245
Tweezer Bottom	Bullish	278	-0.99	36.69	0.000107	0.0005937	-1.19	34.17	0.0000001	0.0000599	-1.32	35.25	0.0000010	0.0000611	-1.58	35.97	0.0000034	0.0000369	-1.59	37.77	0.0000541	0.0006385	-1.59	37.77	0.0000541	0.0006385
Tweezer Top	Bearish	248	-0.83	43.95	0.0653310	0.0163733	-1.19	45.56	0.1822534	0.0036620	-1.21	48.39	0.6567681	0.0083456	-1.44	48.79	0.7509373	0.0089800	-2.44	50.00	1.0000000	0.0011898	-2.44	50.00	1.0000000	0.0011898
Abandoned Baby	Bearish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Abandoned Baby	Bullish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Downside Tasuki Gap	Bearish	11	2.54	63.64	0.5488281	0.1827675	4.16	63.64	0.5488281	0.1410922	4.07	54.55	1.0000000	0.1369414	3.90	63.64	0.5488281	0.1260689	2.59	63.64	0.5488281	0.1205260	2.59	63.64	0.5488281	0.1205260
Evening Doji Star	Bearish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Evening Star	Bearish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Morning Doji Star	Bullish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Morning Star	Bullish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Three Black Crows	Bearish	1	7.08	100.00	1.0000000	NA	0.20	100.00	1.0000000	NA	-5.07	0.00	1.0000000	NA	-3.13	0.00	1.0000000	NA	-3.67	0.00	1.0000000	NA	-3.67	0.00	1.0000000	NA
Three White Soldiers	Bullish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Tri Star	Bullish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Tri Star	Bearish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Upside Tasuki Gap	Bullish	8	3.13	62.50	0.7295625	0.1252979	4.68	62.50	0.7265625	0.0982637	4.76	62.50	0.7265625	0.1128168	5.56	50.00	1.0000000	0.1887086	4.12	50.00	1.0000000	0.4739743	4.12	50.00	1.0000000	0.4739743
Falling Three Methods	Bearish	15	-1.20	33.33	0.3017578	0.0300874	-1.74	20.00	0.0351563	0.0326623	-1.71	26.67	0.1184692	0.1352448	-1.38	26.67	0.1184692	0.3051838	-1.49	40.00	0.6072588	0.5146364	-1.49	40.00	0.6072588	0.5146364
Rising Three Methods	Bullish	13	-0.33	30.77	0.2668457	0.8704763	0.27	30.77	0.2668457	0.8174698	-0.05	30.77	0.2668457	0.9930215	0.58	46.15	1.0000000	0.7719168	2.69	61.54	0.5810547	0.3147833	2.69	61.54	0.5810547	0.3147833

N represents the total number of occurrences. An asterisk (*) indicates that the pattern satisfies conditions 2, 3, and 4 as described in section 3.5. for at least one holding period.

Table 5.2: Empirical results, dataset B

Patterns		Holding period 1 day						Holding period 2 days						Holding period 3 days						Holding period 5 days						Holding period 10 days					
Pattern	Pattern type	N	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.					
Dragonfly Doji	Bullish	820	-1.09	29.43	0.0000000	0.0060124	-0.60	35.04	0.0000000	0.4409166	0.68	39.07	0.0000000	0.8821390	0.84	41.56	0.0000016	0.4491385	1.97	43.07	0.000853	0.0360716	0.0000000	43.07	40.16	0.0000001	0.0000000				
Gravestone Doji	Bearish	761	-3.19	24.61	0.0000000	0.0000000	-3.70	30.30	0.0000000	0.0000000	-4.28	34.12	0.0000000	0.0000000	-4.80	39.13	0.0000000	0.0000000	-6.02	40.16	0.0000000	0.0000000	0.0000000	40.16	39.13	0.0000000	0.0000000				
Hammer*	Bullish	1592	0.54	53.09	0.0148379	0.0003438	0.70	53.72	0.0032921	0.0002904	0.81	53.34	0.083548	0.0000090	0.91	52.90	0.0228280	0.0002418	1.42	52.33	0.0667562	0.0000011	0.0000000	52.33	46.67	0.0000000	0.0000000				
Hanging Man	Bearish	1140	-0.90	46.51	0.0204476	0.0000247	-1.38	44.92	0.0007004	0.0000000	-1.91	45.10	0.0010743	0.0000000	-2.52	43.78	0.0003312	0.0000000	-3.89	45.61	0.0634910	0.0000000	0.0000000	45.61	0.0634910	0.0000000	0.0000000				
Inverted Hammer	Bullish	1256	-0.24	44.51	0.0001092	0.2717881	0.00	44.86	0.0002991	0.9981992	0.02	45.26	0.0008588	0.9296338	0.08	46.53	0.0151662	0.7625988	0.31	46.85	0.0276415	0.0000000	0.0000000	46.85	0.0276415	0.0000000	0.0000000				
Long Lower Shadow	Bullish	10916	0.07	47.23	0.0000000	0.3244385	0.25	48.04	0.0004475	0.0021254	0.49	48.66	0.0054571	0.0000000	0.95	49.10	0.0034570	0.0000000	1.91	49.20	0.0934010	0.0000000	0.0000000	49.20	0.0934010	0.0000000	0.0000000				
Long Upper Shadow	Bearish	11814	-0.39	50.25	0.5935929	0.0000001	-0.57	50.15	0.7543763	0.0000000	-0.79	50.57	0.2203936	0.0000000	-1.27	50.73	0.1132527	0.0000000	-5.90	50.65	0.1605712	0.0000000	0.0000000	50.65	0.1605712	0.0000000	0.0000000				
Marubozu Black	Bearish	4721	-1.93	44.38	0.0000000	0.0000000	-2.23	43.89	0.0000000	0.0000000	-2.19	43.86	0.0000000	0.0000000	-2.09	45.79	0.0000000	0.0000000	-2.87	45.89	0.0000000	0.0000000	0.0000000	45.89	0.0000000	0.0000000	0.0000000				
Marubozu White	Bullish	4684	0.46	43.51	0.0000000	0.0014529	0.74	45.67	0.0000000	0.0000004	1.14	47.50	0.0006614	0.0000000	1.84	48.76	0.0028864	0.0000000	3.52	49.36	0.3884474	0.0000000	0.0000000	49.36	0.3884474	0.0000000	0.0000000				
Shooting Star*	Bearish	1012	0.02	52.87	0.0731160	0.9763777	-0.01	51.28	0.4319592	0.9423773	-0.08	52.57	0.1088511	0.7940076	-0.53	54.42	0.0055260	0.1656529	-1.28	53.79	0.0177760	0.0101988	0.0000000	53.79	0.0177760	0.0000000	0.0101988				
Dark Cloud Cover	Bearish	305	-0.61	46.56	0.2520878	0.0902660	-0.78	47.87	0.4920714	0.0640447	-1.11	48.52	0.6469674	0.0257687	-1.22	51.48	0.6469674	0.0478341	-2.04	48.52	0.6469674	0.0168597	0.0000000	48.52	0.6469674	0.0000000	0.0168597				
Doji Star	Bearish	605	-0.64	50.91	0.6843672	0.0350316	-1.33	48.76	0.5992694	0.0001520	-1.69	45.79	0.0419851	0.0000111	-2.05	44.79	0.0116516	0.0000046	-3.53	44.76	0.0113760	0.0000000	0.0000000	44.76	0.0113760	0.0000000	0.0000000				
Doji Star	Bullish	617	-0.07	53.41	0.0984686	0.8050507	-0.20	48.86	0.6004640	0.5581162	-0.17	48.38	0.4439806	0.6533164	-0.13	48.54	0.4934070	0.7723422	-0.02	50.97	0.6576541	0.964275	0.0000000	50.97	0.6576541	0.0000000	0.964275				
Engulfing	Bearish	4820	-0.61	46.16	0.0000001	0.0000000	-0.94	47.21	0.0001123	0.0000000	-1.22	47.76	0.0020260	0.0000000	-1.80	47.02	0.0004415	0.0000000	-3.06	48.24	0.0163569	0.0000000	0.0000000	48.24	0.0163569	0.0000000	0.0000000				
Engulfing	Bullish	5969	-0.05	45.53	0.0000000	0.53477923	0.07	47.05	0.0000068	0.489608	-0.02	46.13	0.0000000	0.8396804	-0.11	44.93	0.0000000	0.3474392	10.02	44.25	0.0000000	0.1330905	0.0000000	44.25	0.0000000	0.0000000	0.1330905				
Falling Window	Bearish	226	-1.84	34.96	0.0000072	0.0000001	-1.53	46.90	0.3572228	0.0003646	-0.96	49.12	0.8418827	0.0607879	-0.67	51.77	0.6415752	0.2581766	-0.78	53.54	0.3153854	0.2847059	0.0000000	53.54	0.3153854	0.0000000	0.2847059				
Harumi	Bearish	265	-1.23	45.28	0.1402524	0.0049899	-1.75	44.91	0.1100548	0.0005461	-2.22	47.55	0.4610937	0.0001114	-2.68	50.19	1.0000000	0.0002034	-4.94	44.70	0.9963804	0.0000000	0.0000000	44.70	0.9963804	0.0000000	0.0000000				
Harumi	Bullish	297	-0.57	46.46	0.2457926	0.0312414	0.88	49.83	1.0000000	0.8571768	-0.17	48.82	0.7277884	0.7458147	-0.20	46.13	0.2016750	0.7282010	-1.21	39.06	0.8052083	0.0850283	0.0000000	39.06	0.8052083	0.0000000	0.0850283				
Harumi Cross	Bearish	75	2.29	30.67	0.0010795	0.3106688	1.88	42.67	0.2480457	0.4031014	2.10	46.67	0.644639	0.2828525	0.86	44.00	0.3556992	0.7689522	-1.78	48.00	0.8175539	0.9369988	0.0000000	48.00	0.8175539	0.0000000	0.9369988				
Harumi Cross	Bullish	93	2.75	31.18	0.0003658	0.0008846	2.71	35.87	0.0087809	0.0027078	3.11	40.22	0.0757602	0.0019013	3.19	44.57	0.3481413	0.0065371	4.15	54.35	0.4657074	0.0035845	0.0000000	54.35	0.4657074	0.0000000	0.0035845				
Kicking	Bearish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA				
Kicking	Bullish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA				
On Neck*	Bearish	63	3.56	58.73	0.2073679	0.0203224	4.43	60.32	0.1299179	0.0018238	4.49	65.08	0.0225750	0.0016794	4.59	53.97	0.6146550	0.0003801	4.82	63.49	0.4129565	0.0003360	0.0000000	63.49	0.4129565	0.0000000	0.0003360				
Piercing	Bullish	243	-0.94	34.30	0.0000012	0.0651933	-0.90	37.19	0.0000811	0.0862778	-0.62	40.91	0.0055964	0.2533555	0.02	45.87	0.2218698	0.9828131	0.49	52.48	0.4795767	0.3203848	0.0000000	52.48	0.4795767	0.0000000	0.3203848				
Rising Window*	Bullish	167	0.26	43.71	0.1214480	0.6632911	1.01	50.30	1.0000000	0.2034436	1.57	53.89	0.3531378	0.0626935	2.59	56.29	0.1214480	0.0058542	5.83	62.28	0.0018783	0.0000080	0.0000000	62.28	0.0018783	0.0000000	0.0000080				
Tweezer Bottom	Bullish	2999	-0.64	42.51	0.0000000	0.0000002	-0.74	41.26	0.0000000	0.0000000	-0.81	40.53	0.0000000	0.0000001	-0.93	39.57	0.0000000	0.0000001	-0.82	40.49	0.0000000	0.0013710	0.0000000	40.49	0.0000000	0.0000000	0.0013710				
Tweezer Top	Bearish	2395	-0.78	48.60	0.1774451	0.0000001	-1.06	49.31	0.5131983	0.0000000	-1.33	48.48	0.1412136	0.0000000	-1.76	46.59	0.0009191	0.0000000	-2.92	46.08	0.0001378	0.0000000	0.0000000	46.08	0.0001378	0.0000000	0.0000000				
Abandoned Baby	Bearish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA				
Abandoned Baby	Bullish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA				
Downside Tasuki Gap*	Bearish	40	1.69	57.50	0.4295905	0.0485101	2.84	67.50	0.0384773	0.0026617	3.03	65.00	0.0806905	0.0024199	3.14	72.50	0.0064266	0.0030342	2.84	65.00	0.0806905	0.00716509	0.0000000	65.00	0.0806905	0.0000000	0.00716509				
Evening Doji Star	Bearish	6	1.71	66.67	0.6875000	0.5913038	1.59	66.67	0.6875000	0.6458503	-0.56	66.67	1.0000000	0.8450796	-4.68	83.33	0.2187500	0.5586740	-3.82	66.67	0.6875000	0.0917016	0.0000000	66.67	0.6875000	0.0000000	0.0917016				
Evening Star	Bearish	9	-0.11	44.44	1.0000000	0.9525775	-1.07	44.44	1.0000000	0.6501049	-1.69	44.44	1.0000000	0.5288563	-3.14	66.67	0.3078125	0.3234256	-1.51	55.56	1.0000000	0.7755187	0.0000000	55.56	1.0000000	0.0000000	0.7755187				
Morning Doji Star	Bearish	2	1.58	50.00	1.0000000	0.6149445	2.35	50.00	1.0000000	0.5669907	2.45	50.00	1.0000000	0.5581311	0.59	50.00	1.0000000	0.7470009	1.78	50.00	1.0000000	0.7281093	0.0000000	50.00	1.0000000	0.0000000	0.7281093				
Morning Star	Bullish	8	-0.86	25.00	0.2890625	0.6051147	-1.62	37.50	0.7265625	0.4050273	-1.26	37.50	0.7265625	0.5380112	-1.69	37.50	0.7265625	0.4736568	2.70	50.00	1.0000000	0.3201620	0.0000000	50.00	1.0000000	0.0000000	0.3201620				
Three Black Crows	Bearish	11	2.64	45.45	1.0000000	0.2977668	2.32	54.55	1.0000000	0.3432602	0.82	45.45	1.0000000	0.7365437	1.01	45.45	1.0000000	0.7594828	-0.20	45.45	1.0000000	0.9875821	0.0000000	45.45	1.0000000	0.0000000	0.9875821				
Three White Soldiers	Bullish	4	-6.53	0.00	0.1250000	0.1048289	-6.22	25.00	0.6250000	0.2351177	-6.39	25.00	0.6250000	0.3199526	-6.25	25.00	0.6250000	0.2969462	-2.28	50.00	1.0000000	0.5123131	0.0000000	50.00	1.0000000	0.0000000	0.5123131				
Tri Star	Bullish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA				
Tri Star	Bearish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA				
Upside Tasuki Gap	Bullish	36	4.60	66.67	0.0652453	0.0000443	5.76	63.89	0.1324982	0.0001507	7.00	63.89	0.1324982	0.0002072	8.55	58.33	0.0593322	0.0005010	9.47	58.33	0.4053222	0.0003719	0.0000000	58.33	0.4053222	0.0000000	0.0003719				
Falling Three Methods	Bearish	131	0.02	49.62	1.0000000	0.9684072	-0.01	48.09	0.7368761	0.9719648	-0.23	49.62	1.0000000	0.6522821	-0.28																

N represents the total number of occurrences. An asterisk (*) indicates that the pattern satisfies conditions 2, 3, and 4 as described in section 3.5. for at least one holding period.

Table 5.3: Empirical results, dataset C

Patterns		Holding period 1 day						Holding period 2 days						Holding period 3 days						Holding period 5 days						Holding period 10 days					
Pattern	Pattern type	N	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.					
Dragonfly Doji	Bullish	1251	-0.99	34.48	0.0000000	0.0010308	-0.73	38.81	0.0000000	0.1928211	0.20	41.54	0.0000000	0.5520798	1.36	43.40	0.0000036	0.0491040													
Gravestone Doji	Bearish	1205	-2.98	33.38	0.0000000	0.0000000	-3.49	37.05	0.0000000	0.0000000	-3.89	39.16	0.0000000	0.0000000	-4.35	41.83	0.0000000	0.0000000													
Hammer*	Bullish	3196	0.34	51.21	0.1776706	0.0026333	0.49	51.51	0.0919379	0.0006688	0.57	52.23	0.0123665	0.0002633	0.69	51.16	0.1953368	0.0001148	0.0000001												
Hanging Man	Bearish	2169	-0.91	47.73	0.0368540	0.0000002	-3.08	46.67	0.0020848	0.0023146	-3.87	46.94	0.0048109	0.0044882	-4.84	46.27	0.0005713	0.0025836	0.0001197												
Inverted Hammer	Bullish	2785	0.06	44.96	0.0000001	0.6892340	0.13	45.51	0.0000023	0.4325618	3.08	45.22	0.0000005	0.1232780	4.97	45.63	0.0000044	0.1264555	0.0000000												
Long Lower Shadow	Bullish	21285	0.03	47.62	0.0000000	0.5821840	0.38	48.67	0.0001222	0.0007253	0.63	48.98	0.0030696	0.0000014	1.09	48.87	0.0010713	0.0000000	0.0000000												
Long Upper Shadow*	Bearish	25410	-0.47	51.11	0.0004222	0.0000000	-0.60	51.21	0.0001200	0.0000000	-0.70	51.61	0.0000003	0.0000000	-1.09	51.84	0.0000000	0.0000000	0.0000000												
Marubozu Black	Bearish	9453	-1.33	46.93	0.0000000	0.0000000	-1.54	46.37	0.0000000	0.0000000	-1.66	46.23	0.0000000	0.0000000	-1.52	48.10	0.0002584	0.0000000	0.0000000												
Marubozu White	Bullish	8094	0.44	44.28	0.0000000	0.0003525	0.72	46.19	0.0000000	0.0000000	1.06	47.25	0.0000008	0.0000000	1.61	48.07	0.0005440	0.0000000	0.0005566												
Shooting Star*	Bearish	1999	-0.55	51.68	0.1398777	0.0236318	-0.69	50.28	0.8230267	0.0005925	-0.77	51.05	0.3590166	0.0004917	-1.21	52.11	0.0629066	0.0000955	0.0000001												
Dark Cloud Cover	Bearish	535	-0.47	47.66	0.2994458	0.1325815	-0.61	50.47	0.8627231	0.0755705	-0.97	51.40	0.5450376	0.0202775	-1.47	52.53	0.2606674	0.0053725	0.0001062												
Doji Star	Bearish	1148	-0.40	51.92	0.2043836	0.0944972	-0.83	50.61	0.7012311	0.0023063	-1.10	48.00	0.1841099	0.0002705	-1.65	47.82	0.1480920	0.0000049	0.0000000												
Doji Star	Bullish	1144	-0.24	51.27	0.4071576	0.3309158	-0.33	48.73	0.4071576	0.2286376	-0.27	48.38	0.2865271	0.3942377	-0.23	47.68	0.1236613	0.5431483	0.8821999												
Engulfing	Bearish	8814	-0.50	47.09	0.0000000	0.0000000	-0.75	47.91	0.0000880	0.0000000	-1.01	48.47	0.0042733	0.0000000	-1.53	48.02	0.0002311	0.0000000	0.0000000												
Engulfing	Bullish	11646	0.12	44.94	0.0000000	0.3477030	0.20	46.19	0.0000000	0.0712426	0.20	45.51	0.0000000	0.1054099	0.14	44.60	0.0000000	0.3512223	0.333	43.71	0.0000000										
Falling Window	Bearish	417	-1.47	36.69	0.0000001	0.0000013	-1.39	45.08	0.0500049	0.0001611	-0.91	47.96	0.4333556	0.0273839	-0.64	50.12	1.0000000	0.1632845	0.0000000												
Harumi	Bearish	478	-0.25	51.05	0.6806395	0.4317278	-0.63	51.26	0.6149203	0.0867904	-0.96	52.72	0.2528169	0.0256936	-1.49	52.72	0.2528169	0.0058920	0.0000027												
Harumi	Bullish	561	-0.45	44.74	0.0142624	0.0685166	0.15	50.27	0.9327165	0.6541644	0.24	48.84	0.6124466	0.5542033	0.38	47.95	0.3529816	0.3763163	0.8088478												
Harumi Cross	Bearish	90	3.08	31.11	0.0004379	0.1203108	2.88	45.56	0.4607925	0.1560863	3.06	48.89	0.9161289	0.0838248	1.95	46.67	0.5984117	0.4195011	0.8114430												
Harumi Cross*	Bullish	118	4.05	36.44	0.0041237	0.0001915	4.08	42.74	0.1387532	0.0000833	4.47	45.30	0.3552797	0.0003833	4.64	49.57	1.0000000	0.0001094	0.0000088												
Kicking	Bearish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA												
Kicking	Bullish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA												
On Neck*	Bearish	119	2.50	54.62	0.3593583	0.0107829	3.67	57.14	0.1421320	0.0001304	3.71	64.71	0.0017105	0.0001228	3.73	57.63	0.1172129	0.0004470	0.0003166												
Piercing	Bullish	463	-0.71	36.15	0.0000000	0.0359470	-0.57	38.53	0.0000009	0.0975719	-0.39	43.29	0.0044886	0.2877095	0.06	48.70	0.0688625	0.8866920	0.37	54.55	0.0563392	0.4752042									
Rising Bottom*	Bullish	264	0.29	44.70	0.0963804	0.5615332	1.00	50.76	0.8355580	0.0982056	1.46	53.41	0.2954191	0.0250951	2.29	54.92	0.1237292	0.0017174	0.0000001												
Tweezer Bottom	Bullish	5971	-0.53	42.97	0.0000000	0.0001169	-0.60	42.13	0.0000000	0.0000003	-0.64	41.83	0.0000000	0.0000003	-0.72	40.61	0.0000000	0.0001122	0.0000000												
Tweezer Top	Bearish	4197	-0.74	48.92	0.1647561	0.0000000	-0.93	49.70	0.7110442	0.0000000	-1.20	49.08	0.2407424	0.0000000	-1.76	47.55	0.0016231	0.0000000	0.0000000												
Abandoned Baby	Bearish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA												
Abandoned Baby	Bullish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA												
Downside Tasuki Gap	Bearish	87	0.79	55.17	0.3911910	0.2580080	1.40	54.02	0.5202916	0.0044446	1.41	52.87	0.6682850	0.0679641	1.28	57.47	0.1979791	0.2455379	0.7751128												
Evening Doji Star	Bearish	12	0.51	50.00	1.0000000	0.7344157	0.76	58.33	0.7744141	0.6996821	-1.21	75.00	0.1459961	0.6663694	-3.05	75.00	0.1459961	0.5038633	0.3403233												
Evening Star	Bearish	16	-0.25	43.75	0.8036194	0.8388145	-0.42	50.00	1.0000000	0.7716959	-1.43	62.50	0.4544983	0.5205613	-2.03	68.75	0.2101135	0.5401836	0.4331009												
Morning Doji Star	Bullish	5	0.11	60.00	1.0000000	0.9933701	-0.51	60.00	1.0000000	0.8303907	-0.52	60.00	1.0000000	0.8256372	-0.55	60.00	1.0000000	0.7818144	0.9206217												
Morning Star	Bullish	14	-1.03	35.71	0.4239502	0.4089451	-1.97	42.86	0.7905273	0.1685293	-1.46	42.86	0.7905273	0.3158417	-1.76	42.86	0.7905273	0.2779487	0.65	50.00	1.0000000	0.7383006									
Three Black Crows	Bearish	19	2.21	52.63	1.0000000	0.2031991	2.08	57.89	0.6476059	0.2208703	1.48	52.63	1.0000000	0.4109805	0.62	52.63	1.0000000	0.3877012	0.74	57.89	0.6476059	0.314185									
Three White Soldiers	Bullish	9	1.48	44.44	1.0000000	0.6691393	1.21	44.44	1.0000000	0.7562409	0.93	44.44	1.0000000	0.8197336	0.65	44.44	1.0000000	0.8754691	0.2724694												
Tri Star	Bullish	1	5.74	100.00	1.0000000	NA	-0.51	0.00	1.0000000	NA	-5.30	0.00	1.0000000	NA	-8.35	0.00	1.0000000	NA	NA												
Tri Star	Bearish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA												
Upside Tasuki Gap	Bullish	67	1.93	56.72	0.3284321	0.0153925	2.40	55.22	0.4638176	0.0183061	3.09	58.21	0.2215487	0.0098777	4.28	58.21	0.2215487	0.0041541	0.0057917												
Falling Three Methods	Bearish	284	-1.77	49.65	0.9526959	0.0016652	-1.58	48.59	0.6779427	0.0034790	-1.38	47.89	0.5140019	0.0075118	-0.83	48.94	0.7667600	0.1587824	0.4254264												
Rising Three Methods	Bullish	234	0.09	46.15	0.2663894	0.8577317	0.58	49.15	0.8445722	0.2825410	0.88	49.15	0.8445722	0.1473954	1.78	49.79	1.0000000	0.0185833	0.0014563												

N represents the total number of occurrences. An asterisk (*) indicates that the pattern satisfies conditions 2, 3, and 4 as described in section 3.5. for at least one holding period.

Table 5.4: Empirical results, dataset D

Patterns		Holding period 1 day					Holding period 2 days					Holding period 3 days					Holding period 5 days					Holding period 10 days				
Pattern	Pattern type	N	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.
Dragonfly Doji	Bearish	1081	-1.05	37.05	0.000000	0.000860	-1.09	40.67	0.000000	0.001749	-0.88	42.71	0.000019	0.019170	-0.76	43.40	0.0000168	0.0828156	0.27	43.66	0.000365	0.633868	0.27	43.66	0.000365	0.633868
Gravestone Doji	Bearish	1127	-3.02	36.77	0.000000	0.000000	-3.55	40.00	0.000000	0.000000	-3.99	41.81	0.000000	0.000000	-4.45	43.63	0.0000221	0.000000	-5.49	43.30	0.000083	0.000000	-5.49	43.30	0.000083	0.000000
Hammer*	Bullish	3041	0.30	50.43	0.6493894	0.106268	0.46	50.98	0.2914722	0.002422	0.56	51.94	0.0348593	0.0006020	0.69	50.91	0.3262999	0.000267	1.14	50.84	0.3631458	0.000001	1.14	50.84	0.3631458	0.000001
Hanging Man	Bearish	2061	-0.91	48.00	0.000001	0.0737311	-0.93	46.83	0.0043916	0.0029616	-1.37	47.12	0.0007822	0.0054942	-1.99	46.56	0.0019854	0.0033469	-8.77	47.70	0.0395996	0.000316	-8.77	47.70	0.0395996	0.000316
Inverted Hammer	Bullish	2702	0.09	44.86	0.000000	0.5940067	0.35	45.61	0.0000655	0.3712254	3.20	45.32	0.0000112	0.1197084	5.15	45.70	0.0000887	0.1246060	9.89	46.64	0.0005239	0.0029423	9.89	46.64	0.0005239	0.0029423
Long Lower Shadow	Bullish	20222	0.03	47.73	0.000000	0.6177594	0.35	48.76	0.0004589	0.0028484	0.60	49.03	0.0059990	0.0000224	1.03	48.83	0.0009935	0.0000000	2.30	48.75	0.0004030	0.000000	2.30	48.75	0.0004030	0.000000
Long Upper Shadow*	Bearish	24511	-0.47	51.46	0.0000046	0.0000000	-0.59	51.55	0.0000012	0.0000000	-0.69	51.95	0.0000000	0.0000000	-0.99	52.05	0.0000000	0.0000000	-3.79	52.28	0.0000000	0.0001598	-3.79	52.28	0.0000000	0.0001598
Marubozu Black	Bearish	8990	-1.36	47.61	0.0000663	0.0000000	-1.58	46.89	0.0000000	0.0000000	-1.57	46.69	0.0000000	0.0000000	-1.56	48.39	0.0024810	0.0000000	-2.59	48.45	0.0037571	0.0000000	-2.59	48.45	0.0037571	0.0000000
Marubozu White	Bullish	7530	0.45	44.54	0.000000	0.0007338	0.71	46.45	0.0000000	0.0000002	1.04	47.43	0.0000386	0.0000000	1.59	48.26	0.0026226	0.0000000	8.84	48.39	0.0034285	0.0009078	8.84	48.39	0.0034285	0.0009078
Shooting Star*	Bearish	1929	-0.58	51.63	0.1580666	0.0018995	-0.72	50.29	0.8198986	0.0004284	-0.78	51.04	0.3744372	0.0005126	-1.24	51.98	0.0871023	0.000085	-1.97	52.51	0.0297839	0.0000001	-1.97	52.51	0.0297839	0.0000001
Dark Cloud Cover	Bearish	509	-0.42	48.13	0.4249935	0.1966372	-0.58	50.88	0.7229321	0.0955443	-0.97	51.87	0.4249935	0.0264236	-1.52	52.86	0.2136339	0.0058520	-2.91	48.42	0.5049190	0.000898	-2.91	48.42	0.5049190	0.000898
Doji Star	Bearish	1085	-0.42	51.98	0.2022624	0.0940107	-0.83	50.60	0.7156476	0.0036591	-1.08	48.02	0.2022624	0.0005871	-1.63	47.83	0.1625314	0.000162	-3.40	47.96	0.1808743	0.0000000	-3.40	47.96	0.1808743	0.0000000
Doji Star	Bullish	1081	-0.27	50.42	0.8075969	0.2839896	-0.35	48.47	0.3296691	0.2194629	-0.29	48.10	0.223112	0.3756214	-0.23	47.82	0.1613673	0.5615497	-0.05	48.75	0.4286532	0.9536972	-0.05	48.75	0.4286532	0.9536972
Engulfing	Bearish	8395	-0.49	47.39	0.0000018	0.0000000	-0.74	48.19	0.0009387	0.0000000	-0.89	48.80	0.0282976	0.0000000	-1.51	48.20	0.0010818	0.0000000	-2.77	49.37	0.2501613	0.0000000	-2.77	49.37	0.2501613	0.0000000
Engulfing	Bullish	11218	0.13	44.83	0.0000000	0.3370841	0.20	46.11	0.0000000	0.0824663	0.20	45.51	0.0000000	0.1155913	0.14	44.49	0.0000000	0.3513289	5.53	43.67	0.0000000	0.1177485	5.53	43.67	0.0000000	0.1177485
Falling Window	Bearish	381	-1.48	37.27	0.0000008	0.0000044	-1.41	44.62	0.402961	0.002985	-0.96	47.24	0.3055324	0.0280892	-0.73	49.87	1.0000000	0.1348089	-0.71	54.59	0.0813950	0.2341148	-0.71	54.59	0.0813950	0.2341148
Harumi	Bearish	442	-0.22	50.90	0.7392098	0.5024966	-0.59	50.68	0.8120492	0.1260599	-0.90	51.81	0.4755916	0.0453546	-1.39	52.26	0.3661513	0.0140538	-3.28	48.07	0.4461532	0.0000253	-3.28	48.07	0.4461532	0.0000253
Harumi	Bullish	533	-0.48	44.09	0.0071852	0.0636971	0.14	50.09	1.0000000	0.6826110	0.24	48.41	0.4883219	0.5622480	0.39	47.47	0.2606674	0.3925020	-0.11	44.47	0.0119251	0.9097687	-0.11	44.47	0.0119251	0.9097687
Harumi Cross	Bearish	66	3.27	30.30	0.0018582	0.2118391	3.65	46.97	0.7122309	0.0211921	4.75	53.03	0.7122309	0.0211921	4.65	51.52	0.9021585	0.0134929	2.24	51.52	0.9021585	0.4184071	2.24	51.52	0.9021585	0.4184071
Harumi Cross*	Bullish	91	4.78	40.66	0.0929469	0.006119	4.71	45.56	0.4697925	0.0004287	5.18	47.78	0.7520332	0.0001944	5.36	52.22	0.7520332	0.0004071	7.10	61.11	0.0445975	0.0000402	7.10	61.11	0.0445975	0.0000402
Kicking	Bearish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Kicking	Bullish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
On Neck*	Bearish	115	2.55	55.65	0.2630540	0.0118093	3.75	57.39	0.1353504	0.0001496	3.83	65.22	0.0014130	0.0001163	3.88	57.89	0.1109383	0.000395	4.13	62.28	0.0111186	0.0003430	4.13	62.28	0.0111186	0.0003430
Piercing	Bullish	449	-0.75	36.16	0.0000000	0.0309013	-0.62	38.39	0.0000010	0.0793076	-0.44	43.30	0.0052542	0.2410222	0.02	48.88	0.0707310	0.9659903	0.36	54.69	0.0526157	0.4978136	0.36	54.69	0.0526157	0.4978136
Rising Window*	Bullish	233	0.50	45.92	0.2382461	0.3636766	1.40	52.79	0.4318458	0.0317982	1.85	54.51	0.1900004	0.0086112	2.58	55.36	0.1156909	0.0011386	5.55	62.50	0.0001698	0.0000002	5.55	62.50	0.0001698	0.0000002
Tweezer Bottom	Bullish	5693	-0.51	43.28	0.0000000	0.0000631	-0.57	42.52	0.0000000	0.0000026	-0.61	42.15	0.0000000	0.0000026	-0.68	40.84	0.0000000	0.0000613	-0.75	40.53	0.0000000	0.0008837	-0.75	40.53	0.0000000	0.0008837
Tweezer Top	Bearish	3949	-0.73	49.23	0.3396850	0.0000000	-0.91	49.96	0.9746111	0.0000000	-1.20	49.13	0.2792086	0.0000000	-1.79	47.48	0.0016116	0.0000000	-2.97	47.13	0.0003485	0.0000000	-2.97	47.13	0.0003485	0.0000000
Abandoned Baby	Bearish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Abandoned Baby	Bullish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Downside Tasuki Gap	Bearish	76	0.54	53.95	0.5665734	0.4734785	1.00	52.63	0.7310089	0.1533936	1.03	52.63	0.7310089	0.1996336	0.90	56.58	0.3018725	0.4509621	0.33	56.58	0.3018725	0.915994	0.33	56.58	0.3018725	0.915994
Evening Doji Star	Bearish	12	0.51	50.00	1.0000000	0.7344157	0.76	58.33	0.7744141	0.6996821	-1.21	75.00	0.1459961	0.6663694	-3.05	75.00	0.1459961	0.5083633	-6.41	66.67	0.3876953	0.3404233	-6.41	66.67	0.3876953	0.3404233
Evening Star	Bearish	16	-0.25	43.75	0.8036194	0.8388145	-0.42	50.00	1.0000000	0.7716959	-1.43	62.50	0.4549583	0.5205613	-2.03	68.75	0.2101135	0.5401836	-3.90	62.50	0.4549583	0.4331009	-3.90	62.50	0.4549583	0.4331009
Morning Doji Star	Bullish	5	0.11	60.00	1.0000000	0.9933701	-0.51	60.00	1.0000000	0.8303907	-0.52	60.00	1.0000000	0.8256372	-0.55	60.00	1.0000000	0.7818144	0.30	60.00	1.0000000	0.9206217	0.30	60.00	1.0000000	0.9206217
Morning Star	Bullish	14	-1.03	35.71	0.4239502	0.4089451	-1.97	42.86	0.7905273	0.1685293	-1.46	42.86	0.7905273	0.3158417	-1.76	42.86	0.7905273	0.2779487	0.65	50.00	1.0000000	0.7383006	0.65	50.00	1.0000000	0.7383006
Three Black Crows	Bearish	18	1.94	50.00	1.0000000	0.2787911	2.19	55.56	0.8145294	0.2259368	1.85	55.56	0.8145294	0.3281155	2.34	55.56	0.8145294	0.3522467	3.09	61.11	0.4806824	0.2827088	3.09	61.11	0.4806824	0.2827088
Three White Soldiers	Bullish	9	1.48	44.44	1.0000000	0.6691393	1.21	44.44	1.0000000	0.7562409	0.93	44.44	1.0000000	0.8197636	0.62	44.44	1.0000000	0.8754691	4.47	66.67	0.5078125	0.2724694	4.47	66.67	0.5078125	0.2724694
Tri Star	Bullish	1	5.74	100.00	1.0000000	NA	-0.51	0.00	1.0000000	NA	-5.30	0.00	1.0000000	NA	-8.35	0.00	1.0000000	NA	-12.51	0.00	1.0000000	NA	-12.51	0.00	1.0000000	NA
Tri Star	Bearish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Upside Tasuki Gap	Bullish	59	1.77	55.93	0.4349928	0.0444379	2.09	54.24	0.6029232	0.0395991	2.87	57.63	0.2973952	0.0311077	4.10	59.32	0.1925265	0.0114978	5.57	62.07	0.0869489	0.0085824	5.57	62.07	0.0869489	0.0085824
Falling Three Methods	Bearish	269	-1.80	50.56	0.9029745	0.0025922	-1.57	50.19	1.0000000	0.0063419	-1.36	49.07	0.8073758	0.0129439	-0.80	50.19	1.0000000	0.1967201	-0.53	48.33	0.6257559	0.4853976	-0.53	48.33	0.6257559	0.4853976
Rising Three Methods	Bullish	221	0.12	47.06	0.4196093	0.8274034	0.60	50.23</																		

Table 5.5: Empirical results, dataset E

Patterns		Holding period 1 day					Holding period 2 days					Holding period 3 days					Holding period 5 days					Holding period 10 days				
Pattern	Pattern type	N	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Avg. returns (%)	Win rate (%)	Binom. test p-val.	Skew adj. t-test p-val.
Dragonfly Doji	Bullish	38	7.66	28.95	0.0138530	0.1988140	2.72	31.58	0.0335524	0.1756485	1.14	36.84	0.1433067	0.5781912	-0.22	36.84	0.1433067	0.5781912	-0.22	36.84	0.1433067	0.5781912	-0.22	36.84	0.1433067	0.5781912
Gravestone Doji	Bearish	32	-0.46	37.50	0.2153271	0.1086457	-0.39	40.62	0.3770856	0.0245233	-0.34	43.75	0.5066149	0.0445115	-0.44	34.38	0.1101842	0.0858220	-0.69	34.38	0.1101842	0.0858220	-0.69	34.38	0.1101842	0.0858220
Hammer	Bullish	44	1.07	52.27	0.8893958	0.0024331	1.07	50.00	1.0000000	0.0037130	1.10	47.73	0.8803958	0.0188234	1.06	50.00	1.0000000	0.0892822	0.59	45.45	0.6515878	0.0892822	0.59	45.45	0.6515878	0.0892822
Hammer	Bearish	61	0.07	57.38	0.3056774	0.8351011	-0.62	57.38	0.3056774	0.4214757	-1.04	59.02	0.2000314	0.3759841	-1.91	63.93	0.0396170	0.2344106	-2.68	68.85	0.0044440	0.2344106	-2.68	68.85	0.0044440	0.2344106
Inverted Hammer	Bullish	48	-0.24	62.50	0.1114029	0.2808316	-0.26	64.58	0.0594634	0.1333825	-0.16	70.83	0.005152	0.5491574	-0.14	68.75	0.0132833	0.6648952	-0.15	68.75	0.0132833	0.6648952	-0.15	68.75	0.0132833	0.6648952
Long Lower Shadow	Bullish	995	0.05	40.81	0.0000000	0.7592581	-0.10	41.01	0.0000000	0.6946530	-0.08	41.15	0.0000000	0.7117664	-0.07	41.44	0.0000001	0.7535285	-0.03	43.48	0.0000095	0.7535285	-0.03	43.48	0.0000095	0.7535285
Long Upper Shadow	Bearish	1225	-0.02	46.16	0.0078297	0.8352789	-0.07	42.73	0.0000004	0.1327577	0.01	42.76	0.0000005	0.9444278	0.07	41.85	0.0000000	0.6251007	-0.02	41.33	0.0000000	0.6251007	-0.02	41.33	0.0000000	0.6251007
Marubozu Black	Bearish	64	-0.45	32.81	0.0081469	0.4141911	-0.53	32.81	0.0081469	0.3250468	-0.30	29.69	0.0015628	0.6626662	-0.26	32.81	0.0081469	0.8202441	-0.61	33.33	0.011414	0.8202441	-0.61	33.33	0.011414	0.8202441
Marubozu White	Bullish	51	-0.57	35.29	0.0488739	0.5624227	0.18	37.25	0.0919145	0.8797814	1.87	39.22	0.1607796	0.2936048	2.94	41.18	0.2624375	0.1457128	3.17	45.10	0.5758493	0.1457128	3.17	45.10	0.5758493	0.1457128
Shooting Star	Bearish	19	0.31	47.37	1.0000000	0.6822755	0.83	57.89	0.6470659	0.4075478	0.77	42.11	0.6470659	0.5308202	0.81	42.11	0.6470659	0.5494098	1.72	52.63	1.0000000	0.5494098	1.72	52.63	1.0000000	0.5494098
Dark Cloud Cover	Bearish	18	-0.06	44.44	0.8145294	0.6879410	-0.19	27.78	0.0962524	0.1118784	-0.38	33.33	0.2378845	0.0328859	-0.60	50.00	1.0000000	0.0737815	-0.51	50.00	1.0000000	0.0737815	-0.51	50.00	1.0000000	0.0737815
Doji Star*	Bearish	132	0.28	70.45	0.0000029	0.036061	0.27	72.73	0.0000002	0.0001700	0.24	74.24	0.0000000	0.0196081	0.20	71.97	0.0000005	0.0354296	0.19	69.47	0.0000098	0.0354296	0.19	69.47	0.0000098	0.0354296
Doji Star*	Bullish	138	0.22	73.19	0.0000000	0.0005765	0.47	73.91	0.0000000	0.0002785	0.82	74.64	0.0000000	0.0007189	1.04	76.09	0.0000000	0.0023719	1.10	78.99	0.0000000	0.0023719	1.10	78.99	0.0000000	0.0023719
Engulfing	Bearish	285	0.02	46.67	0.2863019	0.8257629	-0.01	43.16	0.0242292	0.9178802	-0.04	47.02	0.3432676	0.7848334	-0.18	47.89	0.5140019	0.3583820	-0.63	46.81	0.3113749	0.3583820	-0.63	46.81	0.3113749	0.3583820
Engulfing	Bullish	325	-0.12	39.38	0.0001537	0.3581306	-0.11	40.62	0.0008472	0.2706465	-0.17	37.23	0.0000048	0.0695615	-0.23	38.58	0.0000466	0.0326838	-0.22	43.21	0.0167061	0.0326838	-0.22	43.21	0.0167061	0.0326838
Falling Window	Bearish	3	1.83	33.33	1.0000000	0.6635955	1.22	33.33	1.0000000	0.7457369	1.80	33.33	1.0000000	0.6629751	2.45	33.33	1.0000000	0.5881953	5.78	66.67	0.0000000	0.5881953	5.78	66.67	0.0000000	0.5881953
Harumi	Bearish	6	0.36	50.00	1.0000000	0.3658427	0.11	66.67	0.6875000	0.7463742	0.01	66.67	0.6875000	0.9788129	-0.14	83.33	0.2187500	0.4248779	-0.02	66.67	0.6875000	0.4248779	-0.02	66.67	0.6875000	0.4248779
Harumi	Bullish	9	1.04	33.33	0.5078125	0.4250763	1.63	44.44	1.0000000	0.1907285	2.18	55.56	1.0000000	0.1058662	2.25	44.44	1.0000000	0.0941200	1.86	55.56	1.0000000	0.0941200	1.86	55.56	1.0000000	0.0941200
Harumi Cross	Bearish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Harumi Cross	Bullish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Kicking	Bearish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Kicking	Bullish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
On Neck	Bearish	1	-0.45	0.00	1.0000000	NA	-0.35	0.00	1.0000000	NA	-0.25	0.00	1.0000000	NA	-0.18	0.00	1.0000000	NA	-0.10	0.00	1.0000000	NA	-0.10	0.00	1.0000000	NA
Piercing	Bullish	22	0.03	40.91	0.5234671	0.8473007	0.14	54.55	0.8318119	0.5048764	0.21	63.64	0.2862787	0.3263936	0.73	72.73	0.0524788	0.0991054	1.93	72.73	0.0524788	0.0991054	1.93	72.73	0.0524788	0.0991054
Rising Window	Bullish	3	0.01	66.67	1.0000000	0.9508956	0.08	66.67	1.0000000	0.3324047	-0.20	0.00	0.2500000	0.2118015	-0.24	33.33	1.0000000	0.3671532	-0.22	33.33	1.0000000	0.3671532	-0.22	33.33	1.0000000	0.3671532
Tweezer Bottom	Bullish	250	-0.03	42.08	0.0127868	0.8484985	0.01	45.95	0.2138841	0.8874108	-0.05	45.56	0.1715017	0.7713902	-0.10	44.19	0.0707962	0.4611007	-0.18	43.19	0.0337304	0.4611007	-0.18	43.19	0.0337304	0.4611007
Tweezer Top	Bearish	201	-0.07	49.75	1.0000000	0.7566588	-0.02	50.25	1.0000000	0.9397224	-0.01	46.50	0.3580031	0.9727908	-0.03	48.50	0.7237710	0.9086113	-0.16	47.98	0.6189710	0.9086113	-0.16	47.98	0.6189710	0.9086113
Abandoned Baby	Bearish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Abandoned Baby	Bullish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Downside Tasuki Gap	Bearish	7	-0.03	16.67	0.2187500	0.9646185	0.18	16.67	0.2187500	0.5416279	0.52	16.67	0.2187500	0.3622282	0.55	33.33	0.6875000	0.3823531	-0.18	20.00	0.3750000	0.3823531	-0.18	20.00	0.3750000	0.3823531
Evening Doji Star	Bearish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Evening Star	Bearish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Morning Doji Star	Bullish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Morning Star	Bullish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Three Black Crows	Bearish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Three White Soldiers	Bullish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Tri Star	Bullish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Tri Star	Bearish	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Upside Tasuki Gap	Bullish	11	-0.03	36.36	0.5488281	0.7247614	-0.09	27.27	0.2263625	0.2725326	-0.18	9.09	0.0117188	0.0041067	-0.21	9.09	0.0117188	0.0063419	-0.25	9.09	0.0117188	0.0063419	-0.25	9.09	0.0117188	0.0063419
Falling Three Methods	Bearish	14	-0.17	28.57	0.1795654	0.1016136	-0.21	14.29	0.0129395	0.0004073	-0.26	14.29	0.0129395	0.0004073	-0.27	7.14	0.0018311	0.0003249	-0.30	7.14	0.0018311	0.0003249	-0.30	7.14	0.0018311	0.0003249
Rising Three Methods	Bullish	11	-0.29	9.09	0.0117188	0.0328372	-0.37	0.00	0.0000766	0.0102923	-0.36	0.00	0.0000766	0.0047540	-0.33	9.09	0.0117188	0.0137250	-0.44	0.00	0.0000766	0.0137250	-0.44	0.00	0.0000766	0.0137250

N represents the total number of occurrences. An asterisk (*) indicates that the pattern satisfies conditions 2, 3, and 4 as described in section 3.5. for at least one holding period.

Chapter 6

Conclusion

The thesis explores the contentious topic of technical analysis within financial markets, particularly in cryptocurrencies. It evaluates the effectiveness of bullish and bearish patterns for predicting market trends. The study provides empirical evidence on the predictive power of these patterns on 5 different datasets. It emphasizes potential limitations and highlights the distinctions between classical financial markets and cryptocurrency markets, such as volatility, liquidity, and regulatory environments, which influence the relevance and performance of these analytical tools. By rigorously testing patterns against predefined, this research aims to guide both academics and investors towards a more informed application of technical analysis, promoting reliability and understanding in this evolving financial landscape.

The analysis identified different useful candlestick patterns for each dataset. In Dataset B, the Hammer Bullish, On Neck Bearish, Rising Window Bullish, Downside Tasuki Gap Bearish, and Shooting Star Bearish (requiring reclassification to bullish) were found effective. In Dataset C, Hammer Bullish, Harami Cross Bullish, On Neck Bearish, Rising Window Bullish, Shooting Star Bearish (suggested for reclassification to bullish), and Long Upper Shadow (suggested for reclassification to bullish) demonstrated utility. The findings for Dataset D mirrored those of Dataset C, with the same patterns identified as useful. In Dataset E, the Doji Star Bullish and Doji Star Bearish patterns consistently showed predictive power. However, this is likely due to stablecoins' inherent price-stabilization mechanisms, where deviations from their peg are corrected by arbitrage. These results reflect stablecoins' unique dynamics rather than the patterns' intrinsic

predictive value, supporting their exclusion from the primary analysis.

Among the most robust patterns identified were Hammer Bullish, On Neck Bearish, Rising Window Bullish, and Shooting Star Bearish (proposed for reclassification as bullish), as they consistently demonstrated predictive potential across multiple datasets. Their recurrence highlights their potential as valuable tools for technical analysis, particularly in the context of cryptocurrency markets. However, their reliability remains uncertain, as none of the patterns showed significance in Dataset A, raising questions about their general applicability and robustness in varying market conditions.

Out of the 41 patterns tested across cryptocurrency datasets A, B, C, and D, only 8 demonstrated some level of usefulness, with 2 of these potentially misclassified. Furthermore, just 4 patterns, one of which may also be misclassified, exhibited more robust results across datasets. These findings underscore the inherent unreliability of employing these patterns in cryptocurrency trading strategies. Additionally, the fact that some patterns produced returns contrary to their intended bullish or bearish designations further undermines the validity of such classifications and calls into question the overall applicability of traditional candlestick labeling in this context.

This thesis makes several key contributions to the field of technical analysis and its application in cryptocurrency markets. By utilizing pattern definitions from TradingView, a widely used and modern platform, the study ensures alignment with the most up-to-date and standardized interpretations of candlestick patterns, enhancing its relevance for contemporary traders and analysts. Additionally, the dataset employed is significantly larger than those in prior studies, encompassing a broad range of cryptocurrencies and providing more comprehensive insights into pattern performance. The research also introduces a distinct trend definition, diverging from the shorter timeframes typically used in earlier works, thereby offering a fresh perspective on how trends can be identified and analyzed. Furthermore, by conducting a separate evaluation of stablecoins, the study addresses the unique characteristics of this asset class, which operates under mechanisms distinct from other cryptocurrencies. This focused analysis contributes to a more nuanced understanding of the applicability of candlestick patterns across different segments of the cryptocurrency market.

Possible extensions of this work could involve expanding the analysis to include

additional datasets, particularly from decentralized exchanges, to better capture the diverse and dynamic nature of cryptocurrency markets. Future research could also explore alternative definitions of trends, as well as variations in holding and exit strategies, to evaluate their influence on the predictive accuracy and reliability of candlestick patterns. Additionally, examining the impact of market seasonality, such as bull and bear phases, on the effectiveness of these patterns could provide valuable insights into their performance under different market conditions. These enhancements would contribute to a deeper and more nuanced understanding of the applicability of candlestick patterns in varying contexts.

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Appendix A

Candlestick patterns definitions

This appendix contains the identification logic for each of the 41 patterns analyzed in this study.

A.1 Preliminary Calculations

The length used for calculating the exponential moving average of body size is given by:

$$\text{length} = 14 \tag{A.1}$$

The threshold for determining the minimum shadow size as a percentage of the body is:

$$\text{shadow_percent} = 5.0\% \tag{A.2}$$

The tolerance for considering upper and lower shadows as equal in size is:

$$\text{shadow_equals_percent} = 100.0\% \tag{A.3}$$

The maximum body size as a percentage of the range for Doji identification is:

$$\text{doji_body_percent} = 5.0\% \tag{A.4}$$

The factor used to determine when shadows dominate the body is:

$$\text{factor} = 2.0 \quad (\text{A.5})$$

Some other pattern specific parametes:

$$\text{long_lower_shadow_percent} = 75\% \quad (\text{A.6})$$

$$\text{long_shadow_percent} = 75\% \quad (\text{A.7})$$

$$\text{marubozu_shadow_percent_bearish} = 5\% \quad (\text{A.8})$$

$$\text{marubozu_shadow_percent_white} = 5\% \quad (\text{A.9})$$

To calculate the high and low of the body, we use:

$$\text{body_high} = \max(\text{open}, \text{close}) \quad (\text{A.10})$$

$$\text{body_low} = \min(\text{open}, \text{close}) \quad (\text{A.11})$$

The size of the body is calculated as:

$$\text{body} = \text{body_high} - \text{body_low} \quad (\text{A.12})$$

The average body size over the specified length is given by:

$$\text{body_avg} = \text{calculate_ema}(\text{body}, \text{length}) \quad (\text{A.13})$$

We determine if the body is smaller or larger than the average body by:

$$\text{small_body} = \text{body} < \text{body_avg} \quad (\text{A.14})$$

$$\text{long_body} = \text{body} > \text{body_avg} \quad (\text{A.15})$$

The sizes of the upper and lower shadows are calculated as follows:

$$\text{up_shadow} = \text{high} - \text{body_high} \quad (\text{A.16})$$

$$\text{dn_shadow} = \text{body_low} - \text{low} \quad (\text{A.17})$$

We check if the upper or lower shadows exceed a certain threshold of body size:

$$\text{has_up_shadow} = \text{up_shadow} > \left(\frac{\text{shadow_percent}}{100} \right) \times \text{body} \quad (\text{A.18})$$

$$\text{has_dn_shadow} = \text{dn_shadow} > \left(\frac{\text{shadow_percent}}{100} \right) \times \text{body} \quad (\text{A.19})$$

To determine if the candlestick closed higher (white) or lower (black) than it opened:

$$\text{white_body} = \text{open} < \text{close} \quad (\text{A.20})$$

$$\text{black_body} = \text{open} > \text{close} \quad (\text{A.21})$$

The total range of the candlestick is calculated as:

$$\text{range} = \text{high} - \text{low} \quad (\text{A.22})$$

We determine if the current candlestick is an inside bar compared to the previous candlestick:

$$\begin{aligned} \text{is_inside_bar} = & (\text{lag}(\text{body_high}, 1) > \text{body_high}) \wedge \\ & \wedge (\text{lag}(\text{body_low}, 1) < \text{body_low}) \end{aligned} \quad (\text{A.23})$$

The midpoint of the candle body is calculated as:

$$\text{body_middle} = \left(\frac{\text{body}}{2} \right) + \text{body_low} \quad (\text{A.24})$$

We determine if the upper and lower shadows are approximately equal in size:

$$\begin{aligned} \text{shadows_equals} = & (\text{up_shadow} = \text{dn_shadow}) \vee \\ & \vee \left(\frac{\text{up_shadow} - \text{dn_shadow}}{\text{dn_shadow}} \times 100 < \text{shadow_equals_percent} \right) \wedge \\ & \wedge \left(\frac{\text{dn_shadow} - \text{up_shadow}}{\text{up_shadow}} \times 100 < \text{shadow_equals_percent} \right) \end{aligned} \quad (\text{A.25})$$

We determine if the candlestick qualifies as a Doji:

$$\begin{aligned} \text{is_doji_body} = & (\text{range} > 0) \wedge \\ & \wedge (\text{body} \leq \text{range} * \frac{\text{doji_body_percent}}{100}) \end{aligned} \quad (\text{A.26})$$

$$\text{doji} = \text{is_doji_body} \wedge \text{shadows_equals} \quad (\text{A.27})$$

The hl2 is calculated as:

$$\text{hl2} = \frac{\text{high} + \text{low}}{2} \quad (\text{A.28})$$

Simple moving average over the last 50 days is given by:

$$\text{priceAvg} = \text{calculate_sma}(\text{close}, 50) \quad (\text{A.29})$$

Up-trend and down-trend is defined as follows:

$$\text{down_trend} = \text{close} < \text{priceAvg} \quad (\text{A.30})$$

$$\text{up_trend} = \text{close} > \text{priceAvg} \quad (\text{A.31})$$

A.2 One day patterns

A.2.1 Dragonfly Doji (Bullish)

$$\text{dragonfly_doji_bullish} = \text{is_doji_body} \wedge (\text{up_shadow} \leq \text{body}) \quad (\text{A.32})$$

A.2.2 Gravestone Doji (Bearish)

$$\text{gravestone_doji_bearish} = \text{is_doji_body} \wedge (\text{dn_shadow} \leq \text{body}) \quad (\text{A.33})$$

A.2.3 Hammer (Bullish)

$$\begin{aligned} \text{hammer_bullish} = & \text{small_body} \wedge \text{body} > 0 \wedge \\ & \text{body_low} > \frac{\text{high} + \text{low}}{2} \wedge \\ & \text{dn_shadow} \geq 2 \times \text{body} \wedge \\ & \neg \text{has_up_shadow} \wedge \text{down_trend} \end{aligned} \quad (\text{A.34})$$

A.2.4 Hanging Man (Bearish)

$$\begin{aligned}\text{hanging_man_bearish} = & \text{small_body} \wedge \text{body} > 0 \wedge \\ & \text{body_low} > \frac{\text{high} + \text{low}}{2} \wedge \\ & \text{dn_shadow} \geq 2 \times \text{body} \wedge \\ & \neg \text{has_up_shadow} \wedge \text{up_trend}\end{aligned}\tag{A.35}$$

A.2.5 Inverted Hammer (Bullish)

$$\begin{aligned}\text{inverted_hammer_bullish} = & \text{small_body} \wedge \text{body} > 0 \wedge \\ & \text{body_high} < \frac{\text{high} + \text{low}}{2} \wedge \\ & \text{up_shadow} \geq 2 \times \text{body} \wedge \\ & \neg \text{has_dn_shadow} \wedge \text{down_trend}\end{aligned}\tag{A.36}$$

A.2.6 Long Lower Shadow (Bullish)

$$\text{long_lower_shadow_bullish} = \text{dn_shadow} > \frac{\text{range}}{100} \times \text{long_lower_shadow_percent}\tag{A.37}$$

A.2.7 Long Upper Shadow (Bearish)

$$\text{long_upper_shadow_bearish} = \text{up_shadow} > \frac{\text{range}}{100} \times \text{long_shadow_percent}\tag{A.38}$$

A.2.8 Marubozu Black (Bearish)

$$\begin{aligned}\text{marubozu_black_bearish} = & \text{black_body} \wedge \text{long_body} \wedge \\ & \text{up_shadow} \leq \frac{\text{marubozu_shadow_percent_bearish}}{100} \times \text{body} \wedge \\ & \text{dn_shadow} \leq \frac{\text{marubozu_shadow_percent_bearish}}{100} \times \text{body}\end{aligned}\tag{A.39}$$

A.2.9 Marubozu White (Bullish)

$$\begin{aligned} \text{marubozu_white_bullish} = & \text{white_body} \wedge \text{long_body} \wedge \\ & \text{up_shadow} \leq \frac{\text{marubozu_shadow_percent_white}}{100} \times \text{body} \wedge \\ & \text{dn_shadow} \leq \frac{\text{marubozu_shadow_percent_white}}{100} \times \text{body} \end{aligned} \quad (\text{A.40})$$

A.2.10 Shooting Star (Bearish)

$$\begin{aligned} \text{shooting_star_bearish} = & \text{small_body} \wedge \text{body} > 0 \wedge \\ & \text{body_high} < \frac{\text{high} + \text{low}}{2} \wedge \\ & \text{up_shadow} \geq 2 \times \text{body} \wedge \\ & \neg \text{has_dn_shadow} \wedge \text{up_trend} \end{aligned} \quad (\text{A.41})$$

A.3 Two day patterns

A.3.1 Dark Cloud Cover (Bearish)

$$\begin{aligned} \text{dark_cloud_cover_bearish} = & \text{lag}(\text{up_trend}, 1) \wedge \\ & \text{lag}(\text{white_body}, 1) \wedge \\ & \text{lag}(\text{long_body}, 1) \wedge \\ & \text{black_body} \wedge \\ & \text{Open} \geq \text{lag}(\text{High}, 1) \wedge \\ & \text{Close} < \text{lag}(\text{body_middle}, 1) \wedge \\ & \text{Close} > \text{lag}(\text{Open}, 1) \end{aligned} \quad (\text{A.42})$$

A.3.2 Doji Star (Bearish)

$$\begin{aligned} \text{doji_star_bearish} = & \text{up_trend} \wedge \\ & \text{lag}(\text{white_body}, 1) \wedge \\ & \text{lag}(\text{long_body}, 1) \wedge \\ & \text{is_doji_body} \wedge \\ & \text{body_low} > \text{lag}(\text{body_high}, 1) \end{aligned} \tag{A.43}$$

A.3.3 Doji Star (Bullish)

$$\begin{aligned} \text{doji_star_bullish} = & \text{down_trend} \wedge \\ & \text{lag}(\text{black_body}, 1) \wedge \\ & \text{lag}(\text{long_body}, 1) \wedge \\ & \text{is_doji_body} \wedge \\ & \text{body_high} < \text{lag}(\text{body_low}, 1) \end{aligned} \tag{A.44}$$

A.3.4 Engulfing (Bearish)

$$\begin{aligned} \text{engulfing_bearish} = & \text{up_trend} \wedge \\ & \text{black_body} \wedge \\ & \text{long_body} \wedge \\ & \text{lag}(\text{white_body}, 1) \wedge \\ & \text{lag}(\text{small_body}, 1) \wedge \\ & \text{Close} \leq \text{lag}(\text{Open}, 1) \wedge \\ & \text{Open} \geq \text{lag}(\text{Close}, 1) \wedge \\ & (\text{Close} < \text{lag}(\text{Open}, 1) \vee \\ & \text{Open} > \text{lag}(\text{Close}, 1)) \end{aligned} \tag{A.45}$$

A.3.5 Engulfing (Bullish)

$$\begin{aligned} \text{engulfing_bullish} = & \text{down_trend} \wedge \\ & \text{white_body} \wedge \\ & \text{long_body} \wedge \\ & \text{lag}(\text{black_body}, 1) \wedge \\ & \text{lag}(\text{small_body}, 1) \wedge \\ & \text{Close} \geq \text{lag}(\text{Open}, 1) \wedge \\ & \text{Open} \leq \text{lag}(\text{Close}, 1) \wedge \\ & (\text{Close} > \text{lag}(\text{Open}, 1) \vee \\ & \text{Open} < \text{lag}(\text{Close}, 1)) \end{aligned} \tag{A.46}$$

A.3.6 Falling Window (Bearish)

$$\begin{aligned} \text{falling_window_bearish} = & \text{lag}(\text{down_trend}, 1) \wedge \\ & \text{lag}(\text{range} \neq 0, 1) \wedge \\ & \text{range} \neq 0 \wedge \\ & \text{High} < \text{lag}(\text{Low}, 1) \end{aligned} \tag{A.47}$$

A.3.7 Harami (Bearish)

$$\begin{aligned} \text{harami_bearish} = & \text{lag}(\text{long_body}, 1) \wedge \\ & \text{lag}(\text{white_body}, 1) \wedge \\ & \text{lag}(\text{up_trend}, 1) \wedge \\ & \text{black_body} \wedge \\ & \text{small_body} \wedge \\ & \text{High} \leq \text{lag}(\text{body_high}, 1) \wedge \\ & \text{Low} \geq \text{lag}(\text{body_low}, 1) \end{aligned} \tag{A.48}$$

A.3.8 Harami (Bullish)

$$\begin{aligned} \text{harami_bullish} = & \text{lag}(\text{long_body}, 1) \wedge \\ & \text{lag}(\text{black_body}, 1) \wedge \\ & \text{lag}(\text{down_trend}, 1) \wedge \\ & \text{white_body} \wedge \\ & \text{small_body} \wedge \\ & \text{High} \leq \text{lag}(\text{body_high}, 1) \wedge \\ & \text{Low} \geq \text{lag}(\text{body_low}, 1) \end{aligned} \tag{A.49}$$

A.3.9 Harami Cross (Bearish)

$$\begin{aligned} \text{harami_cross_bearish} = & \text{lag}(\text{long_body}, 1) \wedge \\ & \text{lag}(\text{white_body}, 1) \wedge \\ & \text{lag}(\text{up_trend}, 1) \wedge \\ & \text{is_doji_body} \wedge \\ & \text{High} \leq \text{lag}(\text{body_high}, 1) \wedge \\ & \text{Low} \geq \text{lag}(\text{body_low}, 1) \end{aligned} \tag{A.50}$$

A.3.10 Harami Cross (Bullish)

$$\begin{aligned} \text{harami_cross_bullish} = & \text{lag}(\text{long_body}, 1) \wedge \\ & \text{lag}(\text{black_body}, 1) \wedge \\ & \text{lag}(\text{down_trend}, 1) \wedge \\ & \text{is_doji_body} \wedge \\ & \text{High} \leq \text{lag}(\text{body_high}, 1) \wedge \\ & \text{Low} \geq \text{lag}(\text{body_low}, 1) \end{aligned} \tag{A.51}$$

A.3.11 Kicking (Bearish)

$$\begin{aligned} \text{kicking_bearish} = & \text{lag}(\text{long_body}, 1) \wedge \\ & \text{lag}(\text{white_body}, 1) \wedge \\ & \text{lag}(\text{up_shadow}, 1) \leq 0.05 \times \text{lag}(\text{body}, 1) \wedge \\ & \text{lag}(\text{dn_shadow}, 1) \leq 0.05 \times \text{lag}(\text{body}, 1) \wedge \\ & \text{long_body} \wedge \\ & \text{black_body} \wedge \\ & \text{up_shadow} \leq 0.05 \times \text{body} \wedge \\ & \text{dn_shadow} \leq 0.05 \times \text{body} \wedge \\ & \text{lag}(\text{Low}, 1) > \text{High} \end{aligned} \tag{A.52}$$

A.3.12 Kicking (Bullish)

$$\begin{aligned} \text{kicking_bullish} = & \text{lag}(\text{long_body}, 1) \wedge \\ & \text{lag}(\text{black_body}, 1) \wedge \\ & \text{lag}(\text{up_shadow}, 1) \leq 0.05 \times \text{lag}(\text{body}, 1) \wedge \\ & \text{lag}(\text{dn_shadow}, 1) \leq 0.05 \times \text{lag}(\text{body}, 1) \wedge \\ & \text{long_body} \wedge \\ & \text{white_body} \wedge \\ & \text{up_shadow} \leq 0.05 \times \text{body} \wedge \\ & \text{dn_shadow} \leq 0.05 \times \text{body} \wedge \\ & \text{lag}(\text{High}, 1) < \text{Low} \end{aligned} \tag{A.53}$$

A.3.13 On Neck (Bearish)

$$\begin{aligned} \text{on_neck_bearish} = & \text{down_trend} \wedge \\ & \text{lag}(\text{black_body}, 1) \wedge \\ & \text{lag}(\text{long_body}, 1) \wedge \\ & \text{white_body} \wedge \\ & \text{small_body} \wedge \\ & \text{range} \neq 0 \wedge \\ & \text{Open} < \text{lag}(\text{Close}, 1) \wedge \\ & |\text{Close} - \text{lag}(\text{Low}, 1)| \leq 0.05 \times \text{body_avg} \end{aligned} \tag{A.54}$$

A.3.14 Piercing (Bullish)

$$\begin{aligned} \text{piercing_bullish} = & \text{lag}(\text{down_trend}, 1) \wedge \\ & \text{lag}(\text{black_body}, 1) \wedge \\ & \text{lag}(\text{long_body}, 1) \wedge \\ & \text{white_body} \wedge \\ & \text{Open} \leq \text{lag}(\text{Low}, 1) \wedge \\ & \text{Close} > \text{lag}(\text{body_middle}, 1) \wedge \\ & \text{Close} < \text{lag}(\text{Open}, 1) \end{aligned} \tag{A.55}$$

A.3.15 Rising Window (Bullish)

$$\begin{aligned} \text{rising_window_bullish} = & \text{lag}(\text{up_trend}, 1) \wedge \\ & (\text{lag}(\text{range}, 1) \neq 0) \wedge \\ & (\text{range} \neq 0) \wedge \\ & \text{Low} > \text{lag}(\text{High}, 1) \end{aligned} \tag{A.56}$$

A.3.16 Tweezer Bottom (Bullish)

$$\begin{aligned} \text{tweezer_bottom_bullish} = & \text{lag}(\text{down_trend}, 1) \wedge \\ & (\neg \text{is_doji_body} \vee (\text{has_up_shadow} \wedge \text{has_dn_shadow})) \wedge \\ & \text{lag}(\text{black_body}, 1) \wedge \\ & \text{lag}(\text{long_body}, 1) \wedge \\ & \text{white_body} \wedge \\ & |\text{Low} - \text{lag}(\text{Low}, 1)| \leq 0.05 \times \text{body_avg} \end{aligned} \tag{A.57}$$

A.3.17 Tweezer Top (Bearish)

$$\begin{aligned} \text{tweezer_top_bearish} = & \text{lag}(\text{up_trend}, 1) \wedge \\ & (\neg \text{is_doji_body} \vee (\text{has_up_shadow} \wedge \text{has_dn_shadow})) \wedge \\ & \text{lag}(\text{white_body}, 1) \wedge \\ & \text{lag}(\text{long_body}, 1) \wedge \\ & \text{black_body} \wedge \\ & |\text{High} - \text{lag}(\text{High}, 1)| \leq 0.05 \times \text{body_avg} \end{aligned} \tag{A.58}$$

A.4 Three day patterns

A.4.1 Abandoned Baby (Bearish)

$$\begin{aligned} \text{abandoned_baby_bearish} = & \text{lag}(\text{up_trend}, 2) \wedge \\ & \text{lag}(\text{white_body}, 2) \wedge \\ & \text{lag}(\text{is_doji_body}, 1) \wedge \\ & \text{lag}(\text{High}, 2) < \text{lag}(\text{Low}, 1) \wedge \\ & \text{black_body} \wedge \\ & \text{lag}(\text{Low}, 1) > \text{High} \end{aligned} \tag{A.59}$$

A.4.2 Abandoned Baby (Bullish)

$$\begin{aligned} \text{abandoned_baby_bullish} = & \text{lag}(\text{down_trend}, 2) \wedge \\ & \text{lag}(\text{black_body}, 2) \wedge \\ & \text{lag}(\text{is_doji_body}, 1) \wedge \\ & \text{lag}(\text{Low}, 2) > \text{lag}(\text{High}, 1) \wedge \\ & \text{white_body} \wedge \\ & \text{lag}(\text{High}, 1) < \text{Low} \end{aligned} \tag{A.60}$$

A.4.3 Downside Tasuki Gap (Bearish)

$$\begin{aligned} \text{downside_tasuki_gap_bearish} = & \text{lag}(\text{long_body}, 2) \wedge \\ & \text{lag}(\text{small_body}, 1) \wedge \\ & \text{down_trend} \wedge \\ & \text{lag}(\text{black_body}, 2) \wedge \\ & \text{lag}(\text{body_high}, 1) < \text{lag}(\text{body_low}, 2) \wedge \\ & \text{black_body} \wedge \\ & \text{white_body} \wedge \\ & \text{body_high} \leq \text{lag}(\text{body_low}, 2) \wedge \\ & \text{body_high} \geq \text{lag}(\text{body_high}, 1) \end{aligned} \tag{A.61}$$

A.4.4 Evening Doji Star (Bearish)

$$\begin{aligned} \text{evening_doji_star_bearish} = & \text{lag}(\text{long_body}, 2) \wedge \\ & \text{lag}(\text{is_doji_body}, 1) \wedge \\ & \text{long_body} \wedge \\ & \text{up_trend} \wedge \\ & \text{lag}(\text{white_body}, 2) \wedge \\ & \text{lag}(\text{body_low}, 1) > \text{lag}(\text{body_high}, 2) \wedge \\ & \text{black_body} \wedge \\ & \text{body_low} \leq \text{lag}(\text{body_middle}, 2) \wedge \\ & \text{body_low} > \text{lag}(\text{body_low}, 2) \wedge \\ & \text{lag}(\text{body_low}, 1) > \text{body_high} \end{aligned} \quad (\text{A.62})$$

A.4.5 Evening Star (Bearish)

$$\begin{aligned} \text{evening_star_bearish} = & \text{lag}(\text{long_body}, 2) \wedge \\ & \text{lag}(\text{small_body}, 1) \wedge \\ & \text{long_body} \wedge \\ & \text{up_trend} \wedge \\ & \text{lag}(\text{white_body}, 2) \wedge \\ & \text{lag}(\text{body_low}, 1) > \text{lag}(\text{body_high}, 2) \wedge \\ & \text{black_body} \wedge \\ & \text{body_low} \leq \text{lag}(\text{body_middle}, 2) \wedge \\ & \text{body_low} > \text{lag}(\text{body_low}, 2) \wedge \\ & \text{lag}(\text{body_low}, 1) > \text{body_high} \end{aligned} \quad (\text{A.63})$$

A.4.6 Morning Doji Star (Bullish)

$$\begin{aligned} \text{morning_doji_star_bullish} = & \text{lag}(\text{long_body}, 2) \wedge \\ & \text{lag}(\text{black_body}, 2) \wedge \\ & \text{lag}(\text{is_doji_body}, 1) \wedge \\ & \text{lag}(\text{body_high}, 1) < \text{lag}(\text{body_low}, 2) \wedge \\ & \text{down_trend} \wedge \\ & \text{white_body} \wedge \\ & \text{long_body} \wedge \\ & \text{body_high} \geq \text{lag}(\text{body_middle}, 2) \wedge \\ & \text{body_high} < \text{lag}(\text{body_high}, 2) \wedge \\ & \text{lag}(\text{body_high}, 1) < \text{body_low} \end{aligned} \tag{A.64}$$

A.4.7 Morning Star (Bullish)

$$\begin{aligned} \text{morning_star_bullish} = & \text{lag}(\text{long_body}, 2) \wedge \\ & \text{lag}(\text{small_body}, 1) \wedge \\ & \text{long_body} \wedge \\ & \text{down_trend} \wedge \\ & \text{lag}(\text{black_body}, 2) \wedge \\ & \text{lag}(\text{body_high}, 1) < \text{lag}(\text{body_low}, 2) \wedge \\ & \text{white_body} \wedge \\ & \text{body_high} \geq \text{lag}(\text{body_middle}, 2) \wedge \\ & \text{body_high} < \text{lag}(\text{body_high}, 2) \wedge \\ & \text{lag}(\text{body_high}, 1) < \text{body_low} \end{aligned} \tag{A.65}$$

A.4.8 Three Black Crows (Bearish)

$$\begin{aligned} \text{three_black_crows_bearish} = & \text{long_body} \wedge \\ & \text{lag}(\text{long_body}, 1) \wedge \\ & \text{lag}(\text{long_body}, 2) \wedge \\ & \text{black_body} \wedge \\ & \text{lag}(\text{black_body}, 1) \wedge \\ & \text{lag}(\text{black_body}, 2) \wedge \\ & \text{Close} < \text{lag}(\text{Close}, 1) \wedge \\ & \text{lag}(\text{Close}, 1) < \text{lag}(\text{Close}, 2) \wedge \quad (\text{A.66}) \\ & \text{Open} > \text{lag}(\text{Close}, 1) \wedge \\ & \text{Open} < \text{lag}(\text{Open}, 1) \wedge \\ & \text{lag}(\text{Open}, 1) > \text{lag}(\text{Close}, 2) \wedge \\ & \text{lag}(\text{Open}, 1) < \text{lag}(\text{Open}, 2) \wedge \\ & \text{have_not_dn_shadow} \wedge \\ & \text{lag}(\text{have_not_dn_shadow}, 1) \wedge \\ & \text{lag}(\text{have_not_dn_shadow}, 2) \end{aligned}$$

A.4.9 Three White Soldiers (Bullish)

$$\begin{aligned} \text{three_white_soldiers_bullish} = & \text{long_body} \wedge \\ & \text{lag}(\text{long_body}, 1) \wedge \\ & \text{lag}(\text{long_body}, 2) \wedge \\ & \text{white_body} \wedge \\ & \text{lag}(\text{white_body}, 1) \wedge \\ & \text{lag}(\text{white_body}, 2) \wedge \\ & \text{Close} > \text{lag}(\text{Close}, 1) \wedge \\ & \text{lag}(\text{Close}, 1) > \text{lag}(\text{Close}, 2) \wedge \\ & \text{Open} < \text{lag}(\text{Close}, 1) \wedge \\ & \text{Open} > \text{lag}(\text{Open}, 1) \wedge \\ & \text{lag}(\text{Open}, 1) < \text{lag}(\text{Close}, 2) \wedge \\ & \text{lag}(\text{Open}, 1) > \text{lag}(\text{Open}, 2) \wedge \\ & \text{have_not_up_shadow} \wedge \\ & \text{lag}(\text{have_not_up_shadow}, 1) \wedge \\ & \text{lag}(\text{have_not_up_shadow}, 2) \end{aligned} \quad (\text{A.67})$$

A.4.10 Tri Star (Bullish)

$$\begin{aligned} \text{tri_star_bullish} = & \text{doji} \wedge \text{lag}(\text{doji}, 1) \wedge \text{lag}(\text{doji}, 2) \wedge \\ & \text{lag}(\text{down_trend}, 2) \wedge \text{lag}(\text{body_gap_down}, 1) \wedge \\ & \text{body_gap_up} \end{aligned} \quad (\text{A.68})$$

A.4.11 Tri Star (Bearish)

$$\begin{aligned} \text{tri_star_bearish} = & \text{doji} \wedge \\ & \text{lag}(\text{doji}, 1) \wedge \\ & \text{lag}(\text{doji}, 2) \wedge \\ & \text{lag}(\text{up_trend}, 2) \wedge \\ & \text{lag}(\text{body_gap_up}, 1) \wedge \\ & \text{body_gap_down} \end{aligned} \quad (\text{A.69})$$

A.4.12 Upside Tasuki Gap (Bullish)

$$\begin{aligned} \text{upside_tasuki_gap_bullish} = & \text{lag}(\text{long_body}, 2) \wedge \\ & \text{lag}(\text{small_body}, 1) \wedge \\ & \text{up_trend} \wedge \\ & \text{lag}(\text{white_body}, 2) \wedge \\ & \text{lag}(\text{body_low}, 1) > \text{lag}(\text{body_high}, 2) \wedge \quad (\text{A.70}) \\ & \text{lag}(\text{white_body}, 1) \wedge \\ & \text{black_body} \wedge \\ & \text{body_low} \geq \text{lag}(\text{body_high}, 2) \wedge \\ & \text{body_low} \leq \text{lag}(\text{body_low}, 1) \end{aligned}$$

A.5 Five day patterns

A.5.1 Falling Three Method (Bearish)

$$\begin{aligned} \text{falling_three_methods_bearish} = & \text{lag}(\text{down_trend}, 4) \wedge \\ & \text{lag}(\text{long_body}, 4) \wedge \text{lag}(\text{black_body}, 4) \wedge \\ & \text{lag}(\text{small_body}, 3) \wedge \text{lag}(\text{white_body}, 3) \wedge \\ & \text{lag}(\text{Open}, 3) > \text{lag}(\text{Low}, 4) \wedge \\ & \text{lag}(\text{Close}, 3) < \text{lag}(\text{High}, 4) \wedge \\ & \text{lag}(\text{small_body}, 2) \wedge \text{lag}(\text{white_body}, 2) \wedge \\ & \text{lag}(\text{Open}, 2) > \text{lag}(\text{Low}, 4) \wedge \\ & \text{lag}(\text{Close}, 2) < \text{lag}(\text{High}, 4) \wedge \\ & \text{lag}(\text{small_body}, 1) \wedge \text{lag}(\text{white_body}, 1) \wedge \\ & \text{lag}(\text{Open}, 1) > \text{lag}(\text{Low}, 4) \wedge \\ & \text{lag}(\text{Close}, 1) < \text{lag}(\text{High}, 4) \wedge \\ & \text{long_body} \wedge \text{black_body} \wedge \\ & \text{Close} < \text{lag}(\text{Close}, 4) \end{aligned} \quad (\text{A.71})$$

A.5.2 Rising Three Method (Bullish)

$$\begin{aligned} \text{rising_three_methods_bullish} = & \text{lag}(\text{up_trend}, 4) \wedge \\ & \text{lag}(\text{long_body}, 4) \wedge \text{lag}(\text{white_body}, 4) \wedge \\ & \text{lag}(\text{small_body}, 3) \wedge \text{lag}(\text{black_body}, 3) \wedge \\ & \text{lag}(\text{Open}, 3) < \text{lag}(\text{High}, 4) \wedge \\ & \text{lag}(\text{Close}, 3) > \text{lag}(\text{Low}, 4) \wedge \\ & \text{lag}(\text{small_body}, 2) \wedge \text{lag}(\text{black_body}, 2) \wedge \\ & \text{lag}(\text{Open}, 2) < \text{lag}(\text{High}, 4) \wedge \\ & \text{lag}(\text{Close}, 2) > \text{lag}(\text{Low}, 4) \wedge \\ & \text{lag}(\text{small_body}, 1) \wedge \text{lag}(\text{black_body}, 1) \wedge \\ & \text{lag}(\text{Open}, 1) < \text{lag}(\text{High}, 4) \wedge \\ & \text{lag}(\text{Close}, 1) > \text{lag}(\text{Low}, 4) \wedge \\ & \text{long_body} \wedge \text{white_body} \wedge \\ & \text{Close} > \text{lag}(\text{Close}, 4) \end{aligned} \tag{A.72}$$

Appendix B

Lists of Cryptocurrencies in Datasets A, B, C, D and E.

Table B.1: Dataset A

Symbol	Name	Market cap.
BTC-USD	Bitcoin	1.917T
ETH-USD	Ethereum	409.638B
XRP-USD	XRP	127.974B
BNB-USD	BNB	96.430B
SOL-USD	Solana	91.425B
DOGE-USD	Dogecoin	45.634B
ADA-USD	Cardano	30.891B
TRX-USD	TRON	21.06B
AVAX-USD	Avalanche	15.705B
LINK-USD	Chainlink	14.125B
TON11419-USD	Toncoin	13.704B
SUI20947-USD	Sui	13.04B
SHIB-USD	Shiba Inu	12.468B
XLM-USD	Stellar	10.928B
DOT-USD	Polkadot	10.72B
HBAR-USD	Hedera	10.028B
BCH-USD	Bitcoin Cash	8.664B

Table B.2: Dataset B

Symbol	Name	Market cap.
BTC-USD	Bitcoin	1.917T
ETH-USD	Ethereum	409.638B
XRP-USD	XRP	127.974B
BNB-USD	BNB	96.430B
SOL-USD	Solana	91.425B
DOGE-USD	Dogecoin	45.634B
ADA-USD	Cardano	30.891B
TRX-USD	TRON	21.06B
AVAX-USD	Avalanche	15.705B
LINK-USD	Chainlink	14.125B
TON11419-USD	Toncoin	13.704B
SUI20947-USD	Sui	13.04B
SHIB-USD	Shiba Inu	12.468B
XLM-USD	Stellar	10.928B
DOT-USD	Polkadot	10.72B
HBAR-USD	Hedera	10.028B
BCH-USD	Bitcoin Cash	8.664B
LEO-USD	UNUS SED LEO	8.555B
UNI7083-USD	Uniswap	7.553B
LTC-USD	Litecoin	7.333B
PEPE24478-USD	Pepe	6.902B
NEAR-USD	NEAR Protocol	6.264B
APT21794-USD	Aptos	5.921B
BGB-USD	Bitget Token	5.862B
ICP-USD	Internet Computer	4.844B
AAVE-USD	Aave	4.658B
CRO-USD	Cronos	4.197B
MNT27075-USD	Mantle	4.04B
ETC-USD	Ethereum Classic	3.974B
VET-USD	VeChain	3.755B
OM-USD	MANTRA	3.602B
XMR-USD	Monero	3.472B

Continued on next page

Symbol	Name	Market cap.
TAO22974-USD	Bittensor	3.418B
ARB11841-USD	Arbitrum	3.233B
FET-USD	Artificial Superintelligence Alliance	3.181B
KAS-USD	Kaspa	3.102B
ENA-USD	Ethena	3.077B
FIL-USD	Filecoin	3.04B
ALGO-USD	Algorand	2.722B
FTM-USD	Fantom	2.713B
OKB-USD	OKB	2.691B
ATOM-USD	Cosmos	2.617B
STX4847-USD	Stacks	2.604B
VIRTUAL-USD	Virtuals Protocol	2.487B
OP-USD	Optimism	2.476B
ONDO-USD	Ondo	2.43B
IMX10603-USD	Immutable	2.385B
TIA22861-USD	Celestia	2.295B
BONK-USD	Bonk	2.156B
INJ-USD	Injective	2.094B
GRT6719-USD	The Graph	2.021B
THETA-USD	Theta Network	1.994B
WIF-USD	dogwifhat	1.964B
SEI-USD	Sei	1.787B
WLD-USD	Worldcoin	1.787B
RUNE-USD	THORChain	1.769B
JASMY-USD	JasmyCoin	1.637B
FLOKI-USD	FLOKI	1.62B
FLR-USD	Flare	1.499B
MKR-USD	Maker	1.473B
LDO-USD	Lido DAO	1.469B
FTN-USD	Fasttoken	1.452B
BBTC31369-USD	BounceBit BTC	1.389B
BEAM28298-USD	Beam	1.346B
SAND-USD	The Sandbox	1.342B

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Symbol	Name	Market cap.
PYTH-USD	Pyth Network	1.321B
QNT-USD	Quant	1.299B
KCS-USD	KuCoin Token	1.292B
BRETT29743-USD	Brett (Based)	1.289B
GALA-USD	Gala	1.276B
RAY-USD	Raydium	1.267B
ENS-USD	Ethereum Name Service	1.247B
EOS-USD	EOS	1.236B
XTZ-USD	Tezos	1.231B
HNT-USD	Helium	1.184B
JUP29210-USD	Jupiter	1.142B
ZBU-USD	Zeebu	1.128B
GT-USD	GateToken	1.116B
AERO29270-USD	Aerodrome Finance	1.108B
FLOW-USD	Flow	1.106B
STRK22691-USD	Starknet	1.095B
AR-USD	Arweave	1.095B
AIOZ-USD	AIOZ Network	1.092B
DYDX-USD	dYdX (Native)	1.057B
XDC-USD	XDC Network	1.059B
BSV-USD	Bitcoin SV	1.057B
IOTA-USD	IOTA	1.056B
CORE23254-USD	Core	1.024B
BTT-USD	BitTorrent(New)	1.024B
NEO-USD	Neo	1.007B
CRV-USD	Curve DAO Token	984.387M
FLZ-USD	Fellaz	977.298M
AXS-USD	Axie Infinity	953.685M
EGLD-USD	MultiversX	946.134M
MANA-USD	Decentraland	932.366M
FTT-USD	FTX Token	911.413M
MATIC-USD	Polygon	906.335M
NEXO-USD	Nexo	883.144M

Continued on next page

Symbol	Name	Market cap.
APE18876-USD	ApeCoin	863.233M
JTO-USD	Jito	818.162M
ZEC-USD	Zcash	815.92M
AKT-USD	Akash Network	807.599M
PENDLE-USD	Pendle	802.098M
CHZ-USD	Chiliz	782.452M
CFX-USD	Conflux	759.533M
W-USD	Wormhole	749.661M
RON14101-USD	Ronin	741.845M
POPCAT28782-USD	Popcat (SOL)	736.818M
SUPER8290-USD	SuperVerse	731.993M
SNX-USD	Synthetix	733.639M
MINA-USD	Mina	729.026M
CAKE-USD	PancakeSwap	718.496M
COMP5692-USD	Compound	701.895M
XEC-USD	eCash	688.608M
GNO-USD	Gnosis	683.216M
ZK24091-USD	ZKsync	664.322M
DOG30933-USD	Dog (Runes)	652.793M
NOT-USD	Notcoin	651.373M
XAUT-USD	Tether Gold	649.675M
AXL17799-USD	Axelar	649.217M
CHEX-USD	Chintai	645.043M
AMP-USD	Amp	645.218M
ZRO26997-USD	LayerZero	612.706M
SPX28081-USD	SPX6900	607.475M
RSR-USD	Reserve Rights	591.686M
LUNC-USD	Terra Classic	584.61M
ROSE-USD	Oasis	580.014M
ORDI-USD	ORDI	561.418M
BLUR-USD	Blur	551.582M
BDX-USD	Beldex	538.466M
DEXE-USD	DeXe	537.875M

Continued on next page

Symbol	Name	Market cap.
MEW30126-USD	cat in a dogs world	535.832M
1INCH-USD	1inch Network	533.185M
CHEEL-USD	Cheelee	530.15M
TURBO-USD	Turbo	524.046M
PAXG-USD	PAX Gold	522.828M
MGC29839-USD	Meta Games Coin	500.917M
CTC-USD	Creditcoin	496.607M
TWT-USD	Trust Wallet Token	487.794M
CKB-USD	Nervos Network	481.269M
TEL-USD	Telcoin	482.317M
LPT-USD	Livepeer	469.301M
KSM-USD	Kusama	463.042M
KAVA-USD	Kava	454.912M
ASTR-USD	Astar	446.988M
GIGA30063-USD	Gigachad (gigachadsolana.com)	439.326M
DASH-USD	Dash	441.9M
TFUEL-USD	Theta Fuel	429.903M
BOME-USD	BOOK OF MEME	429.226M
ETHFI-USD	ether.fi	427.754M
SNEK25264-USD	Snek	422.244M
HOT2682-USD	Holo	415.317M
VRSC-USD	VerusCoin	413.505M
ATH30083-USD	Aethir	411.247M
QUBIC-USD	Qubic	399.127M
ZIL-USD	Zilliqa	397.072M
CVX-USD	Convex Finance	393.342M
WOO-USD	WOO	392.589M
ZEN-USD	Horizen	390.205M
ZRX-USD	0x Protocol	389.502M
ENJ-USD	Enjin Coin	387.289M
SUSHI-USD	SushiSwap	386.486M
PRIME23711-USD	Echelon Prime	386.152M
IO29835-USD	io.net	379.013M

Continued on next page

Symbol	Name	Market cap.
GMT18069-USD	GMT	370.153M
JST-USD	JUST	367.755M
CELO-USD	Celo	361.935M
ETHW-USD	EthereumPoW	361.157M
ONE3945-USD	Harmony	361.718M
WEMIX-USD	WEMIX	360.091M
GLM-USD	Golem	355.658M
ANKR-USD	Ankr	349.357M
BORG-USD	SwissBorg	348.084M
MEME28301-USD	Memecoin	346.553M
PEPECOIN-USD	PepeCoin	347.115M
ARKM-USD	Arkham	341.502M
IOTX-USD	IoTeX	338.356M
CPOOL-USD	Clearpool	333.85M
ID21846-USD	SPACE ID	332.847M
AEVO-USD	Aevo	332.432M
ZETA-USD	ZetaChain	328.98M
MRS21178-USD	Metars Genesis	328.712M
BAT-USD	Basic Attention Token	326.511M
TRAC-USD	OriginTrail	324.556M
MANTA-USD	Manta Network	320.103M
SFP-USD	SafePal	322.44M
SC-USD	Siacoin	323.194M
DYM-USD	Dymension	322.018M
ETHDYDX-USD	dYdX (ethDYDX)	321.399M
QTUM-USD	Qtum	322.086M
OSMO-USD	Osmosis	320.671M
ELF-USD	aelf	319.037M
MX-USD	MX Token	315.962M
MWC-USD	MimbleWimbleCoin	314.377M
RVN-USD	Ravencoin	309.403M
DSYNC-USD	Destra Network	303.595M
TRIBE-USD	Tribe	299.43M

Continued on next page

Symbol	Name	Market cap.
LUNA20314-USD	Terra	295.708M
XCH-USD	Chia	295.711M
BTG-USD	Bitcoin Gold	294.65M
KDA-USD	Kadena	294.252M
MASK8536-USD	Mask Network	291.544M
GAS-USD	Gas	288.811M
YFI-USD	yearn.finance	287M
GMX11857-USD	GMX	285.66M
ALT29073-USD	Altlayer	282.252M
ORBR-USD	Orbler	283.216M
RLB-USD	Rollbit Coin	278.773M
T-USD	Threshold	275.099M
LRC-USD	Loopring	273.2M
DRIFT31278-USD	Drift	272.187M
METIS-USD	Metis	268.38M
RBTC-USD	Rootstock Smart Bitcoin	267.707M
XRD-USD	Radix	266.475M
XYO-USD	XYO	265.386M
SKL-USD	SKALE	265.217M
DCR-USD	Decred	259.674M
WILD-USD	Wilder World	259.655M
COW19269-USD	CoW Protocol	259.194M
BICO-USD	Biconomy	255.679M
0X0-USD	0x0.ai	255.824M
BZR-USD	Bazaars	245.903M
SSV-USD	ssv.network	244.424M
POLYX-USD	Polymesh	245.074M
RPL-USD	Rocket Pool	240.173M
PAAL-USD	PAAL AI	245.045M
FLUX-USD	Flux	225.84M
ANDY29879-USD	ANDY (ETH)	225.262M
CETUS-USD	Cetus Protocol	225.145M
ZKJ-USD	Polyhedra Network	223.613M

Continued on next page

Symbol	Name	Market cap.
VTHO-USD	VeThor Token	222.932M
COTI-USD	COTI	219.978M
UMA-USD	UMA	219.359M
TAI20605-USD	TARS AI	220.34M
FXS-USD	Frax Share	218.337M
BAND-USD	Band Protocol	218.343M
XEM-USD	NEM	218.135M
GLMR-USD	Moonbeam	216.843M
ILV-USD	Illuvium	215.495M
PEOPLE-USD	ConstitutionDAO	214.239M
BLAST28480-USD	Blast	211.058M
CHR-USD	Chromia	210.466M
HONEY22850-USD	Hivemapper	208.435M
EDU24613-USD	Open Campus	206.786M
YGG-USD	Yield Guild Games	206.09M
SUN-USD	Sun [New]	204.985M
ONT-USD	Ontology	202.074M
DGB-USD	DigiByte	201.029M
PIXEL29335-USD	Pixels	200.539M
BIGTIME-USD	Big Time	200.734M
GOMINING-USD	Go ining	200.231M
ACH-USD	Alchemy Pay	199.468M
VANRY-USD	Vanar Chain	199.381M
XAI28933-USD	Xai	197.124M
AUDIO-USD	Audius	196.013M
CSPR-USD	Casper	195.794M
VVS-USD	VVS Finance	193.556M
LCX-USD	LCX	191.452M
PONKE-USD	Ponke	191.686M
ORCA-USD	Orca	191.649M
WOLF30902-USD	Landwolf 0x67	190.566M
APU30008-USD	Apu Apustaja	188.401M
UPC-USD	UPCX	186.687M

Continued on next page

Symbol	Name	Market cap.
BITCOIN25220-USD	HarryPotterObamaSonic10Inu (ERC-20)	186.465M
SXP-USD	Solar	183.996M
NPC27960-USD	Non-Playable Coin	181.251M
BB30746-USD	BounceBit	181.09M
STORJ-USD	Storj	180.485M
XVG-USD	Verge	180.281M
MPLX-USD	Metaplex	178.46M
ACX22620-USD	Across Protocol	177.478M
DAG-USD	Constellation	174.018M
ZENT-USD	Zentry	171.878M
VELO-USD	Velo	174.093M
AGI-USD	Delysium	169.934M
ZIG-USD	Signal	172.643M
ICX-USD	ICON	171.06M
HT-USD	Huobi Token	169.962M
VELO20435-USD	Velodrome Finance	169.416M
SOLO-USD	Sologenic	168.528M
XNO-USD	Nano	168.538M
HEART-USD	Humans.ai	167.382M
CFG-USD	Centrifuge	167.088M
SLP-USD	Smooth Love Potion	165.958M
WAVES-USD	Waves	165.139M
CGPT-USD	ChainGPT	162.73M
BASEDAI-USD	BasedAI	163.971M
SNT-USD	Status	162.96M
SAGA30372-USD	Saga	163.011M
MYTH-USD	Mythos	162.125M
ABT-USD	Arcblock	160.657M
LQTY-USD	Liquity	161.275M
CVC-USD	Civic	160.485M
BNX23635-USD	BinaryX	158.709M
BAL-USD	Balancer	158.353M
TRB-USD	Tellor	156.844M

Continued on next page

Symbol	Name	Market cap.
DEGEN30096-USD	Degen	156.042M
ALPH-USD	Alephium	152.59M
STRAX30168-USD	Stratis [New]	148.225M
ZANO-USD	Zano	147.898M

Table B.3: Dataset C

Symbol	Name	Market cap.
BTC-USD	Bitcoin	1.917T
ETH-USD	Ethereum	409.638B
XRP-USD	XRP	127.974B
BNB-USD	BNB	96.430B
SOL-USD	Solana	91.425B
DOGE-USD	Dogecoin	45.634B
ADA-USD	Cardano	30.891B
TRX-USD	TRON	21.06B
AVAX-USD	Avalanche	15.705B
LINK-USD	Chainlink	14.125B
TON11419-USD	Toncoin	13.704B
SUI20947-USD	Sui	13.04B
SHIB-USD	Shiba Inu	12.468B
XLM-USD	Stellar	10.928B
DOT-USD	Polkadot	10.72B
HBAR-USD	Hedera	10.028B
BCH-USD	Bitcoin Cash	8.664B
LEO-USD	UNUS SED LEO	8.555B
UNI7083-USD	Uniswap	7.553B
LTC-USD	Litecoin	7.333B
PEPE24478-USD	Pepe	6.902B
NEAR-USD	NEAR Protocol	6.264B
APT21794-USD	Aptos	5.921B
BGB-USD	Bitget Token	5.862B
ICP-USD	Internet Computer	4.844B
AAVE-USD	Aave	4.658B
CRO-USD	Cronos	4.197B
MNT27075-USD	Mantle	4.04B
ETC-USD	Ethereum Classic	3.974B
VET-USD	VeChain	3.755B
OM-USD	MANTRA	3.602B
XMR-USD	Monero	3.472B

Continued on next page

Symbol	Name	Market cap.
TAO22974-USD	Bittensor	3.418B
ARB11841-USD	Arbitrum	3.233B
FET-USD	Artificial Superintelligence Alliance	3.181B
KAS-USD	Kaspa	3.102B
ENA-USD	Ethena	3.077B
FIL-USD	Filecoin	3.04B
ALGO-USD	Algorand	2.722B
FTM-USD	Fantom	2.713B
OKB-USD	OKB	2.691B
ATOM-USD	Cosmos	2.617B
STX4847-USD	Stacks	2.604B
VIRTUAL-USD	Virtuals Protocol	2.487B
OP-USD	Optimism	2.476B
ONDO-USD	Ondo	2.43B
IMX10603-USD	Immutable	2.385B
TIA22861-USD	Celestia	2.295B
BONK-USD	Bonk	2.156B
INJ-USD	Injective	2.094B
GRT6719-USD	The Graph	2.021B
THETA-USD	Theta Network	1.994B
WIF-USD	dogwifhat	1.964B
SEI-USD	Sei	1.787B
WLD-USD	Worldcoin	1.787B
RUNE-USD	THORChain	1.769B
JASMY-USD	JasmyCoin	1.637B
FLOKI-USD	FLOKI	1.62B
FLR-USD	Flare	1.499B
MKR-USD	Maker	1.473B
LDO-USD	Lido DAO	1.469B
FTN-USD	Fasttoken	1.452B
BBTC31369-USD	BounceBit BTC	1.389B
BEAM28298-USD	Beam	1.346B
SAND-USD	The Sandbox	1.342B

Continued on next page

Symbol	Name	Market cap.
PYTH-USD	Pyth Network	1.321B
QNT-USD	Quant	1.299B
KCS-USD	KuCoin Token	1.292B
BRETT29743-USD	Brett (Based)	1.289B
GALA-USD	Gala	1.276B
RAY-USD	Raydium	1.267B
ENS-USD	Ethereum Name Service	1.247B
EOS-USD	EOS	1.236B
XTZ-USD	Tezos	1.231B
HNT-USD	Helium	1.184B
JUP29210-USD	Jupiter	1.142B
ZBU-USD	Zeebu	1.128B
GT-USD	GateToken	1.116B
AERO29270-USD	Aerodrome Finance	1.108B
FLOW-USD	Flow	1.106B
STRK22691-USD	Starknet	1.095B
AR-USD	Arweave	1.095B
AIOZ-USD	AIOZ Network	1.092B
DYDX-USD	dYdX (Native)	1.057B
XDC-USD	XDC Network	1.059B
BSV-USD	Bitcoin SV	1.057B
IOTA-USD	IOTA	1.056B
CORE23254-USD	Core	1.024B
BTT-USD	BitTorrent(New)	1.024B
NEO-USD	Neo	1.007B
CRV-USD	Curve DAO Token	984.387M
FLZ-USD	Fellaz	977.298M
AXS-USD	Axie Infinity	953.685M
EGLD-USD	MultiversX	946.134M
MANA-USD	Decentraland	932.366M
FTT-USD	FTX Token	911.413M
MATIC-USD	Polygon	906.335M
NEXO-USD	Nexo	883.144M

Continued on next page

Symbol	Name	Market cap.
APE18876-USD	ApeCoin	863.233M
JTO-USD	Jito	818.162M
ZEC-USD	Zcash	815.92M
AKT-USD	Akash Network	807.599M
PENDLE-USD	Pendle	802.098M
CHZ-USD	Chiliz	782.452M
CFX-USD	Conflux	759.533M
W-USD	Wormhole	749.661M
RON14101-USD	Ronin	741.845M
POPCAT28782-USD	Popcat (SOL)	736.818M
SUPER8290-USD	SuperVerse	731.993M
SNX-USD	Synthetix	733.639M
MINA-USD	Mina	729.026M
CAKE-USD	PancakeSwap	718.496M
COMP5692-USD	Compound	701.895M
XEC-USD	eCash	688.608M
GNO-USD	Gnosis	683.216M
ZK24091-USD	ZKsync	664.322M
DOG30933-USD	Dog (Runes)	652.793M
NOT-USD	Notcoin	651.373M
XAUT-USD	Tether Gold	649.675M
AXL17799-USD	Axelar	649.217M
CHEX-USD	Chintai	645.043M
AMP-USD	Amp	645.218M
ZRO26997-USD	LayerZero	612.706M
SPX28081-USD	SPX6900	607.475M
RSR-USD	Reserve Rights	591.686M
LUNC-USD	Terra Classic	584.61M
ROSE-USD	Oasis	580.014M
ORDI-USD	ORDI	561.418M
BLUR-USD	Blur	551.582M
BDX-USD	Beldex	538.466M
DEXE-USD	DeXe	537.875M

Continued on next page

Symbol	Name	Market cap.
MEW30126-USD	cat in a dogs world	535.832M
1INCH-USD	1inch Network	533.185M
CHEEL-USD	Cheelee	530.15M
TURBO-USD	Turbo	524.046M
PAXG-USD	PAX Gold	522.828M
MGC29839-USD	Meta Games Coin	500.917M
CTC-USD	Creditcoin	496.607M
TWT-USD	Trust Wallet Token	487.794M
CKB-USD	Nervos Network	481.269M
TEL-USD	Telcoin	482.317M
LPT-USD	Livepeer	469.301M
KSM-USD	Kusama	463.042M
KAVA-USD	Kava	454.912M
ASTR-USD	Astar	446.988M
GIGA30063-USD	Gigachad (gigachadsolana.com)	439.326M
DASH-USD	Dash	441.9M
TFUEL-USD	Theta Fuel	429.903M
BOME-USD	BOOK OF MEME	429.226M
ETHFI-USD	ether.fi	427.754M
SNEK25264-USD	Snek	422.244M
HOT2682-USD	Holo	415.317M
VRSC-USD	VerusCoin	413.505M
ATH30083-USD	Aethir	411.247M
QUBIC-USD	Qubic	399.127M
ZIL-USD	Zilliqa	397.072M
CVX-USD	Convex Finance	393.342M
WOO-USD	WOO	392.589M
ZEN-USD	Horizen	390.205M
ZRX-USD	0x Protocol	389.502M
ENJ-USD	Enjin Coin	387.289M
SUSHI-USD	SushiSwap	386.486M
PRIME23711-USD	Echelon Prime	386.152M
IO29835-USD	io.net	379.013M

Continued on next page

Symbol	Name	Market cap.
GMT18069-USD	GMT	370.153M
JST-USD	JUST	367.755M
CELO-USD	Celo	361.935M
ETHW-USD	EthereumPoW	361.157M
ONE3945-USD	Harmony	361.718M
WEMIX-USD	WEMIX	360.091M
GLM-USD	Golem	355.658M
ANKR-USD	Ankr	349.357M
BORG-USD	SwissBorg	348.084M
MEME28301-USD	Memecoin	346.553M
PEPECOIN-USD	PepeCoin	347.115M
ARKM-USD	Arkham	341.502M
IOTX-USD	IoTeX	338.356M
CPOOL-USD	Clearpool	333.85M
ID21846-USD	SPACE ID	332.847M
AEVO-USD	Aevo	332.432M
ZETA-USD	ZetaChain	328.98M
MRS21178-USD	Metars Genesis	328.712M
BAT-USD	Basic Attention Token	326.511M
TRAC-USD	OriginTrail	324.556M
MANTA-USD	Manta Network	320.103M
SFP-USD	SafePal	322.44M
SC-USD	Siacoin	323.194M
DYM-USD	Dymension	322.018M
ETHDYDX-USD	dYdX (ethDYDX)	321.399M
QTUM-USD	Qtum	322.086M
OSMO-USD	Osmosis	320.671M
ELF-USD	aelf	319.037M
MX-USD	MX Token	315.962M
MWC-USD	MimbleWimbleCoin	314.377M
RVN-USD	Ravencoin	309.403M
DSYNC-USD	Destra Network	303.595M
TRIBE-USD	Tribe	299.43M

Continued on next page

Symbol	Name	Market cap.
LUNA20314-USD	Terra	295.708M
XCH-USD	Chia	295.711M
BTG-USD	Bitcoin Gold	294.65M
KDA-USD	Kadena	294.252M
MASK8536-USD	Mask Network	291.544M
GAS-USD	Gas	288.811M
YFI-USD	yearn.finance	287M
GMX11857-USD	GMX	285.66M
ALT29073-USD	Altlayer	282.252M
ORBR-USD	Orbler	283.216M
RLB-USD	Rollbit Coin	278.773M
T-USD	Threshold	275.099M
LRC-USD	Loopring	273.2M
DRIFT31278-USD	Drift	272.187M
METIS-USD	Metis	268.38M
RBTC-USD	Rootstock Smart Bitcoin	267.707M
XRD-USD	Radix	266.475M
XYO-USD	XYO	265.386M
SKL-USD	SKALE	265.217M
DCR-USD	Decred	259.674M
WILD-USD	Wilder World	259.655M
COW19269-USD	CoW Protocol	259.194M
BICO-USD	Biconomy	255.679M
0X0-USD	0x0.ai	255.824M
BZR-USD	Bazaars	245.903M
SSV-USD	ssv.network	244.424M
POLYX-USD	Polymesh	245.074M
RPL-USD	Rocket Pool	240.173M
PAAL-USD	PAAL AI	245.045M
FLUX-USD	Flux	225.84M
ANDY29879-USD	ANDY (ETH)	225.262M
CETUS-USD	Cetus Protocol	225.145M
ZKJ-USD	Polyhedra Network	223.613M

Continued on next page

Symbol	Name	Market cap.
VTHO-USD	VeThor Token	222.932M
COTI-USD	COTI	219.978M
UMA-USD	UMA	219.359M
TAI20605-USD	TARS AI	220.34M
FXS-USD	Frax Share	218.337M
BAND-USD	Band Protocol	218.343M
XEM-USD	NEM	218.135M
GLMR-USD	Moonbeam	216.843M
ILV-USD	Illuvium	215.495M
PEOPLE-USD	ConstitutionDAO	214.239M
BLAST28480-USD	Blast	211.058M
CHR-USD	Chromia	210.466M
HONEY22850-USD	Hivemapper	208.435M
EDU24613-USD	Open Campus	206.786M
YGG-USD	Yield Guild Games	206.09M
SUN-USD	Sun [New]	204.985M
ONT-USD	Ontology	202.074M
DGB-USD	DigiByte	201.029M
PIXEL29335-USD	Pixels	200.539M
BIGTIME-USD	Big Time	200.734M
GOMINING-USD	Go ining	200.231M
ACH-USD	Alchemy Pay	199.468M
VANRY-USD	Vanar Chain	199.381M
XAI28933-USD	Xai	197.124M
AUDIO-USD	Audius	196.013M
CSPR-USD	Casper	195.794M
VVS-USD	VVS Finance	193.556M
LCX-USD	LCX	191.452M
PONKE-USD	Ponke	191.686M
ORCA-USD	Orca	191.649M
WOLF30902-USD	Landwolf 0x67	190.566M
APU30008-USD	Apu Apustaja	188.401M
UPC-USD	UPCX	186.687M

Continued on next page

Symbol	Name	Market cap.
BITCOIN25220-USD	HarryPotterObamaSonic10Inu (ERC-20)	186.465M
SXP-USD	Solar	183.996M
NPC27960-USD	Non-Playable Coin	181.251M
BB30746-USD	BounceBit	181.09M
STORJ-USD	Storj	180.485M
XVG-USD	Verge	180.281M
MPLX-USD	Metaplex	178.46M
ACX22620-USD	Across Protocol	177.478M
DAG-USD	Constellation	174.018M
ZENT-USD	Zentry	171.878M
VELO-USD	Velo	174.093M
AGI-USD	Delysium	169.934M
ZIG-USD	Signal	172.643M
ICX-USD	ICON	171.06M
HT-USD	Huobi Token	169.962M
VELO20435-USD	Velodrome Finance	169.416M
SOLO-USD	Sologenic	168.528M
XNO-USD	Nano	168.538M
HEART-USD	Humans.ai	167.382M
CFG-USD	Centrifuge	167.088M
SLP-USD	Smooth Love Potion	165.958M
WAVES-USD	Waves	165.139M
CGPT-USD	ChainGPT	162.73M
BASEDAI-USD	BasedAI	163.971M
SNT-USD	Status	162.96M
SAGA30372-USD	Saga	163.011M
MYTH-USD	Mythos	162.125M
ABT-USD	Arcblock	160.657M
LQTY-USD	Liquity	161.275M
CVC-USD	Civic	160.485M
BNX23635-USD	BinaryX	158.709M
BAL-USD	Balancer	158.353M
TRB-USD	Tellor	156.844M

Continued on next page

Symbol	Name	Market cap.
DEGEN30096-USD	Degen	156.042M
ALPH-USD	Alephium	152.59M
STRAX30168-USD	Stratis [New]	148.225M
ZANO-USD	Zano	147.898M
JOE-USD	JOE	147.608M
WAXP-USD	WAX	147.859M
BORA-USD	BORA	146.884M
LSK-USD	Lisk	146.595M
ANYONE-USD	ANYONE Protocol	144.21M
POND-USD	Marlin	143.705M
CELR-USD	Celer Network	143.538M
TAIKO-USD	Taiko	141.606M
XVS-USD	Venus	140.783M
IOST-USD	IOST	141.32M
C98-USD	Coin98	141.488M
ANT-USD	Aragon	139.692M
BANANA28066-USD	Banana Gun	139.813M
API3-USD	API3	137.803M
OZO-USD	Ozone Chain	136.787M
NOS-USD	Nosana	136.303M
SHDW16868-USD	Shadow Token	136.15M
RLC-USD	iExec RLC	135.58M
KUB-USD	Bitkub Coin	135.285M
MERL-USD	Merlin Chain	135.182M
ULTIMA-USD	Ultima	134.439M
GEAR16360-USD	Gearbox Protocol	133.96M
PORTAL29555-USD	Portal	133.523M
IQ-USD	IQ	132.595M
FIDA-USD	Solana Name Service	131.911M
ERG-USD	Ergo	131.791M
OLAS-USD	Autonolas	129.805M
POWR-USD	Powerledger	128.834M
CTSI-USD	Cartesi	127.435M

Continued on next page

Symbol	Name	Market cap.
SDEX-USD	SmarDex	125.826M
ONG3217-USD	Ontology Gas	124.989M
AI28846-USD	Sleepless AI	124.146M
OAS-USD	Oasys	123.94M
KEEP-USD	Keep Network	122.89M
AURORA14803-USD	Aurora	121.867M
AUCTION-USD	Bounce Token	119.785M
GPU-USD	Node AI	118.875M
PYR-USD	Vulcan Forged (PYR)	118.279M
OMI19075-USD	ECOMI	117.324M
COQ-USD	Coq Inu	116.983M
HIVE-USD	Hive	116.479M
CUDOS-USD	CUDOS	116.365M
XYM-USD	Symbol	115.139M
MAGIC14783-USD	Treasure	115.135M
SCRT-USD	Secret	115.686M
NEURAL-USD	NeuralAI	114.197M
NMR-USD	Numeraire	114.647M
PRO-USD	Propy	113.916M
NTRN26680-USD	Neutron	113.91M
SPELL-USD	Spell Token	113.292M
RIO-USD	Realio Network	113.394M
DENT-USD	Dent	112.786M
ORAI-USD	Oraichain	112.812M
CYBER24781-USD	Cyber	112.291M
H2O19091-USD	H2O DAO	112.183M
MOVR-USD	Moonriver	111.738M
PUNDIX-USD	Pundi X (New)	111.99M
MVL-USD	MVL	111.134M
AZERO-USD	Aleph Zero	111.275M
TRU7725-USD	TrueFi	111.026M
DKA-USD	dKargo	107.487M
VERUM-USD	Verum Coin	107.339M

Continued on next page

Symbol	Name	Market cap.
BONE11865-USD	Bone ShibaSwap	106.863M
SYN12147-USD	Synapse	103.982M
HASHAI-USD	HashAI	103.949M
HIGH-USD	Highstreet	103.104M
MLK-USD	MiL.k	102.216M
WIN-USD	WINKLink	102.54M
DODO-USD	DODO	101.832M
AITECH-USD	Solidus Ai Tech	101.725M
OX29530-USD	OX Coin	101.256M
ACA-USD	Acala Token	101.083M
OXT-USD	Orchid	100.475M
SHIBTC-USD	ShibaBitcoin	100.18M
PROM-USD	Prom	99.731M
THE23335-USD	THENA	99.673M
NMT29447-USD	NetMind Token	99.579M
AGLD-USD	Adventure Gold	99.36M
RIF-USD	Rootstock Infrastructure Framework	99.398M
HFT22461-USD	Hashflow	99.113M
IAG-USD	IAGON	98.602M
ALU-USD	Altura	98.104M
RSC27054-USD	ResearchCoin	97.882M
ORBS-USD	Orbs	97.278M
BTRST-USD	Braintrust	96.846M
MED-USD	MediBloc	96.809M
CTK4807-USD	Shentu	96.778M
NAKA-USD	Nakamoto Games	96.715M
ALI16876-USD	Artificial Liquid Intelligence	95.943M
KNC-USD	Kyber Network Crystal v2	95.66M
OMNI30315-USD	Omni Network	94.453M
PHA-USD	Phala Network	94.205M
STEEM-USD	Steem	93.626M
ARK-USD	Ark	93.425M
WZRD30491-USD	Bitcoin Wizards	93.379M

Continued on next page

Symbol	Name	Market cap.
LMWR-USD	LimeWire	92.659M
ZCX-USD	Unizen	91.421M
LON-USD	Tokenlon Network Token	91.641M
NYM-USD	NYM	91.051M
DUSK-USD	Dusk	90.766M
MAV-USD	Maverick Protocol	90.263M
MOBILE-USD	Helium Mobile	90.405M
STPT-USD	STP	89.792M
DAR-USD	Mines of Dalarnia	89.773M
SFUND-USD	Seedify.fund	89.808M
MODE-USD	Mode	89.128M
AIC-USD	AI Companions	88.616M
MTL-USD	Metal DAO	86.349M
PCI-USD	Paycoin	87.358M
ACE28674-USD	Fusionist	86.959M
MOB-USD	MobileCoin	87.053M
SYS-USD	Syscoin	87.097M
QI9288-USD	BENQI	87.124M
TLOS-USD	Telos	86.662M
REQ-USD	Request	86.177M
ARDR-USD	Ardor	85.68M
RSS3-USD	RSS3	85.421M
DESO-USD	Decentralized Social	84.341M
HOOK-USD	Hooked Protocol	83.831M
AIAT-USD	AI Analysis Token	83.84M
SLERF-USD	SLERF	83.636M
BMX-USD	BitMart Token	83.55M
GAME31246-USD	GameBuild	83.156M
ISLM-USD	Islamic Coin	82.893M
CLV-USD	CLV	82.818M
BNT-USD	Bancor	81.355M
RACA-USD	RACA	81.322M
STTON-USD	bemo staked TON	80.181M

Continued on next page

Symbol	Name	Market cap.
ADS11036-USD	Alkimi	80.468M
PRCL-USD	Parcl	80.388M
COREUM-USD	Coreum	79.459M
RARE11294-USD	SuperRare	78.804M
LISTA-USD	Lista DAO	78.688M
DPI-USD	DeFi Pulse Index	77.963M
RISE15257-USD	EverRise	77.922M
DIA-USD	DIA	77.593M
MPL-USD	Maple	77.456M
APEX19843-USD	ApeX Protocol	77.073M
PNG-USD	Pangolin	77.254M
MBOX-USD	MOBOX	76.87M
PHB-USD	Phoenix	76.669M
TLM-USD	Alien Worlds	76.424M
TKO-USD	Toko Token	76.636M
TON-USD	Tokamak Network	75.863M
ALICE-USD	MyNeighborAlice	75.707M
BAKE-USD	BakeryToken	75.215M
REZ-USD	Renzo	74.954M
AVA-USD	AVA (Travala)	74.545M
SURE-USD	inSure DeFi	74.119M
NFP28778-USD	NFPrompt	72.914M
ALPHA-USD	Stella	73.477M
SAUCE-USD	SaucerSwap	73.004M
NKN-USD	NKN	72.897M
MICHI30943-USD	michi (SOL)	73.202M
OGN-USD	Origin Protocol	72.815M
LKY-USD	Luckycoin	72.349M
KARRAT-USD	KARRAT	72.161M
ARPA-USD	ARPA	72.081M
FORTH-USD	Ampleforth Governance Token	72.155M
PALM28567-USD	PaLM AI	72.315M
HIFI23037-USD	Hifi Finance	71.966M

Continued on next page

Symbol	Name	Market cap.
STMX-USD	StormX	71.47M
RETARDIO-USD	RETARDIO	70.754M
CTXC-USD	Cortex	70.821M
LOOM-USD	Loom Network	70.482M
AL-USD	ArchLoot	70.311M
NGL29709-USD	Entangle	70.048M
RDNT-USD	Radiant Capital	68.994M
VR-USD	Victoria VR	68.667M
QKC-USD	QuarkChain	67.4M
MBX18895-USD	MARBLEX	67.542M
CET-USD	CoinEx Token	66.24M
DEVVE-USD	Devve	64.88M
GODS-USD	Gods Unchained	65.797M
DAO-USD	DAO Maker	65.59M
MYRIA22289-USD	Myria	65.542M
BADGER-USD	Badger DAO	65.118M
XPLA-USD	XPLA	64.974M
UTK-USD	xMoney	64.856M
STIK-USD	Staika	65.103M
UQC-USD	Uquid Coin	64.023M
RAD-USD	Radworks	63.847M
XCN18679-USD	Onyxcoin	64.787M
APX-USD	APX	63.661M
WEN29175-USD	Wen	63.052M
MUMU30285-USD	Mumu the Bull (SOL)	63.008M
ATLAS-USD	Star Atlas	62.658M
LTO-USD	LTO Network	62.888M
DEGO-USD	Dego Finance	62.282M
POLY-USD	Polymath	61.873M
MYRO-USD	Myro	61.539M
PRQ-USD	PARSIQ	60.817M
VENOM-USD	Venom	60.554M
WOJAK-USD	Wojak	60.19M

Continued on next page

Symbol	Name	Market cap.
STG18934-USD	Stargate Finance	60.113M
AERGO-USD	Aergo	59.937M
BFC7817-USD	Bifrost	60.143M
ERN-USD	Ethernity	59.795M
LEVER-USD	LeverFi	59.538M
DEAI-USD	Zero1 Labs	59.509M
GTC10052-USD	Gitcoin	59.355M
RARI-USD	RARI	59.056M
FORT20622-USD	Forta	58.726M
POKT-USD	Pocket Network	58.541M
TNSR-USD	Tensor	58.283M
ACS23195-USD	Access Protocol	57.72M
LAT-USD	PlatON	57.525M
TOKE-USD	Tokemak	57.003M
HAI-USD	Hacken Token	56.411M
FX-USD	Function X	56.11M
MBL-USD	MovieBloc	56.245M
CBK-USD	Cobak Token	55.801M
COL20672-USD	Clash of Lilliput	55.643M
VSG-USD	Vector Smart Gas	54.841M
ETN-USD	Electroneum	54.695M
QRL-USD	Quantum Resistant Ledger	53.608M
ATA-USD	Automata Network	53.088M
DIONE-USD	Dione Protocol	53.048M
PERP-USD	Perpetual Protocol	52.702M
VCNT-USD	ViciCoin	52.715M
META-USD	Metadium	52.532M
LIT6833-USD	Litentry	52.107M
ALEX-USD	ALEX Lab	52.443M
REI19819-USD	REI Network	52.093M
DIMO-USD	DIMO	52.271M
TORN-USD	Tornado Cash	52.075M
GHX-USD	GamerCoin	51.106M

Continued on next page

Symbol	Name	Market cap.
GEMS31750-USD	Gems	51.212M
GHST-USD	Aavegotchi	51.112M
LOOKS-USD	LooksRare	50.868M
MLN-USD	Enzyme	50.403M
ELA-USD	Elastos	50.023M
TOKEN28299-USD	TokenFi	49.476M
CSWAP29780-USD	ChainSwap	49.756M
HOPPY30859-USD	Hoppy	49.397M
BWB31503-USD	Bitget Wallet Token	48.732M
MANEKI30912-USD	MANEKI	48.608M
OMG-USD	OMG Network	48.48M
DNT-USD	district0x	48.372M
DEP-USD	DEAPcoin	48.304M
GEL-USD	Gelato	48.007M
POLIS11213-USD	Star Atlas DAO	47.907M
NCT-USD	PolySwarm	47.537M
FUN-USD	FUNToken	47.78M
DATA-USD	Streamr	47.606M
MIN12787-USD	Minswap	47.096M
MAPO-USD	MAP Protocol	47.035M
ALCX-USD	Alchemix	46.851M
MOTHER-USD	Mother Iggy	46.73M
MCT16946-USD	Metacraft	46.596M
MCADE-USD	Metacade	46.368M
NAVX-USD	NAVI Protocol	46.386M
EUL-USD	Euler	46.19M
FLM-USD	Flamingo	45.9M
OORT29331-USD	OORT	45.928M
STRD-USD	Stride	45.394M
GFI13967-USD	Goldfinch	44.738M
WAN-USD	Wanchain	44.822M
MAVIA-USD	Heroes of Mavia	43.731M
NULS-USD	NULS	43.844M

Continued on next page

Symbol	Name	Market cap.
KLV-USD	Klever Coin	43.123M
IDEX-USD	IDEX	42.396M
EWT-USD	Energy Web Token	42.552M
NUM-USD	Numbers Protocol	42.473M
SD-USD	Stader	42.097M
BOSON-USD	Boson Protocol	41.606M
ROOT28479-USD	The Root Network	41.495M
COS-USD	Contentos	41.209M
GST16352-USD	Green Satoshi Token (SOL)	41.21M
FIS-USD	StaFi	40.947M
ICE27650-USD	Ice Open Network	40.684M
SWEAT-USD	Sweat Economy	40.597M
GROK28394-USD	Grok	40.575M
VRA-USD	Veracity	40.528M
BEL-USD	Bella Protocol	40.056M
MASA-USD	Masa	39.891M
GEOD-USD	GEODNET	39.983M
LOKA-USD	League of Kingdoms Arena	39.933M
MDT-USD	Measurable Data Token	39.592M
EL-USD	ELYSIA	39.633M
KMD-USD	Komodo	39.361M
POLS-USD	Polkastarter	39.332M
VOXEL-USD	Voxies	38.827M
PAID-USD	PAID	37.4M
CELL-USD	Cellframe	36.966M
RBN-USD	Ribbon Finance	36.819M
LINA7102-USD	Linear Finance	34.811M
BLZ-USD	Bluzelle	33.225M
WRX-USD	WazirX	32.777M

Table B.4: Dataset D

Symbol	Name	Market cap.
LEO-USD	UNUS SED LEO	8.555B
UNI7083-USD	Uniswap	7.553B
LTC-USD	Litecoin	7.333B
PEPE24478-USD	Pepe	6.902B
NEAR-USD	NEAR Protocol	6.264B
APT21794-USD	Aptos	5.921B
BGB-USD	Bitget Token	5.862B
ICP-USD	Internet Computer	4.844B
AAVE-USD	Aave	4.658B
CRO-USD	Cronos	4.197B
MNT27075-USD	Mantle	4.04B
ETC-USD	Ethereum Classic	3.974B
VET-USD	VeChain	3.755B
OM-USD	MANTRA	3.602B
XMR-USD	Monero	3.472B
TAO22974-USD	Bittensor	3.418B
ARB11841-USD	Arbitrum	3.233B
FET-USD	Artificial Superintelligence Alliance	3.181B
KAS-USD	Kaspa	3.102B
ENA-USD	Ethena	3.077B
FIL-USD	Filecoin	3.04B
ALGO-USD	Algorand	2.722B
FTM-USD	Fantom	2.713B
OKB-USD	OKB	2.691B
ATOM-USD	Cosmos	2.617B
STX4847-USD	Stacks	2.604B
VIRTUAL-USD	Virtuals Protocol	2.487B
OP-USD	Optimism	2.476B
ONDO-USD	Ondo	2.43B
IMX10603-USD	Immutable	2.385B
TIA22861-USD	Celestia	2.295B
BONK-USD	Bonk	2.156B

Continued on next page

Symbol	Name	Market cap.
INJ-USD	Injective	2.094B
GRT6719-USD	The Graph	2.021B
THETA-USD	Theta Network	1.994B
WIF-USD	dogwifhat	1.964B
SEI-USD	Sei	1.787B
WLD-USD	Worldcoin	1.787B
RUNE-USD	THORChain	1.769B
JASMY-USD	JasmyCoin	1.637B
FLOKI-USD	FLOKI	1.62B
FLR-USD	Flare	1.499B
MKR-USD	Maker	1.473B
LDO-USD	Lido DAO	1.469B
FTN-USD	Fasttoken	1.452B
BBTC31369-USD	BounceBit BTC	1.389B
BEAM28298-USD	Beam	1.346B
SAND-USD	The Sandbox	1.342B
PYTH-USD	Pyth Network	1.321B
QNT-USD	Quant	1.299B
KCS-USD	KuCoin Token	1.292B
BRETT29743-USD	Brett (Based)	1.289B
GALA-USD	Gala	1.276B
RAY-USD	Raydium	1.267B
ENS-USD	Ethereum Name Service	1.247B
EOS-USD	EOS	1.236B
XTZ-USD	Tezos	1.231B
HNT-USD	Helium	1.184B
JUP29210-USD	Jupiter	1.142B
ZBU-USD	Zeebu	1.128B
GT-USD	GateToken	1.116B
AERO29270-USD	Aerodrome Finance	1.108B
FLOW-USD	Flow	1.106B
STRK22691-USD	Starknet	1.095B
AR-USD	Arweave	1.095B

Continued on next page

Symbol	Name	Market cap.
AIOZ-USD	AIOZ Network	1.092B
DYDX-USD	dYdX (Native)	1.057B
XDC-USD	XDC Network	1.059B
BSV-USD	Bitcoin SV	1.057B
IOTA-USD	IOTA	1.056B
CORE23254-USD	Core	1.024B
BTT-USD	BitTorrent(New)	1.024B
NEO-USD	Neo	1.007B
CRV-USD	Curve DAO Token	984.387M
FLZ-USD	Fellaz	977.298M
AXS-USD	Axie Infinity	953.685M
EGLD-USD	MultiversX	946.134M
MANA-USD	Decentraland	932.366M
FTT-USD	FTX Token	911.413M
MATIC-USD	Polygon	906.335M
NEXO-USD	Nexo	883.144M
APE18876-USD	ApeCoin	863.233M
JTO-USD	Jito	818.162M
ZEC-USD	Zcash	815.92M
AKT-USD	Akash Network	807.599M
PENDLE-USD	Pendle	802.098M
CHZ-USD	Chiliz	782.452M
CFX-USD	Conflux	759.533M
W-USD	Wormhole	749.661M
RON14101-USD	Ronin	741.845M
POPCAT28782-USD	Popcat (SOL)	736.818M
SUPER8290-USD	SuperVerse	731.993M
SNX-USD	Synthetix	733.639M
MINA-USD	Mina	729.026M
CAKE-USD	PancakeSwap	718.496M
COMP5692-USD	Compound	701.895M
XEC-USD	eCash	688.608M
GNO-USD	Gnosis	683.216M

Continued on next page

Symbol	Name	Market cap.
ZK24091-USD	ZKsync	664.322M
DOG30933-USD	Dog (Runes)	652.793M
NOT-USD	Notcoin	651.373M
XAUT-USD	Tether Gold	649.675M
AXL17799-USD	Axelar	649.217M
CHEX-USD	Chintai	645.043M
AMP-USD	Amp	645.218M
ZRO26997-USD	LayerZero	612.706M
SPX28081-USD	SPX6900	607.475M
RSR-USD	Reserve Rights	591.686M
LUNC-USD	Terra Classic	584.61M
ROSE-USD	Oasis	580.014M
ORDI-USD	ORDI	561.418M
BLUR-USD	Blur	551.582M
BDX-USD	Beldex	538.466M
DEXE-USD	DeXe	537.875M
MEW30126-USD	cat in a dogs world	535.832M
1INCH-USD	1inch Network	533.185M
CHEEL-USD	Cheelee	530.15M
TURBO-USD	Turbo	524.046M
PAXG-USD	PAX Gold	522.828M
MGC29839-USD	Meta Games Coin	500.917M
CTC-USD	Creditcoin	496.607M
TWT-USD	Trust Wallet Token	487.794M
CKB-USD	Nervos Network	481.269M
TEL-USD	Telcoin	482.317M
LPT-USD	Livepeer	469.301M
KSM-USD	Kusama	463.042M
KAVA-USD	Kava	454.912M
ASTR-USD	Astar	446.988M
GIGA30063-USD	Gigachad (gigachadsolana.com)	439.326M
DASH-USD	Dash	441.9M
TFUEL-USD	Theta Fuel	429.903M

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Symbol	Name	Market cap.
BOME-USD	BOOK OF MEME	429.226M
ETHFI-USD	ether.fi	427.754M
SNEK25264-USD	Snek	422.244M
HOT2682-USD	Holo	415.317M
VRSC-USD	VerusCoin	413.505M
ATH30083-USD	Aethir	411.247M
QUBIC-USD	Qubic	399.127M
ZIL-USD	Zilliqa	397.072M
CVX-USD	Convex Finance	393.342M
WOO-USD	WOO	392.589M
ZEN-USD	Horizen	390.205M
ZRX-USD	0x Protocol	389.502M
ENJ-USD	Enjin Coin	387.289M
SUSHI-USD	SushiSwap	386.486M
PRIME23711-USD	Echelon Prime	386.152M
IO29835-USD	io.net	379.013M
GMT18069-USD	GMT	370.153M
JST-USD	JUST	367.755M
CELO-USD	Celo	361.935M
ETHW-USD	EthereumPoW	361.157M
ONE3945-USD	Harmony	361.718M
WEMIX-USD	WEMIX	360.091M
GLM-USD	Golem	355.658M
ANKR-USD	Ankr	349.357M
BORG-USD	SwissBorg	348.084M
MEME28301-USD	Memecoin	346.553M
PEPECOIN-USD	PepeCoin	347.115M
ARKM-USD	Arkham	341.502M
IOTX-USD	IoTeX	338.356M
CPOOL-USD	Clearpool	333.85M
ID21846-USD	SPACE ID	332.847M
AEVO-USD	Aevo	332.432M
ZETA-USD	ZetaChain	328.98M

Continued on next page

Symbol	Name	Market cap.
MRS21178-USD	Metars Genesis	328.712M
BAT-USD	Basic Attention Token	326.511M
TRAC-USD	OriginTrail	324.556M
MANTA-USD	Manta Network	320.103M
SFP-USD	SafePal	322.44M
SC-USD	Siacoin	323.194M
DYM-USD	Dymension	322.018M
ETHDYDX-USD	dYdX (ethDYDX)	321.399M
QTUM-USD	Qtum	322.086M
OSMO-USD	Osmosis	320.671M
ELF-USD	aelf	319.037M
MX-USD	MX Token	315.962M
MWC-USD	MimbleWimbleCoin	314.377M
RVN-USD	Ravencoin	309.403M
DSYNC-USD	Destra Network	303.595M
TRIBE-USD	Tribe	299.43M
LUNA20314-USD	Terra	295.708M
XCH-USD	Chia	295.711M
BTG-USD	Bitcoin Gold	294.65M
KDA-USD	Kadena	294.252M
MASK8536-USD	Mask Network	291.544M
GAS-USD	Gas	288.811M
YFI-USD	yearn.finance	287M
GMX11857-USD	GMX	285.66M
ALT29073-USD	Altlayer	282.252M
ORBR-USD	Orbler	283.216M
RLB-USD	Rollbit Coin	278.773M
T-USD	Threshold	275.099M
LRC-USD	Loopring	273.2M
DRIFT31278-USD	Drift	272.187M
METIS-USD	Metis	268.38M
RBTC-USD	Rootstock Smart Bitcoin	267.707M
XRD-USD	Radix	266.475M

Continued on next page

Symbol	Name	Market cap.
XYO-USD	XYO	265.386M
SKL-USD	SKALE	265.217M
DCR-USD	Decred	259.674M
WILD-USD	Wilder World	259.655M
COW19269-USD	CoW Protocol	259.194M
BICO-USD	Biconomy	255.679M
0X0-USD	0x0.ai	255.824M
BZR-USD	Bazaars	245.903M
SSV-USD	ssv.network	244.424M
POLYX-USD	Polymesh	245.074M
RPL-USD	Rocket Pool	240.173M
PAAL-USD	PAAL AI	245.045M
FLUX-USD	Flux	225.84M
ANDY29879-USD	ANDY (ETH)	225.262M
CETUS-USD	Cetus Protocol	225.145M
ZKJ-USD	Polyhedra Network	223.613M
VTHO-USD	VeThor Token	222.932M
COTI-USD	COTI	219.978M
UMA-USD	UMA	219.359M
TAI20605-USD	TARS AI	220.34M
FXS-USD	Frax Share	218.337M
BAND-USD	Band Protocol	218.343M
XEM-USD	NEM	218.135M
GLMR-USD	Moonbeam	216.843M
ILV-USD	Illuvium	215.495M
PEOPLE-USD	ConstitutionDAO	214.239M
BLAST28480-USD	Blast	211.058M
CHR-USD	Chromia	210.466M
HONEY22850-USD	Hivemapper	208.435M
EDU24613-USD	Open Campus	206.786M
YGG-USD	Yield Guild Games	206.09M
SUN-USD	Sun [New]	204.985M
ONT-USD	Ontology	202.074M

Continued on next page

Symbol	Name	Market cap.
DGB-USD	DigiByte	201.029M
PIXEL29335-USD	Pixels	200.539M
BIGTIME-USD	Big Time	200.734M
GOMINING-USD	Go ining	200.231M
ACH-USD	Alchemy Pay	199.468M
VANRY-USD	Vanar Chain	199.381M
XAI28933-USD	Xai	197.124M
AUDIO-USD	Audius	196.013M
CSPR-USD	Casper	195.794M
VVS-USD	VVS Finance	193.556M
LCX-USD	LCX	191.452M
PONKE-USD	Ponke	191.686M
ORCA-USD	Orca	191.649M
WOLF30902-USD	Landwolf 0x67	190.566M
APU30008-USD	Apu Apustaja	188.401M
UPC-USD	UPCX	186.687M
BITCOIN25220-USD	HarryPotterObamaSonic10Inu (ERC-20)	186.465M
SXP-USD	Solar	183.996M
NPC27960-USD	Non-Playable Coin	181.251M
BB30746-USD	BounceBit	181.09M
STORJ-USD	Storj	180.485M
XVG-USD	Verge	180.281M
MPLX-USD	Metaplex	178.46M
ACX22620-USD	Across Protocol	177.478M
DAG-USD	Constellation	174.018M
ZENT-USD	Zentry	171.878M
VELO-USD	Velo	174.093M
AGI-USD	Delysium	169.934M
ZIG-USD	Signalys	172.643M
ICX-USD	ICON	171.06M
HT-USD	Huobi Token	169.962M
VELO20435-USD	Velodrome Finance	169.416M
SOLO-USD	Sologenic	168.528M

Continued on next page

Symbol	Name	Market cap.
XNO-USD	Nano	168.538M
HEART-USD	Humans.ai	167.382M
CFG-USD	Centrifuge	167.088M
SLP-USD	Smooth Love Potion	165.958M
WAVES-USD	Waves	165.139M
CGPT-USD	ChainGPT	162.73M
BASEDAI-USD	BasedAI	163.971M
SNT-USD	Status	162.96M
SAGA30372-USD	Saga	163.011M
MYTH-USD	Mythos	162.125M
ABT-USD	Arcblock	160.657M
LQTY-USD	Liquity	161.275M
CVC-USD	Civic	160.485M
BNX23635-USD	BinaryX	158.709M
BAL-USD	Balancer	158.353M
TRB-USD	Tellor	156.844M
DEGEN30096-USD	Degen	156.042M
ALPH-USD	Alephium	152.59M
STRAX30168-USD	Stratis [New]	148.225M
ZANO-USD	Zano	147.898M
JOE-USD	JOE	147.608M
WAXP-USD	WAX	147.859M
BORA-USD	BORA	146.884M
LSK-USD	Lisk	146.595M
ANYONE-USD	ANyONe Protocol	144.21M
POND-USD	Marlin	143.705M
CELR-USD	Celer Network	143.538M
TAIKO-USD	Taiko	141.606M
XVS-USD	Venus	140.783M
IOST-USD	IOST	141.32M
C98-USD	Coin98	141.488M
ANT-USD	Aragon	139.692M
BANANA28066-USD	Banana Gun	139.813M

Continued on next page

Symbol	Name	Market cap.
API3-USD	API3	137.803M
OZO-USD	Ozone Chain	136.787M
NOS-USD	Nosana	136.303M
SHDW16868-USD	Shadow Token	136.15M
RLC-USD	iExec RLC	135.58M
KUB-USD	Bitkub Coin	135.285M
MERL-USD	Merlin Chain	135.182M
ULTIMA-USD	Ultima	134.439M
GEAR16360-USD	Gearbox Protocol	133.96M
PORTAL29555-USD	Portal	133.523M
IQ-USD	IQ	132.595M
FIDA-USD	Solana Name Service	131.911M
ERG-USD	Ergo	131.791M
OLAS-USD	Autonolas	129.805M
POWR-USD	Powerledger	128.834M
CTSI-USD	Cartesi	127.435M
SDEX-USD	SmarDex	125.826M
ONG3217-USD	Ontology Gas	124.989M
AI28846-USD	Sleepless AI	124.146M
OAS-USD	Oasys	123.94M
KEEP-USD	Keep Network	122.89M
AURORA14803-USD	Aurora	121.867M
AUCTION-USD	Bounce Token	119.785M
GPU-USD	Node AI	118.875M
PYR-USD	Vulcan Forged (PYR)	118.279M
OMI19075-USD	ECOMI	117.324M
COQ-USD	Coq Inu	116.983M
HIVE-USD	Hive	116.479M
CUDOS-USD	CUDOS	116.365M
XYM-USD	Symbol	115.139M
MAGIC14783-USD	Treasure	115.135M
SCRT-USD	Secret	115.686M
NEURAL-USD	NeuralAI	114.197M

Continued on next page

Symbol	Name	Market cap.
NMR-USD	Numeraire	114.647M
PRO-USD	Propy	113.916M
NTRN26680-USD	Neutron	113.91M
SPELL-USD	Spell Token	113.292M
RIO-USD	Realio Network	113.394M
DENT-USD	Dent	112.786M
ORAI-USD	Oraichain	112.812M
CYBER24781-USD	Cyber	112.291M
H2O19091-USD	H2O DAO	112.183M
MOVR-USD	Moonriver	111.738M
PUNDIX-USD	Pundi X (New)	111.99M
MVL-USD	MVL	111.134M
AZERO-USD	Aleph Zero	111.275M
TRU7725-USD	TrueFi	111.026M
DKA-USD	dKargo	107.487M
VERUM-USD	Verum Coin	107.339M
BONE11865-USD	Bone ShibaSwap	106.863M
SYN12147-USD	Synapse	103.982M
HASHAI-USD	HashAI	103.949M
HIGH-USD	Highstreet	103.104M
MLK-USD	MiL.k	102.216M
WIN-USD	WINKLink	102.54M
DODO-USD	DODO	101.832M
AITECH-USD	Solidus Ai Tech	101.725M
OX29530-USD	OX Coin	101.256M
ACA-USD	Acala Token	101.083M
OXT-USD	Orchid	100.475M
SHIBTC-USD	ShibaBitcoin	100.18M
PROM-USD	Prom	99.731M
THE23335-USD	THENA	99.673M
NMT29447-USD	NetMind Token	99.579M
AGLD-USD	Adventure Gold	99.36M
RIF-USD	Rootstock Infrastructure Framework	99.398M

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Symbol	Name	Market cap.
HFT22461-USD	Hashflow	99.113M
IAG-USD	IAGON	98.602M
ALU-USD	Altura	98.104M
RSC27054-USD	ResearchCoin	97.882M
ORBS-USD	Orbs	97.278M
BTRST-USD	Braintrust	96.846M
MED-USD	MediBloc	96.809M
CTK4807-USD	Shentu	96.778M
NAKA-USD	Nakamoto Games	96.715M
ALI16876-USD	Artificial Liquid Intelligence	95.943M
KNC-USD	Kyber Network Crystal v2	95.66M
OMNI30315-USD	Omni Network	94.453M
PHA-USD	Phala Network	94.205M
STEEM-USD	Steem	93.626M
ARK-USD	Ark	93.425M
WZRD30491-USD	Bitcoin Wizards	93.379M
LMWR-USD	LimeWire	92.659M
ZCX-USD	Unizen	91.421M
LON-USD	Tokenlon Network Token	91.641M
NYM-USD	NYM	91.051M
DUSK-USD	Dusk	90.766M
MAV-USD	Maverick Protocol	90.263M
MOBILE-USD	Helium Mobile	90.405M
STPT-USD	STP	89.792M
DAR-USD	Mines of Dalarnia	89.773M
SFUND-USD	Seedify.fund	89.808M
MODE-USD	Mode	89.128M
AIC-USD	AI Companions	88.616M
MTL-USD	Metal DAO	86.349M
PCI-USD	Paycoin	87.358M
ACE28674-USD	Fusionist	86.959M
MOB-USD	MobileCoin	87.053M
SYS-USD	Syscoin	87.097M

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Symbol	Name	Market cap.
QI9288-USD	BENQI	87.124M
TLOS-USD	Telos	86.662M
REQ-USD	Request	86.177M
ARDR-USD	Ardor	85.68M
RSS3-USD	RSS3	85.421M
DESO-USD	Decentralized Social	84.341M
HOOK-USD	Hooked Protocol	83.831M
AIAT-USD	AI Analysis Token	83.84M
SLERF-USD	SLERF	83.636M
BMX-USD	BitMart Token	83.55M
GAME31246-USD	GameBuild	83.156M
ISLM-USD	Islamic Coin	82.893M
CLV-USD	CLV	82.818M
BNT-USD	Bancor	81.355M
RACA-USD	RACA	81.322M
STTON-USD	bemo staked TON	80.181M
ADS11036-USD	Alkimi	80.468M
PRCL-USD	Parcl	80.388M
COREUM-USD	Coreum	79.459M
RARE11294-USD	SuperRare	78.804M
LISTA-USD	Lista DAO	78.688M
DPI-USD	DeFi Pulse Index	77.963M
RISE15257-USD	EverRise	77.922M
DIA-USD	DIA	77.593M
MPL-USD	Maple	77.456M
APEX19843-USD	ApeX Protocol	77.073M
PNG-USD	Pangolin	77.254M
MBOX-USD	MOBOX	76.87M
PHB-USD	Phoenix	76.669M
TLM-USD	Alien Worlds	76.424M
TKO-USD	Toko Token	76.636M
TON-USD	Tokamak Network	75.863M
ALICE-USD	MyNeighborAlice	75.707M

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Symbol	Name	Market cap.
BAKE-USD	BakeryToken	75.215M
REZ-USD	Renzo	74.954M
AVA-USD	AVA (Travala)	74.545M
SURE-USD	inSure DeFi	74.119M
NFP28778-USD	NFPrompt	72.914M
ALPHA-USD	Stella	73.477M
SAUCE-USD	SaucerSwap	73.004M
NKN-USD	NKN	72.897M
MICHI30943-USD	michi (SOL)	73.202M
OGN-USD	Origin Protocol	72.815M
LKY-USD	Luckycoin	72.349M
KARRAT-USD	KARRAT	72.161M
ARPA-USD	ARPA	72.081M
FORTH-USD	Ampleforth Governance Token	72.155M
PALM28567-USD	PaLM AI	72.315M
HIFI23037-USD	Hifi Finance	71.966M
STMX-USD	StormX	71.47M
RETARDIO-USD	RETARDIO	70.754M
CTXC-USD	Cortex	70.821M
LOOM-USD	Loom Network	70.482M
AL-USD	ArchLoot	70.311M
NGL29709-USD	Entangle	70.048M
RDNT-USD	Radiant Capital	68.994M
VR-USD	Victoria VR	68.667M
QKC-USD	QuarkChain	67.4M
MBX18895-USD	MARBLEX	67.542M
CET-USD	CoinEx Token	66.24M
DEVVE-USD	Devve	64.88M
GODS-USD	Gods Unchained	65.797M
DAO-USD	DAO Maker	65.59M
MYRIA22289-USD	Myria	65.542M
BADGER-USD	Badger DAO	65.118M
XPLA-USD	XPLA	64.974M

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Symbol	Name	Market cap.
UTK-USD	xMoney	64.856M
STIK-USD	Staika	65.103M
UQC-USD	Uquid Coin	64.023M
RAD-USD	Radworks	63.847M
XCN18679-USD	Onyxcoin	64.787M
APX-USD	APX	63.661M
WEN29175-USD	Wen	63.052M
MUMU30285-USD	Mumu the Bull (SOL)	63.008M
ATLAS-USD	Star Atlas	62.658M
LTO-USD	LTO Network	62.888M
DEGO-USD	Dego Finance	62.282M
POLY-USD	Polymath	61.873M
MYRO-USD	Myro	61.539M
PRQ-USD	PARSIQ	60.817M
VENOM-USD	Venom	60.554M
WOJAK-USD	Wojak	60.19M
STG18934-USD	Stargate Finance	60.113M
AERGO-USD	Aergo	59.937M
BFC7817-USD	Bifrost	60.143M
ERN-USD	Ethernity	59.795M
LEVER-USD	LeverFi	59.538M
DEAI-USD	Zero1 Labs	59.509M
GTC10052-USD	Gitcoin	59.355M
RARI-USD	RARI	59.056M
FORT20622-USD	Forta	58.726M
POKT-USD	Pocket Network	58.541M
TNSR-USD	Tensor	58.283M
ACS23195-USD	Access Protocol	57.72M
LAT-USD	PlatON	57.525M
TOKE-USD	Tokemak	57.003M
HAI-USD	Hacken Token	56.411M
FX-USD	Function X	56.11M
MBL-USD	MovieBloc	56.245M

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Symbol	Name	Market cap.
CBK-USD	Cobak Token	55.801M
COL20672-USD	Clash of Lilliput	55.643M
VSG-USD	Vector Smart Gas	54.841M
ETN-USD	Electroneum	54.695M
QRL-USD	Quantum Resistant Ledger	53.608M
ATA-USD	Automata Network	53.088M
DIONE-USD	Dione Protocol	53.048M
PERP-USD	Perpetual Protocol	52.702M
VCNT-USD	ViciCoin	52.715M
META-USD	Metadium	52.532M
LIT6833-USD	Litentry	52.107M
ALEX-USD	ALEX Lab	52.443M
REI19819-USD	REI Network	52.093M
DIMO-USD	DIMO	52.271M
TORN-USD	Tornado Cash	52.075M
GHX-USD	GamerCoin	51.106M
GEMS31750-USD	Gems	51.212M
GHST-USD	Aavegotchi	51.112M
LOOKS-USD	LooksRare	50.868M
MLN-USD	Enzyme	50.403M
ELA-USD	Elastos	50.023M
TOKEN28299-USD	TokenFi	49.476M
CSWAP29780-USD	ChainSwap	49.756M
HOPPY30859-USD	Hoppy	49.397M
BWB31503-USD	Bitget Wallet Token	48.732M
MANEKI30912-USD	MANEKI	48.608M
OMG-USD	OMG Network	48.48M
DNT-USD	district0x	48.372M
DEP-USD	DEAPcoin	48.304M
GEL-USD	Gelato	48.007M
POLIS11213-USD	Star Atlas DAO	47.907M
NCT-USD	PolySwarm	47.537M
FUN-USD	FUNToken	47.78M

Continued on next page

Symbol	Name	Market cap.
DATA-USD	Streamr	47.606M
MIN12787-USD	Minswap	47.096M
MAPO-USD	MAP Protocol	47.035M
ALCX-USD	Alchemix	46.851M
MOTHER-USD	Mother Iggy	46.73M
MCT16946-USD	Metacraft	46.596M
MCAD-USD	Metacade	46.368M
NAVX-USD	NAVI Protocol	46.386M
EUL-USD	Euler	46.19M
FLM-USD	Flamingo	45.9M
OORT29331-USD	OORT	45.928M
STRD-USD	Stride	45.394M
GFI13967-USD	Goldfinch	44.738M
WAN-USD	Wanchain	44.822M
MAVIA-USD	Heroes of Mavia	43.731M
NULS-USD	NULS	43.844M
KLV-USD	Klever Coin	43.123M
IDEX-USD	IDEX	42.396M
EWT-USD	Energy Web Token	42.552M
NUM-USD	Numbers Protocol	42.473M
SD-USD	Stader	42.097M
BOSON-USD	Boson Protocol	41.606M
ROOT28479-USD	The Root Network	41.495M
COS-USD	Contentos	41.209M
GST16352-USD	Green Satoshi Token (SOL)	41.21M
FIS-USD	StaFi	40.947M
ICE27650-USD	Ice Open Network	40.684M
SWEAT-USD	Sweat Economy	40.597M
GROK28394-USD	Grok	40.575M
VRA-USD	Verasity	40.528M
BEL-USD	Bella Protocol	40.056M
MASA-USD	Masa	39.891M
GEOD-USD	GEODNET	39.983M

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Symbol	Name	Market cap.
LOKA-USD	League of Kingdoms Arena	39.933M
MDT-USD	Measurable Data Token	39.592M
EL-USD	ELYSIA	39.633M
KMD-USD	Komodo	39.361M
POLS-USD	Polkastarter	39.332M
VOXEL-USD	Voxies	38.827M
PAID-USD	PAID	37.4M
CELL-USD	Cellframe	36.966M
RBN-USD	Ribbon Finance	36.819M
LINA7102-USD	Linear Finance	34.811M
BLZ-USD	Bluzelle	33.225M
WRX-USD	WazirX	32.777M

Table B.5: Dataset E

Symbol	Name	Market cap.
USDT-USD	Tether USDt	140.553B
USDC-USD	USD Coin	42.213B
USDE29470-USD	Ethena USDe	5.93B
DAI-USD	Dai	5.366B
FDUSD-USD	First Digital USD	1.725B
USDD-USD	USDD	745.552M
FRAX-USD	Frax	645.449M
TUSD-USD	TrueUSD	496.829M
PYUSD-USD	PayPal USD	496.617M
USDY29256-USD	Ondo US Dollar Yield	448.436M
USDJ-USD	USDJ	149.485M
EURS-USD	STASIS EURO	129.194M
USDB29599-USD	USDB	111.583M
USTC-USD	TerraClassicUSD	103.524M
USDP-USD	Pax Dollar	93.848M
EURC-USD	EURC	89.958M
USDX6651-USD	USDX [Kava]	79.904M
CRVUSD-USD	crvUSD	77.063M
BUSD-USD	BUSD	68.32M
LUSD-USD	Liquity USD	61.953M
GUSD-USD	Gemini Dollar	61.963M
AEUR-USD	Anchored Coins AEUR	56.254M
SBD-USD	Steem Dollars	42.377M